

Abstract

The next generation of gravitational-wave (GW) observatories, Laser Interferometer Space Antenna (LISA) and the Einstein Telescope (ET), will jointly probe compact objects with masses ranging from a few to $\sim 10^7 M_\odot$ over the frequency interval extending from 10^{-4} to 10^4 Hz. This unprecedented observational window makes it essential to identify promising GW sources, determine which of them may emit a detectable signal and characterize its physical properties. Among the most compelling candidates are intermediate-mass black hole (IMBH) binaries, whose inspiral GW emission would offer unique insights into this still poorly constrained population of massive black holes (MBHs).

This thesis investigates the dynamical evolution of IMBH binaries embedded in stellar backgrounds, with particular emphasis on the coalescence timescale and the time evolution of the binary separation and eccentricity. To this end, both the Quinlan–Sesana hybrid model and Ketju direct N-body simulations are employed. While the former approach provides an efficient framework for reconstructing the evolution of BH binaries once the stellar background properties and the binary initial conditions are specified, the Ketju module, implemented into GADGET-4, modeling SMBHs and stars as point particles during close encounters through an algorithmically regularized integrator, accurately captures the orbital decay driven by stellar scattering with unprecedented spatial resolution. The analysis explores different stellar backgrounds, including nucleated dwarf galaxies and observationally motivated high-redshift stellar clumps and galaxy models with enhanced stellar densities resulting from cosmic size evolution. The obtained results confirm that IMBH binaries embedded in nucleated dwarf galaxies can merge within a few Myr. More remarkably, they show that even non-nucleated high-redshift dwarf galaxies and stellar clumps may drive binary coalescence within a Hubble time, identifying these systems as promising sources of detectable gravitational waves. Finally, the reconstructed binary evolution is finally exploited to compute the associated GW signals, providing observational predictions for the LISA and Einstein Telescope era.

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INTRODUCTION

Massive black holes play a fundamental role in the formation and evolution of galaxies. Observations have established that they are ubiquitously found at the centers of nearby galaxies (Kormendy and Richstone 1995), while tight empirical correlations between their masses and the properties of the host galactic bulges suggest a close co-evolution of black holes and galaxies (Magorrian et al. 1998). Moreover, MBHs are believed to be common in galaxy centres at all cosmic epochs; a piece of evidence for their existence, from low to high redshift, comes from the transient activity of galactic nuclei (AGN) over the hosts lifetime (see, e.g., Jahnke (2025)).

Black holes (BHs) span a wide range of masses ($\sim 10 - 10^9 M_\odot$) and various formation and accretion models have been proposed based on their mass. At the two extremes of the black-hole mass spectrum lie stellar-mass black holes and supermassive black holes (SMBHs). Stellar evolution theory predicts that stellar-mass BHs may be the remnants of massive stars with either Zero Age Main Sequence mass $M_{ZAMS} \lesssim 140 M_\odot$ or $M_{ZAMS} \gtrsim 260 M_\odot$; otherwise, they undergo fully disruption and end their life as Pair Instability Super Novae, leaving no remnant behind (Farmer et al. 2019). In recent years, the properties of this population have been significantly constrained through the detection of gravitational waves from merging compact binaries (Abbott et al. 2016), as well as by advances in microlensing techniques capable of identifying isolated BH candidates (Bennett et al. 2002). On the opposite side ($\sim 10^6 - 10^9 M_\odot$), instead, Pulsar timing arrays has already probed the upper mass range of SMBHs population (Taylor 2021), whose most accepted formation model interpret them as the final stage of a secular accretion onto initially lighter black hole seeds (see, e.g., Peterson (1997)).

While substantial progress has been made in detecting and characterizing both stellar-mass BHs and SMBHs, the observational confirmation and detailed investigation of intermediate-mass black holes (IMBHs), lying in the mass range $\sim 10^2 - 10^5 M_\odot$, remain significantly more challenging. Although an increasing number of candidates have been identified in the local Universe, particularly in dwarf galaxies, through accretion signatures (see, e.g., Reines et al. (2013)), their sphere of influence—conventionally defined as the sphere enclosing a stellar mass equal to twice that of the central BH (Sesana and Khan 2015), within which the BH gravitational potential dominates—is typically too small for reliable stellar or gas dynamical measurements, especially at higher redshift. Nevertheless, the discovery of high-redshift quasars (see, e.g., Bañados et al. (2018)) points to the existence of BH seeds with masses comparable to those of IMBHs already in the early Universe, strengthening the idea that even the nuclei of less massive galaxies harbor MBHs. Therefore, it is common practice to infer the central BH mass by extrapolating the empirical correlation between BH mass and the stellar mass of the host galaxy’s bulge, even for quiescent galactic nuclei (see, e.g., Schutte et al. (2019); Maiolino et al. (2024)).

Figure 1 illustrates three distinct theoretical pathways for IMBH formation, highlighting the number of BHs expected to form through each channel, their corresponding mass ranges, and the redshift range over which each formation mechanism is expected to be most efficient. Each formation channel proceeds as follows (Greene et al. 2020):

Pop III. Sufficiently massive primordial stars, commonly referred to as Population III (Pop III) stars, may possess enough gravitational binding energy to withstand the energy released during the photodisintegration-induced collapse of their cores, ultimately leading to the formation of a BH remnant (Fryer et al. 2001).

Direct collapse. At high redshift, metal-free gas clouds may undergo nearly monolithic gravitational collapse without significant fragmentation, in contrast to what is typically observed in the present-day Universe, owing to the inefficient cooling of neutral hydrogen, and lead to the formation of IMBHs (Lodato and Natarajan 2006).

Gravitational runaway. Runaway stellar collisions in high-density clusters may trigger a sequence of stellar mergers, progressively producing more massive stars that can undergo further collisions. The final product of this repeated mergers process may eventually collapse into an IMBH (see, e.g., Rodriguez et al. (2019); Devecchi and Volonteri (2009)).

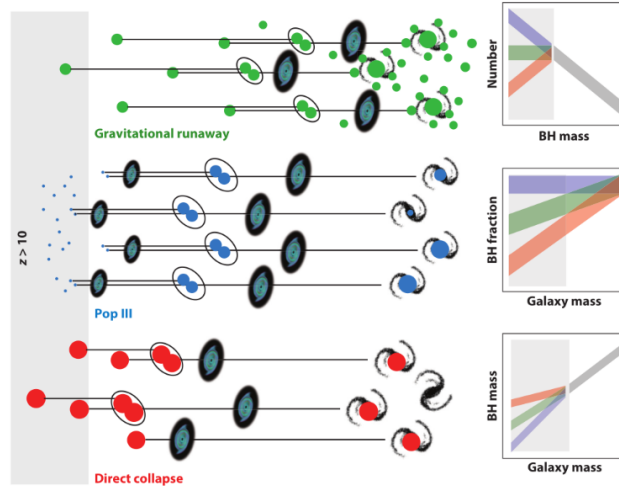


Figure 1: Schematic representation of different BH seeding channels, from their formation and subsequent growth through mergers and accretion to the present day: Pop III stars (blue), direct collapse (red), and gravitational runaway (green). The corresponding present-day BH mass functions, occupation fractions, and BH–galaxy scaling relations are shown on the right. Picture from Greene et al. (2020).

Within the framework of the hierarchical growth scenario of galaxies (White and Rees 1978), the formation of IMBH binaries is expected to occur naturally as a consequence of mergers between their host stellar systems (Maiolino et al. 2024). These systems are among the most powerful sources of gravitational waves (GWs) in the Universe (Maggiore 2007), making them prime targets for current and future GW observatories. In particular, the upcoming Laser Interferometer Space Antenna (LISA), a joint ESA/NASA space mission, will probe the low-frequency GW regime, from $\sim 10^{-4}$ to $\sim 10^{-1}$ Hz (Amaro-Seoane, Audley, et al. 2017; Amaro-Seoane, Andrews, et al. 2023), providing access to compact-object binaries with masses of $\sim 10^4 - 10^7, M_{\odot}$ and enabling the detection of their mergers out to redshift $z \sim 20$ (Colpi et al. 2024). At higher frequencies, the ground-based Einstein Telescope (ET) will complement LISA by extending the accessible frequency and mass ranges towards lower-mass systems, being expected to detect IMBH binaries with masses of $\sim 10^2 - 10^3, M_{\odot}$ out to redshifts of a few (Abac et al. 2026). Therefore, current and future GW observatories may provide a unique opportunity to unveil the population of IMBHs through their gravitational-wave emission, which may be detectable even when their electromagnetic signatures are too faint to be observed.

In light of this perspective, this thesis investigates the time evolution of the main orbital parameters of IMBH binaries, with particular focus on the semimajor axis a and eccentricity e , and examines how BH masses, stellar background properties, and the binary initial condition influence their dynamical evolution and coalescence timescale through a series of numerical experiments. To this end, the evolution of IMBH binaries is studied using two complementary approaches: the numerical integration of the differential equations for the binary separation and eccentricity provided by the hybrid model developed by Quinlan (1996) and extended by Sesana (2010), and high-resolution, gas-free N-body simulations of mergers between BHs and their stellar hosts performed with the hybrid tree-regularized code KETJU (Mannerkoski et al. 2023).

The suite of numerical experiments presented in Khan and Holley-Bockelmann (2021) and Khan, Javed, et al. (2024) on both nucleated and non-nucleated dwarf galaxies have highlighted the crucial role of the stellar background density, particularly in the vicinity of the BHB sphere of influence, in driving fast and efficient mergers. In particular, high-density stellar environments ensure an adequate supply of low-angular-momentum stars, thereby sustaining efficient hardening via stellar slingshot (see, e.g., Mikkola and Valtonen (1992)), while in low-density environments, the insufficient replenishment of such orbits may lead to binary stalling, giving rise to the so-called "last-parsec problem" (Milosavljević 2003). The evidence that different stellar environments produce peculiar pathways to coalescence highlights the need to investigate the dynamical evolution of IMBH binaries across a representative range of realistic stellar hosts. The first class considered here comprises dwarf galaxies, whose most commonly adopted definition relies on an upper limit of $M_V > -18$ in absolute (V)-band magnitude Grebel et al. (2003). The observation of these systems even at high redshifts (see, e.g., Nguyen et al. (2018); (Kikuchihara et al. 2020)) makes them plausible stellar hosts at the time of IMBH pairing, whose GW emission may be probed by present-day observatories. Although N-body simulations have shown that IMBH binaries embedded in non-nucleated dwarf galaxies have coalescence times exceeding the Hubble time, and despite the fact that most dwarf galaxies harbor a nuclear star cluster (see, e.g., Neumayer et al. (2020)), we artificially evolve non-nucleated dwarf galaxies back to redshift z according to the cosmic size evolution law derived by Ono et al. (2023). Together with the observed increase in the BH-to-bulge mass ratio towards higher redshifts as shown in Figure 2, this effect is expected to partially compensate for the absence of a nuclear star cluster, potentially allowing the binary to coalesce within the present-day Universe.

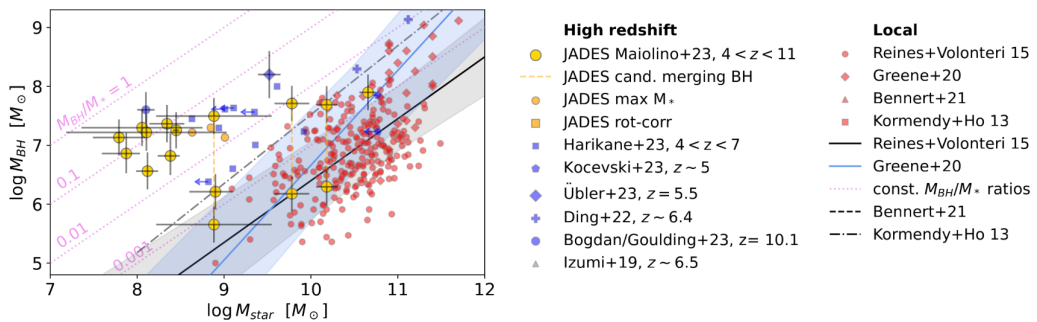


Figure 2: Black hole mass as a function of host galaxy stellar mass. Golden circles denote the JADES results, blue symbols high- z JWST measurements, grey triangles high- z QSOs, and red symbols local galaxies. Dashed orange lines connect candidate merging BHs. Picture from Maiolino et al. (2024).

In recent years, several claims have been made regarding the possible presence of IMBHs in Galactic and extragalactic globular clusters (GCs). Although these detections are not always conclusive, dynamical measurements and signatures of accretion in the X-ray band have

provided indications of central IMBHs in a number of old stellar clusters (e.g., Gebhardt et al. (2005); H. Baumgardt (2017)). At the same time, observations of galaxies at very high redshift are revealing stellar systems with properties that may provide insights into the possible progenitors of present-day GCs. An example of this is constituted by the "Misty Moons" (Nakane et al. 2026) and the "Bullet arc" (Bradač et al. 2025), two strongly lensed galaxies at a photometric redshift of $z_{\text{phot}} \sim 11 - 12$ and spectroscopic redshift $z_{\text{spec}} \sim 11$ respectively, which are resolved into several massive stellar clumps. This suggests that star formation in the early universe may have been clump-dominated, with such stellar clumps potentially representing the progenitors of the more compact and less massive GCs observed in the local Universe, before undergoing substantial mass loss due to tidal interactions with their host galaxies. Figure 3 shows the size–mass relation of local star clusters, star-forming clumps, and dwarf galaxies, providing a direct comparison between the properties of local GCs and high- z stellar clumps.

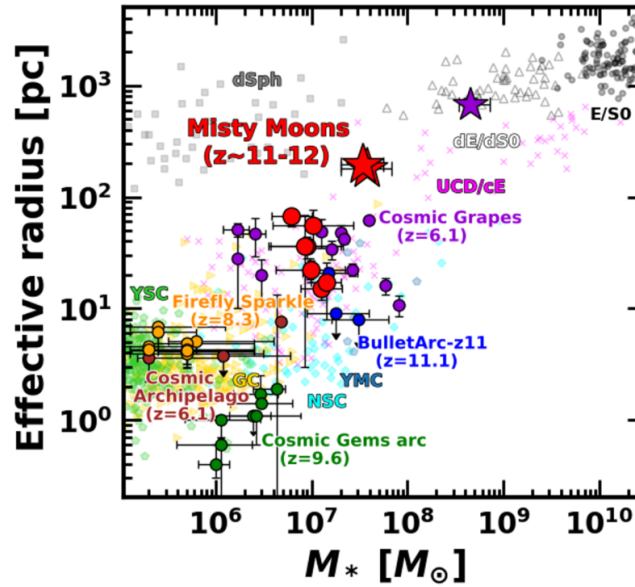


Figure 3: Size–mass relation. Stars denote the entire systems, while circles denote individual stellar clumps. Other symbols denote local Globular Clusters, young massive star clusters, and Nuclear Star Clusters. Picture from Nakane et al. (2026).

Mindful of these considerations, we also include stellar-clump-like backgrounds in our suite of numerical experiments, adopting IMBH masses consistent with the empirical scaling relation shown in Figure 2.

The structure of this thesis is outlined as follows.

In chapter 1, we provide a brief overview of the main physical mechanisms promoting the shrinking of BH binaries, including violent relaxation, dynamical friction, stellar scattering by low-angular-momentum stars, and GW emission.

In chapter 2, it will be introduced the self-consistent hybrid model developed by Quinlan (1996) and Sesana (2010). We then present the evolutionary tracks obtained for the adopted sample of host galaxies and explore the relevant parameter space, highlighting how different environmental properties and initial conditions affect the binary evolution.

In chapter 3 are reported the main features of the codes used to generate equilibrium N-body initial conditions and carry out the numerical simulations. The chapter then presents the results obtained by combining the KETJU evolution, which follows the BH pair from early merger to the stellar-scattering-dominated regime, with the hybrid model used to track its subsequent evolution towards final coalescence.

Finally, the main results of this work are summarized, and possible future improvements are reported in the concluding chapter.

PHYSICAL MECHANISMS OF BINARY HARDENING

In the "bottom-up" scenario, in which low-mass structures form first and progressively more massive systems assemble through hierarchical clustering and mergers, MBH binaries are expected to form as a consequence of mergers involving stellar systems or galaxies embedding such compact objects. This chapter reviews the main physical mechanisms driving the hardening of MBH binaries, from the initial merger of their hosts to the final coalescence. We first discuss violent relaxation, which efficiently converts the orbital energy of the merging systems into disordered kinetic energy of the newly formed system. Then, dynamical friction is considered. It acts on massive components such as MBHs and drags them toward the centre of the merger remnant, eventually leading to the formation of a gravitationally bound binary. The subsequent evolution is governed by three-body interactions with low-angular momentum stars: we discuss the gravitational slingshot mechanism, through which stellar ejections extract energy and angular momentum from the binary and promote its hardening, as well as the role of loss-cone refilling. Finally, we derive the key equations describing the binary orbital evolution once gravitational-wave emission becomes the dominant mechanism governing the orbital decay.

1.1 Galaxy Merger and Violent Relaxation

When two galaxies approach each other with sufficiently low relative velocity and small impact parameter, the encounter may result in a galaxy merger (see, e.g., A. Toomre and J. Toomre (1972); Cimatti et al. (2019)). Galaxy mergers can be broadly classified according to the physical properties of the systems involved. Of particular relevance here are dissipationless, or dry, mergers, which involve gas-poor galaxies and are characterized by negligible radiative energy losses, so that the total energy of the system is approximately conserved (Khochfar and Silk 2009). Numerical N -body simulations have shown that collisions between galaxies of comparable mass, commonly referred to "major mergers", generally result in a single, virialized remnant (see, e.g. (Barnes and Hernquist 1992); Hilz et al. (2012)). The formation of this remnant proceeds through a series of relaxation processes driven by the rapidly varying gravitational potential during the encounter, which redistributes the energy and angular momentum of the system and drives the initially out-of-equilibrium configuration toward a new equilibrium state (see, e.g. Lynden-Bell (1967); Saslaw (1970)).

In the scientific literature, several approaches have been proposed to define and derive the timescale of violent relaxation. Here, we follow the approach of Lynden-Bell (1967) and define it as the characteristic oscillation period of the newly formed galaxy around the equilibrium configuration established by the virial condition

$$2K + W = 0, \quad (1.1)$$

holding in its barycentric inertial reference system S_{CM} , with K and W standing for the total kinetic and gravitational potential energy of the galaxy remnant respectively. We therefore first characterize the equilibrium state attained by the system following the relaxation process.

Let us consider two galaxies, each with mass M_i and total energy $E_i = K_i + W_i$. In an ideal dissipationless merger, the total mass and energy are conserved, so the resulting remnant has mass $M = M_1 + M_2$ and energy $E = E_1 + E_2 + E_{\text{orb}}$, where E_{orb} accounts for the orbital energy of the pair, including both its ordered kinetic and mutual gravitational contributions. Following violent relaxation on a timescale T_r , the remnant settles into an equilibrium configuration that can be characterized by the virial velocity dispersion σ_v and virial radius r_v , defined by

$$\sigma_v^2 = \frac{2K}{M}, \quad r_v = -\frac{GM^2}{W}, \quad (1.2)$$

which provide proxies for the characteristic stellar velocity dispersion and size of the galaxy in equilibrium, thus representing key information on the final stage reached by the merging galaxies.

By combining the virial condition (1.1)—which applies both to the progenitor galaxies prior to the interaction and to the remnant after the violent relaxation timescale T_r —with energy conservation, one obtains

$$\sigma_v^2 = \frac{M_1\sigma_{v,1}^2 + M_2\sigma_{v,2}^2 - 2E_{\text{orb}}}{M_1 + M_2} \quad (1.3)$$

and

$$\frac{(M_1 + M_2)^2}{r_v} = \frac{M_1^2}{r_{v,1}} + \frac{M_2^2}{r_{v,2}} - \frac{2E_{\text{orb}}}{G}, \quad (1.4)$$

where $\sigma_{v,i}$ and $r_{v,i}$ denote the virial velocity dispersion and virial radius, respectively, of the colliding systems (Cimatti et al. 2019). From Equations (1.3) and (1.4), it follows that parabolic encounters between identical progenitors, defined by $E_{\text{orb}} = 0$, which are the configuration considered here to carry out N -body numerical experiments, give rise to a galaxy remnant having

$$M = 2M_1 = 2M_2, \quad r_v = 2r_{v,1} = 2r_{v,2}, \quad \sigma_v = \sigma_{v,1} = \sigma_{v,2}. \quad (1.5)$$

Therefore, before the merger to occur, each progenitor galaxy, which is assumed to be made of N stars of masses m_i and velocities $\mathbf{v}_i$, is individually virialized and satisfies the virial equilibrium condition

$$2 \sum_{i=1}^N \frac{1}{2} m_i v_i^2 - \frac{1}{2} \sum_{i=1}^N \sum_{j \neq i}^N \frac{G m_i m_j}{|\mathbf{x}_i - \mathbf{x}_j|} = 2K + W = 0. \quad (1.6)$$

This condition is temporarily violated once the collision begins, as the newly formed remnant is driven far from equilibrium by the coherent motion inherited from the relative orbital motion of the progenitors. Consequently, during violent relaxation, Equation (1.1) is no longer satisfied at every instant; rather, the system obeys the time-dependent virial theorem,

$$\frac{1}{2} \ddot{I} = 2K + W, \quad (1.7)$$

where I denotes the polar moment of inertia.

To quantitatively characterize the level of virialization of any system evolving towards equilibrium, it is useful to introduce the so-called virial ratio

$$Q \equiv \frac{2K}{|W|}. \quad (1.8)$$

Figure 1.1, and particularly the green curves, illustrates how the virial ratio Q oscillates about its equilibrium value, $Q = 1$, while the amplitude of these oscillations rapidly decreases with time as the ordered kinetic energy is progressively redistributed into random stellar motions. This behaviour, also observed in numerical N -body simulations of galaxy mergers (see, e.g., Nipoti et al. (2007)), indicates that the system progressively approaches virial equilibrium and that most of the virialization process takes place in the vicinity of the final equilibrium configuration.

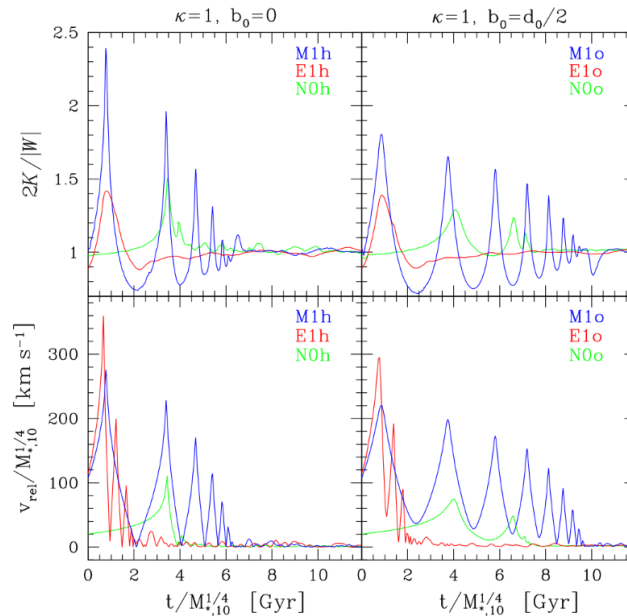


Figure 1.1: Time evolution of the virial ratio (top panels) and of the barycentric relative speed (bottom panels) for the two sets of galaxy merger simulations: head-on (left panels) and off-centre (right panels). Blue, red, and green curves refer to MOND, equivalent Newtonian, and purely baryonic Newtonian simulations, respectively. Picture from Nipoti et al. (2007).

We are now in a position to introduce the violent relaxation timescale T_r and derive an estimate for it. To this end, it is useful to start from the physical mechanism underlying violent

relaxation and consider the evolution of the energy ϵ_* of an individual star moving in a time-dependent gravitational potential $\psi(\mathbf{x}, t)$. We first show that the stellar energy can change only as a consequence of an explicit time dependence of the gravitational potential. Thus, if the potential is chosen to vanish at infinity, the energy ϵ_* of a star with mass m_* and velocity $\mathbf{v}$ is given by

$$\epsilon_* = \frac{1}{2}m_*v^2 + m_*\psi(\mathbf{x}, t). \quad (1.9)$$

Differentiating both sides of Equation (1.9) with respect to time yields

$$\frac{d\epsilon_*}{dt} = m_* \left[\mathbf{v} \cdot \dot{\mathbf{v}} + \nabla\psi \cdot \mathbf{v} + \frac{\partial\psi}{\partial t} \right]. \quad (1.10)$$

For a star moving under the gravitational potential ψ , its acceleration satisfies $\dot{\mathbf{v}} = -\nabla\psi$ and the previous equation reduces to

$$\frac{d\epsilon}{dt} = \frac{\partial\psi}{\partial t}, \quad (1.11)$$

where $\epsilon = \epsilon_*/m_*$ is the stellar energy per unit mass. Equation (1.11) shows that individual stars continuously gain and lose orbital energy as a consequence of the rapidly varying gravitational field. This naturally motivates defining the violent relaxation timescale T_r as the timescale over which the energy change of a typical star becomes comparable to its initial energy (Lynden-Bell 1967), i.e.

$$T_r = \left\langle \frac{(d\epsilon/dt)^2}{\epsilon^2} \right\rangle^{-1/2} = \left\langle \frac{(\partial\psi/\partial t)^2}{\epsilon^2} \right\rangle^{-1/2}. \quad (1.12)$$

As discussed above, during violent relaxation, the remnant progressively approaches virial equilibrium set by $Q = 1$, as indicated by the damped oscillations of the virial ratio around this value. However, in this regime $\ddot{I}$ remains small but non-zero and a mean star experiences changes in both kinetic and gravitational energies in a coupled manner, approximately satisfying

$$\frac{1}{2}m_*v^2 \sim -\frac{1}{4}m_*\psi, \quad (1.13)$$

as dictated by Equation (1.6).

By substituting Equation (1.13) into Equation (1.9), we can express the specific stellar energy solely in terms of the gravitational potential and, mindful of Equation (1.12), obtain the following equivalent form for the violent relaxation timescale:

$$T_r \equiv \frac{3}{4} \left\langle \frac{\psi^2}{\dot{\psi}^2} \right\rangle^{1/2}. \quad (1.14)$$

Dimensional analysis suggests that the polar moment of inertia can be expressed in terms of the total mass M and the square of the instantaneous characteristic radius $R(t)$ as

$$I(t) = \lambda^2 M R^2(t), \quad (1.15)$$

where λ^2 is a dimensionless structural factor of order unity that encodes the instantaneous mass distribution and is determined by the geometrical configuration of the masses m_i .

Let us consider small radial perturbations of the system about its virial equilibrium configuration. In this regime, the instantaneous characteristic radius of the system can be expressed as a small deviation from its equilibrium value, which we identify with the gravitational radius. Therefore, to the first order,

$$R(t) \approx r_v + \delta R, \quad (1.16)$$

with $|\delta R| \ll r_v$ representing the small radial displacement from equilibrium. Under this approximation, linearizing Equation (1.1), written in terms of σ_v and r_v , with respect to the perturbation yields the equation

$$2\lambda^2 r_v \delta \ddot{R} - \frac{GM}{r_v^2} \delta R, \quad (1.17)$$

which we recognize to be that of a simple harmonic oscillator, with angular frequency

$$\omega = \left(\frac{GM}{2\lambda^2 r_v^3} \right)^{1/2} \quad (1.18)$$

It is worth of noticing that, introducing the galaxy mean density $\bar{\rho} = 3M/(4\pi r_v^3)$, the latter quantity can be expressed as:

$$\omega = (2\pi G \bar{\rho})^{1/2}. \quad (1.19)$$

Since the gravitational potential scales as $\psi \sim 1/r$, its time variation is governed by the same characteristic timescale as the radial oscillations, i.e. $\sim \omega^{-1}$. Plugging this scaling into the definition of the violent relaxation timescale in Equation (1.14) returns

$$T_r \simeq \frac{3}{4\omega} = \frac{3}{\sqrt{32\pi G \bar{\rho}}}. \quad (1.20)$$

Thus, the violent relaxation timescale is directly related to the characteristic dynamical timescale of the newly formed system and is expected to be of the order of its free-fall timescale $t_{ff} \propto 1/\sqrt{G\bar{\rho}}$.

To obtain an order-of-magnitude estimate of this timescale for the merger of two identical non-nucleated dwarf galaxies, we may consider In this case, the violent relaxation timescale is expected to be of the order of ..., demonstrating that this mechanism can redistribute the orbital energy of the progenitor galaxies on a much shorter timescale than that associated with two-body relaxation t_{2b} . The latter will be introduced in § 1.2 as a key process responsible for replenishing the so-called binary "loss cone".

1.2 Two-Body Relaxation and Dynamical Friction

Galaxies are multi-component systems in dynamical equilibrium, where different stellar populations, gaseous phases, dust, and dark matter coexist (Cimatti et al. 2019). These environments host an enormous number of stars, which makes it impossible to track the orbit of each element. Nevertheless, by abandoning the idea of characterizing the time evolution of each degree of freedom in the phase-space (PS), a probabilistic description of the stellar population is still possible in terms of the one-particle distribution function (DF) $f(\mathbf{x}, \mathbf{v}, t)$. This can be interpreted as the probability density of finding a star at a given point $\mathbf{w} = (\mathbf{x}, \mathbf{v})$ in phase space at time t . It is conventionally normalized so that

$$\rho(\mathbf{x}, t) = \int_{\mathbb{R}^3} d^3\mathbf{v} f(\mathbf{x}, \mathbf{v}, t), \quad (1.21)$$

where $\rho(\mathbf{x}, t)$ is the stellar density providing a smooth approximation to the actual, discrete distribution of N stars.

In the collisionless approximation, DF is conserved along the orbits of the gravitational potential $\Phi(\mathbf{x}, t)$ associated with the stellar density $\rho(\mathbf{x}, t)$. In this regime, it thus obeys the so-called Collisionless Boltzmann Equation (CBE),

$$\frac{df}{dt} \equiv \frac{\partial f}{\partial t} + \frac{d\mathbf{x}}{dt} \cdot \frac{\partial f}{\partial \mathbf{x}} + \frac{d\mathbf{v}}{dt} \cdot \frac{\partial f}{\partial \mathbf{v}} = 0. \quad (1.22)$$

Equation (3.2) is expected to hold as long as the effects of repeated gravitational encounters between stars remain negligible over the age of the system. Although individual encounters are typically weak, their cumulative effect can progressively redistribute energy and angular momentum among stars, causing their trajectories to deviate from those drawn by the smooth potential Φ and DF to diffuse in PS (Binney and Tremaine 2011).

The characteristic timescale over which these cumulative encounters significantly alter stellar trajectories is the two-body relaxation time t_{2b} , which provides a natural criterion for assessing the validity of the collisionless approximation. Systems whose age is shorter than their two-body relaxation time, i.e. $t_{age} < t_{2b}$, are conventionally referred to as collisionless; otherwise, they will be referred to as collisional (Ciotti 2021).

The approach of Chandrasekhar (1943) for estimating t_{2b} relies on the Impulsive Approximation. It consists of two main assumptions:

1. Stellar encounters are treated as independent events, so that correlations between consecutive encounters and simultaneous multiple interactions are neglected.
2. The gravitational influence of the remaining $N-2$ stars is neglected during each encounter, reducing the problem to a sequence of isolated hyperbolic two-body interactions.

Let us work under the simplifying assumption of a spatially homogeneous distribution of field stars with isotropic velocity field. The DF $f(\mathbf{x}_f, \mathbf{v}_f)$ of such system can be expressed as the product of the position-independent number density n_f and probability density function $g(v_f)$ in the velocity space,

$$f(\mathbf{x}_f, \mathbf{v}_f) = n_f g(v_f). \quad (1.23)$$

Two-body encounters between a test particle of mass m_t and a field star of mass m_f induce small changes in an orbital property P of the test particle. Although each interaction produces only a small perturbation, the cumulative effect of many encounters causes P to be diffusive. This process is quantified by the diffusion coefficient $D(P)$, which measures the rate at which P changes per unit time. Owing to the Impulsive Approximation, it can be written as

$$D(P) \equiv 2\pi n_f \int_{\mathbb{R}^3} d^3\mathbf{v}_f \|\mathbf{v}_t - \mathbf{v}_f\| g(v_f) \int_0^\infty db \langle P \rangle b, \quad (1.24)$$

where $\langle P \rangle$ denotes the angle-averaged value of the orbital property P , and b is the impact parameter of the hyperbolic encounter (Ciotti 2021). Since significant deflections of the original orbits are expected beyond t_{2b} , corresponding to the onset of the collisional regime, a useful indicator for determining t_{2b} may be the orbital property $P = \Delta\mathbf{v}_{t\perp}$, that is the component of the test-particle velocity change perpendicular to the initial relative velocity with respect to the field particle. However, under the assumption of isotropy, the first-order diffusion coefficients of the velocity components vanish (Binney and Tremaine 2011). Consequently, the leading non-vanishing contribution is represented by the second-order diffusion coefficient, hence, making the quantity $P = \|\Delta\mathbf{v}_{t\perp}\|^2$ the relevant orbital parameter for our analysis. By taking advantage of the small-angle scattering approximation (see, e.g., Meneghetti (2022)),

$$\|\Delta\mathbf{v}_{t\perp}\| \simeq \frac{2Gm_f}{vb}, \quad (1.25)$$

with v denoting the modulus of the relative velocity between m_t and m_f at infinity, and Equation (1.24) becomes

$$D(\|\Delta\mathbf{v}_{t\perp}\|^2) \simeq 8\pi G^2 m_f^2 n_f \int_{\mathbb{R}^3} d^3\mathbf{v}_f \frac{g(v_f)}{v} \int_0^\infty \frac{db}{b}. \quad (1.26)$$

The integral in Equation (1.26) exhibits two logarithmic divergences. The former, known as ultraviolet (UV) divergence, arises in the limit $b \rightarrow 0$ and results from extending the small-angle scattering approximation to encounters with sufficiently small impact parameters, where

it breaks down. The latter, the infrared (IR) divergence, arises in the limit $b \rightarrow \infty$ and is a consequence of modelling the stellar system as infinite and homogeneous. The UV divergence is avoided by replacing the small-angle approximated solution in Equation (1.25) with the exact two-body scattering solution; mathematically, this operation is equivalent to impose a lower cutoff to the impact parameter equals $b_{\pi/2}(v)$, corresponding to the impact parameter associated with a deflection angle of $\pi/2$. The IR divergence cannot be eliminated and thus we have to set a fiducial upper cutoff $b_{\max}$.

The resulting diffusion coefficient can then be written as

$$D(\|\Delta\mathbf{v}_{t\perp}\|^2) \sim 8\pi G^2 n_f m_f^2 \int_{\mathbb{R}^3} \frac{g(v_f) \ln \Lambda}{v} d^3\mathbf{v}_f, \quad (1.27)$$

where $\ln(\Lambda) = \ln(b_{\max}/b_{\pi/2})$, is the so-called Coulomb logarithm. Although it depends on test-field particle relative velocity through $b_{\pi/2}(v)$, corresponding to a deflection angle of $\pi/2$, by defining the velocity-weighted Coulomb logarithm $\ln(\bar{\Lambda})$ so that

$$\int_{\mathbb{R}^3} \frac{g(v_f) \ln \Lambda}{\|\mathbf{v}_t - \mathbf{v}_f\|} d^3\mathbf{v}_f \equiv \Psi(v_t) \ln \bar{\Lambda}, \quad (1.28)$$

with

$$\Psi(v_t) = \int_{\mathbb{R}^3} \frac{g(v_f)}{\|\mathbf{v}_t - \mathbf{v}_f\|} d^3\mathbf{v}_f, \quad (1.29)$$

being the Rosenbluth potential, Equation (1.27) turns into

$$D(\|\Delta\mathbf{v}_{t\perp}\|^2) \sim 8\pi G^2 n_f m_f^2 \Psi(v_t) \ln \bar{\Lambda}. \quad (1.30)$$

One of the major simplifications introduced by the assumption of isotropy is that $\Psi(v_t)$ can be expressed as

$$\Psi(v_t) = \int_{\mathbb{R}^3} \frac{g(v_f)}{\|\mathbf{v}_t - \mathbf{v}_f\|} d^3\mathbf{v}_f \equiv \frac{\Xi(v_t)}{v_t} + 4\pi \int_{v_t}^{\infty} g(v_f) v_f dv_f, \quad (1.31)$$

where

$$\Xi(v_t) = 4\pi \int_0^{v_t} g(v_f) v_f^2 dv_f \quad (1.32)$$

is normalized to unity as a consequence of the normalization of velocity-space distribution function $g(v_f)$.

Finally, if one defines the two-body relaxation time t_{2b} as the ratio between the orbital parameter $P = \|\Delta\mathbf{v}_{t\perp}\|^2$ and the associated diffusion coefficient, it turns out that

$$t_{2b} \equiv \frac{v_t^2(-\infty)}{D(\Delta E_{t\perp})} \sim \frac{v_t^3}{8\pi G^2 n_f m_f^2 \ln \bar{\Lambda}}, \quad (1.33)$$

holding in the high velocity limit from Equation (1.31). As discussed above, two-body encounters redistribute the test particle's kinetic energy, increasing its velocity component perpendicular to the initial direction of motion. In this sense, two-body interactions act as a heating mechanism in the transverse direction. Conversely, dynamical friction produces a systematic loss of kinetic energy along the direction of motion and behaving as cooling process (see, e.g., Chandrasekhar (1943); Binney and Tremaine (2011)). This results in the sinking of massive objects, such as an intruding secondary BH, and N -body simulations by Holger Baumgardt et al. (2006) have demonstrated that dynamical friction efficiently drives the secondary BH down to a_0 , defined as the radius enclosing a stellar mass equal to twice that of M_2 , that is

$$M_*(< a_0) = 2M_2. \quad (1.34)$$

In order to derive an estimate for the dynamical friction timescale t_{df} , we follow the same approach and notation adopted above and first calculate the diffusion coefficient associated with the orbital property $P = \Delta \mathbf{v}_t$ for a test particle of mass m_t moving through a stellar field composed of particles of mass m_f , i.e.

$$\frac{d\mathbf{v}_t}{dt} = D[\Delta \mathbf{v}_t] = -4\pi G^2 m_t m_f \ln \Lambda \int d^3 \mathbf{v}_f f(\mathbf{v}_f) \frac{\mathbf{v}_t - \mathbf{v}_f}{|\mathbf{v}_t - \mathbf{v}_f|^3}. \quad (1.35)$$

Assuming an isotropic velocity field, Equation (1.35) simplifies to

$$\frac{d\mathbf{v}_t}{dt} = -16\pi^2 G^2 m_t m_f \ln \Lambda \left[\int_0^{v_t} dv_f v_f^2 f(v_f) \right] \frac{\mathbf{v}_t}{v_t^3}, \quad (1.36)$$

known as Chandrasekhar's dynamical friction formula (Chandrasekhar 1943). Equation (1.36) shows that the effect of dynamical friction is to systematically decrease the test particle's velocity through an acceleration directed antiparallel to its velocity vector.

By following an analogous derivation to that used to estimate the two-body relaxation timescale t_{2b} in Equation (1.33), the dynamical friction timescale t_{df} can be expressed as

$$t_{df} \equiv \frac{\|\mathbf{v}_t(-\infty)\|}{\|D(\Delta \mathbf{v}_{t\parallel})\|} \sim \frac{v_t^3}{4\pi G^2 n_f m_f (m_f + m_t) \ln(\bar{\Lambda}) \Xi(v_t)}, \quad (1.37)$$

We conclude this Section by deriving, under physically motivated assumptions, an analytic expression for the dynamical friction timescale t_{df} of an intruding secondary BH of mass m_t , which sinks towards the center of a stellar background characterized by a Maxwellian velocity distribution and a power-law density profile

$$\rho(r) = \rho_0 \left(\frac{r}{r_0} \right)^{-\gamma} \quad (1.38)$$

and hosting a primary BH of mass m_f .

For a Maxwellian velocity distribution, the integral over the stellar velocity distribution appearing in Equation (1.36) can be evaluated analytically, yielding

$$\frac{d\mathbf{v}_t}{dt} = -4\pi \ln \Lambda \left[\operatorname{erf}(X) - \frac{2X}{\sqrt{\pi}} e^{-X^2} \right] \frac{G^2 m_t \rho(r)}{v_t^3} \mathbf{v}_t \quad (1.39)$$

where

$$\operatorname{erf}(X) = \frac{2}{\sqrt{\pi}} \int_0^X dt e^{-t^2} \quad (1.40)$$

is the so-called error function (Binney and Tremaine 2011). Here, $X \equiv v_c/(\sqrt{2}\sigma)$ is a dimensionless parameter that combines the circular velocity v_c of the test particle with the velocity dispersion σ of the stellar background.

Let us now consider the orbital decay of the intruding BH along quasi-circular orbits after it enters the primary BH's sphere of influence. In this region, the gravitational potential is dominated by the primary BH of mass m_f , so that the angular momentum of the secondary BH and the dynamical friction force acting on it are given by

$$\tilde{L} = m_t \sqrt{G m_f r} \quad (1.41)$$

and

$$\tilde{F} = m_t \frac{dv_t}{dt} = -C(\Lambda, X) \frac{G^2 m_t^2 \rho(r)}{v_c^2} \quad (1.42)$$

respectively. Here, $C(\Lambda, X)$ is the dimensionless factor implicitly defined through the Equations (1.39) and (1.42).

At this point, we can exploit the fact that, within the primary BH's sphere of influence, the circular velocity can be approximated as

$$v_c^2 = \frac{Gm_f}{r}. \quad (1.43)$$

Under these assumptions, the second cardinal equation of dynamics takes the form

$$\frac{d\tilde{L}}{dt} = \tilde{F}r, \quad (1.44)$$

and leads to the differential equation for $r = r(t)$ upon substituting the expressions given in Equations (1.42), (1.41) and (1.43) (see, e.g., Holger Baumgardt et al. (2006); Gualandris et al. (2022)):

$$r^{-5/2} \frac{dr}{dt} = -2C(\Lambda, X(r)) \frac{\sqrt{G}m_t}{m_f^{3/2}} \rho(r). \quad (1.45)$$

Finally, we note that Equation (1.45) admits an analytic solution for an isotropic power-law density profile. In this case,

$$X = \sqrt{\gamma - 1} \quad (1.46)$$

and is therefore independent of radius. Integrating the differential equation subject to the initial condition $r(t=0) = r_0 < r_{inf}$, we obtain

$$r(t) = \begin{cases} r_0 \left[1 - 2C(\Lambda, X) \frac{\sqrt{G}m_t}{m_f^{3/2}} \left(\gamma - \frac{3}{2} \right) \rho_0 r_0^{3/2} t \right]^{\frac{1}{\gamma-3/2}} & \text{if } 1 < \gamma \leq 3 \wedge \gamma \neq \frac{3}{2} \\ r_0 \exp \left[-2C(\Lambda, X) \frac{\sqrt{G}m_t}{m_f^{3/2}} \rho_0 r_0^{3/2} t \right] & \text{if } \gamma = \frac{3}{2} \end{cases} \quad (1.47)$$

Equation (1.47) applies to an isotropic, power-law stellar background, or more generally to a stellar distribution that approaches such a density profile within the sphere of influence of the central BH and provides an estimate of the duration of the final stage of dynamical friction, from r_{inf} down to a_0 . Indeed, setting the reference radius r_0 and density ρ_0 to their respective values at the influence radius, and defining t_{fric} such that $r(t_{fric}) = a_0$, we obtain

$$t_{fric} = \begin{cases} \frac{m_f^{3/2}}{2C(\Lambda, X)\sqrt{G}m_t\rho_{inf}r_{inf}^\gamma} \frac{r_{inf}^{\gamma-3/2} - a_0^{\gamma-3/2}}{\gamma - \frac{3}{2}}, & \text{if } \gamma \neq \frac{3}{2}, \\ \frac{m_f^{3/2}}{2C(\Lambda, X)\sqrt{G}m_t\rho_{inf}r_{inf}^{3/2}} \ln \left(\frac{r_{inf}}{a_0} \right), & \text{if } \gamma = \frac{3}{2}. \end{cases} \quad (1.48)$$

From a_0 onward, dynamical friction, which has driven the orbital decay and pairing of the BHs up to this point, becomes inefficient and the further hardening of the binary becomes primarily driven by individual encounters with low-angular-momentum stars that are unbound to the binary and can extract energy and angular momentum from it. We address this shrinking mechanism in § 1.3.

1.3 Black Hole Binary–Star Three-Body Scattering

As discussed in § 1.2 and § 1.1, during a dry, major merger, the remnant evolves towards virial equilibrium on the violent relaxation timescale T_r , while the MBHs initially located at the

centers of the progenitor galaxies sink towards the center of the remnant through dynamical friction on a timescale t_{fric} . As their separation decreases, the MBHs eventually cross each other sphere of influence and form a gravitationally bound pair at a separation of order a_0 , defined as the binary separation at which the enclosed stellar mass equals the secondary BH mass M_2 (Holger Baumgardt et al. 2006). At this stage, the binary separation is sufficiently small that distant stars mainly perturb the binary center of mass, with negligible effect on its orbital energy (Quinlan 1996). Further hardening is therefore driven predominantly by low angular momentum stars passing in the vicinity of the binary and extracting both energy and angular momentum from it. Each time such a star—belonging to the so-called binary "loss cone", as we will discuss in more detail in § 1.4—passes sufficiently close to the binary, a three-body encounter takes place. Under suitable conditions, this interaction produces hyper-velocity stars (see, e.g., Quinlan (1996); Sesana, Haardt, et al. (2006)) through the so-called stellar slingshot mechanism (Mikkola and Valtonen 1992). In this section, we review the main physical aspects of three-body scattering and the stellar slingshot mechanism, which are key processes driving the hardening of MBHBs.

We begin by considering the so-called "two-body problem" and reviewing some of its key results, which will be useful in understanding the physics of the stellar slingshot. Adopting the same notation as in the previous Section, let us consider a test particle and a field particle, with masses m_t and m_f , respectively, interacting solely through gravity. Introducing the total mass of the system, $M = m_t + m_f$, and the reduced mass, $\mu = m_t m_f / M$, the total energy in the mass centre frame S_{CM} is given by

$$E = \frac{1}{2}\mu\|\mathbf{v}\|^2 - \frac{Gm_f m_t}{r}, \quad (1.49)$$

where $r(t) \equiv \|\mathbf{x}_t - \mathbf{x}_f\|$ and $\mathbf{v}(t) \equiv \mathbf{v}_t - \mathbf{v}_f$ denote particles relative separation and velocity. For two unbound particles moving along hyperbolic orbits in S_{CM} , energy conservation, as expressed by Equation (1.49), evaluated in the asymptotic limit $r(-\infty) = r(+\infty) \rightarrow +\infty$, yields

$$\|\mathbf{v}(-\infty)\| = \|\mathbf{v}(+\infty)\|. \quad (1.50)$$

Although it is not conserved along the entire orbit, Equation (1.50) shows that the magnitude of the relative velocity—i.e., the velocity of the test particle as measured by an observer comoving with the field particle—is the same at the two asymptotic limits. Therefore, the encounter results only in a rotation of the velocity vector $\mathbf{v}$ and the corresponding velocity change $\Delta\mathbf{v} = \mathbf{v}(+\infty) - \mathbf{v}(-\infty)$ can be conveniently decomposed as the sum of the components perpendicular and parallel to the initial relative velocity $\mathbf{v}(-\infty)$ as follows:

$$\mathbf{v}(+\infty) = \mathbf{v}(-\infty) + \Delta\mathbf{v}_{\parallel} + \Delta\mathbf{v}_{\perp}. \quad (1.51)$$

Substituting the expression for $\mathbf{v}(+\infty)$ given in Equation (1.51) into Equation (1.50) leads to the identity

$$\|\Delta\mathbf{v}_{\parallel}\|^2 + \|\Delta\mathbf{v}_{\perp}\|^2 + \langle\Delta\mathbf{v}_{\parallel}, \mathbf{v}(-\infty)\rangle = 0. \quad (1.52)$$

A first inspection of Equation (1.52) reveals that a necessary condition for it to hold is that the scalar product be negative. This observation will be crucial for assessing whether a binary BH–star interaction results in binary hardening or softening (Ciotti 2021).

The results derived above impose a constraint on the relative velocity of the interacting particles in a specific inertial frame. However, the interaction can be described in any inertial frame S_0 . Indeed, while Equation (1.50) shows that both the total energy and the individual mechanical energies of the particles are conserved between the two asymptotic limits, the close encounter generally induces changes in the individual energies E_t and E_f in S_0 .

Between the two configurations at infinite relative separation, the test particle undergoes a net variation in mechanical energy,

$$\Delta E_t = \frac{1}{2}m_t[|\mathbf{v}_t(+\infty)|^2 - |\mathbf{v}_t(-\infty)|^2] = -\Delta E_f, \quad (1.53)$$

which is exactly compensated by the corresponding variation in the mechanical energy of the field particle, such that

$$\Delta E = \Delta E_t + \Delta E_f = 0. \quad (1.54)$$

By combining the velocity composition law

$$\mathbf{v}_t(t) = \mathbf{v}_{\text{CM}} + \frac{\mu}{m_t}\mathbf{v}(t) \quad (1.55)$$

with the conservation of the center-of-mass velocity vector $\mathbf{v}_{\text{CM}}$, which holds for isolated systems, Equation (1.53) can be written as (Ciotti 2021)

$$\Delta E_t = \mu \langle \Delta \mathbf{v}, \mathbf{v}_{\text{CM}} \rangle. \quad (1.56)$$

We are now in a position to address three-body scattering involving a gravitationally bound BH binary and an incoming star. The dynamics of this encounter can be understood by considering the limiting case of strongly asymmetric binary in which $M_1 \gg M_2 \gg m_*$ so that both the star and the secondary BH, having masses m_* and M_2 , can travel along nearly Keplerian orbits around M_1 with slightly different radii and, consequently, possess slightly different Keplerian velocities. The secondary BH acts as the field particle, while the star plays the role of the test particle. The primary BH serves only to establish the relative velocity between the two bodies. As a result, the center of mass motion of this gravitationally unbound pair is essentially determined by the orbit of M_2 around M_1 . Therefore, the approximation

$$\mathbf{x}_{\text{CM}} \simeq \mathbf{x}_f, \quad \mathbf{v}_{\text{CM}} \simeq \mathbf{v}_f, \quad (1.57)$$

applies and the mass center will move locally at constant velocity if the observation time is short compared to the orbital period, thus making negligible its acceleration.

Hereafter, by analogy with the previous notation, we identify $M_2 \equiv m_f$ and $m_* \equiv m_t$.

Binary hardening. When the star is placed outside the secondary BH orbit, the Keplerian velocity profile imposes $\|\mathbf{v}_f(-\infty)\| = \|\mathbf{v}_{\text{CM}}\| > \|\mathbf{v}_t(-\infty)\|$. Therefore, at point where m_f surpasses m_t and they are both located on the same radius, $\mathbf{v}(-\infty)$ is nearly antiparallel to $\mathbf{v}_{\text{CM}}$ and $\Delta \mathbf{v}_{\parallel}$ can be read as the projected vector along $\mathbf{v}_{\text{CM}}$ rather than $\mathbf{v}(-\infty)$. Taking advantage of Equation (1.52), it follows that

$$0 < \langle \Delta \mathbf{v}_{\parallel}, \mathbf{v}_{\text{CM}} \rangle = \langle \Delta \mathbf{v}, \mathbf{v}_{\text{CM}} \rangle = \Delta E_t / \mu. \quad (1.58)$$

Equation (1.58) shows that in the reference system centered on M_1 the star gains orbital energy and is scattered onto a wider orbit. On the other hand, the secondary BH experiences an equal and opposite change in orbital energy, leading to a reduction of the binary separation and, consequently, to binary hardening.

It is worth noticing that if $\Delta E_t > |E_t(-\infty)|$, which occurs at very high circular velocity of the binary BHs, stellar ejection occurs; in this regime “hypervelocity stars” (HVSs) travelling along nearly radial orbits are produced (see, e.g., Quinlan (1996); Sesana, Haardt, et al. (2006)).

Binary softening. Contrary to the previous case, binary softening takes place when the star is placed inside the secondary BH orbit. Now $\mathbf{v}(-\infty)$ is nearly parallel to $\mathbf{v}_{\text{CM}}$ at the crossover point and by repeating the same arguments used above it follows

$$E_t/\mu = \langle \Delta \mathbf{v}, \mathbf{v}_{\text{CM}} \rangle = \langle \Delta \mathbf{v}_{\parallel}, \mathbf{v}_{\text{CM}} \rangle < 0. \quad (1.59)$$

On one hand the star loses energy to the binary, it is pushed inward and may undergo tidal disruption due to primary BH (Lightman and Shapiro 1977); on the other hand the binary gains energy and its semimajor axis increases, resulting in binary softening.

Although our discussion has so far focused on a specific three-body scattering configuration, the results can be readily generalized by observing that the distinction between binary hardening and softening is determined by the relative magnitude of the BH binary's orbital velocity and the velocity of the incoming loss-cone star.

The binary BH embedded in an isotropic stellar background with constant velocity dispersion σ is expected to undergo three-body scatterings with loss-cone stars that will approach the galactic centre where the binary lives with velocity of the order of σ . Because the specific binding energy of a keplerian with total mass $M = M_1 + M_2$, reduced mass μ and semimajor axis a is

$$E_b = \frac{G\mu}{2a} \quad (1.60)$$

and the specific energy of the loss-cone star is of the order of σ^2 , the velocity criterion can be conveniently reformulated in terms of the specific energy. When $E_b \ll \sigma^2$, the binary is said to be soft, and repeated stellar encounters tend to further decrease its binding energy. On the other hand, when $E_b \gg \sigma^2$, the binary is hard, and stellar encounters, on average, increase its binding energy. This behavior is well summarized by Heggie's law (see, e.g., D. C. Heggie (1975); Douglas C. Heggie and Hut (1993)).

Heggie's law. – *On average, hard binaries get harder and soft binaries get softer.*

By extrapolation of the two limiting cases, one can define the "hardening radius" a_h as the binary separation at which binary passes from the soft to the hard regime. Despite its expected weak dependence on the BH mass ratio, throughout this work we adopt the definition proposed by Quinlan (1996), unless otherwise specified:

$$a_h = \frac{GM_2}{4\sigma^2}. \quad (1.61)$$

At this stage, it should be clear that stellar scattering plays a fundamental role in driving the binary toward the regime in which GWs emission dominates its evolution. Nevertheless, one may wonder how a binary can evolve from the soft to the hard regime. The key point is that most of the work for this evolution to continue in the the direction of binary hardening is done by dynamical friction with the stellar background rather than individual encounters, as explained in Section 1.2.

The discussion then turns to the loss-cone refilling problem, whose efficient replenishment is essential to ensure a sustained supply of low angular momentum stars capable of driving the continued hardening of the binary through stellar ejection.

1.4 Stellar Scattering and Loss Cone Problem

In § 1.3 we have been concerned in describing the physics of three-body scattering, pointing out its relevance to the orbital decay of the BH binary. In the following Section, instead, we

will be explaining which stars can effectively exchange energy and angular momentum with the binary, the rate at which these close encounters take place and how all of this enters the self-consistent model for MBHBs evolution developed by Quinlan (1996) and Sesana (2010) that will be presented in Chapter 2.

Let $a(t)$ be the binary semimajor axis. Numerical experiments demonstrated that only stars intersecting the binary orbit can efficiently extract energy and angular momentum from when ejected at higher velocity (Mikkola and Valtonen 1992). This class of stars of the stellar background populates the so-called “binary loss cone” and comprises both low-energy stars and high-energy, low angular momentum stars. Under the assumption of Keplerian orbits, the former have small mean radius ($\langle r \rangle < a$) and specific energy $E_* < -GM/a$, while latter spend most of their orbital period $P(E_*)$ far from the binary but a close encounter can occur at the pericenter r_P whenever $r_P < a$.

The size of the binary loss cone, J_{lc} , in specific angular momentum space, hereafter J-space, is a function of E_* and can be determined from the maximum and minimum angular momentum values allowed at a given specific energy. At the configuration of closest approach, the star radial velocity vanishes and the specific energy in terms of r_P is

$$E_* = \frac{J^2}{2r_P^2} - \frac{GM}{r_P}. \quad (1.62)$$

From Equation (1.62) follows that the minimum value of J , below which the star enters the binary loss cone, is

$$J_{lc}(E_*) \equiv \left[2 \left(E_* + \frac{GM}{a} \right) \right]^{1/2} a, \quad (1.63)$$

which can be approximated to $\sim \sqrt{2GMa}$ as the vast majority of stars orbit far from the binary and have a specific angular energy $|E_*| \ll GM/a$.

Besides, stars of fixed specific energy E_* , other than a minimum of the specific angular momentum $J_{min} = J_{lc}(E_*)$, possess a maximum set by the chain of inequalities

$$E_* \geq \frac{J^2}{2r^2} - \frac{GM}{r} \equiv V_{eff}(r) \geq V_{eff,min} = -\frac{G^2 M^2}{2J^2}, \quad (1.64)$$

holding because of the non-negative behavior of the radial kinetic term. This constrain poses an upper limit to J , that is

$$J_{max}(E_*) = \frac{GM}{\sqrt{2|E_*|}}, \quad (1.65)$$

corresponding to the specific angular momentum of a circular orbit of specific energy E_* (Capozziello and De Laurentis 2008).

Once the binary has depleted, on a dynamical timescale $P(E_*)$, the population of stars that are bound to it and intersect its orbit—most of which, hence, reside within the binar sphere of influence—further hardening through stellar slingshot interactions depends on the rate at which the binary loss cone is replenished by stellar diffusion in both energy and angular momentum space, allowing new stars to enter the loss cone (Lightman and Shapiro 1977). This diffusion arises from stellar encounters that have been proven to be efficient in exchanging angular momentum and results in a Random Walk in J-space (see, e.g., Milosavljević (2003); Merritt (2013)). Determining the rate at which this process occurs is crucial for assessing its role in the Quinlan–Sesana hybrid model.

Let us define j_2 the rms angular momentum transferred in one orbital period and $J_{max}(E_*)$ the same quantity associated with the timescale t_{2b} . Only stars with $J_{lc} \leq J \leq J_{lc} + j_2$ can be scattered into the loss cone over one orbital period and the kind of loss cone replenishing

mechanism, along with the associated timescale, are set by the dimensionless quantity

$$q(E) \equiv \frac{j_2^2}{J_{min}^2}. \quad (1.66)$$

Taking Figure 1.2 as a reference, $q(E_*)$ enables us to distinguish between two extreme scenarios:

Empty Loss Cone. This condition is represented by picture on the left in Figure 1.2(b). When $q(E_*) \ll 1$ the RW step size is very small compared to the loss cone size, making the process ascribable to a diffusion problem with timescale t_{2b} . Therefore, stars are removed from the loss cone faster than the rate at which it is fed, hence, remaining nearly empty.

Full Loss Cone. This condition is represented by the picture on the right in Figure 1.2(b). The loss cone is smaller than the RW step size and even high J stars can be scattered into the loss cone by many, weak stellar encounters taking place during one orbital period. Thus, the rate at which loss cone stars are expelled is lower than the rate of refilling, causing the loss cone to remain almost full.

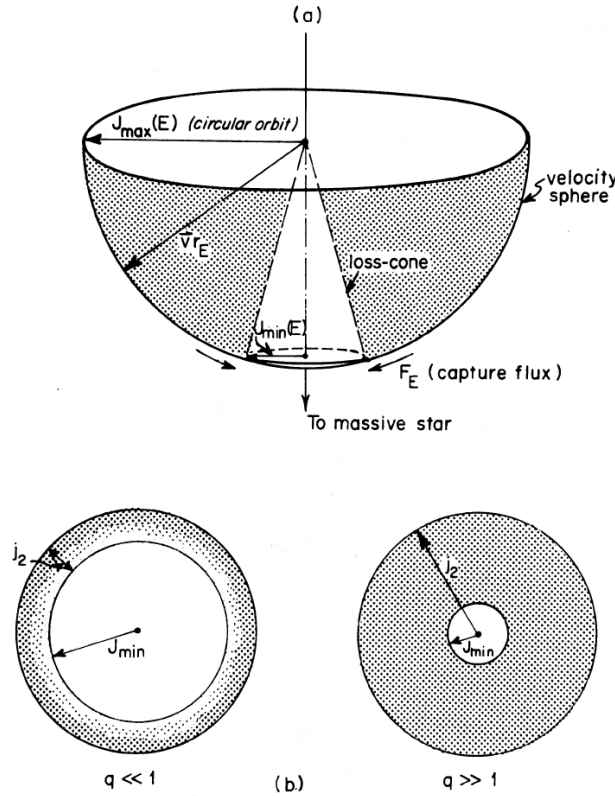


Figure 1.2: Fig. 1(a).—Velocity distribution of stars with fixed energy E ; J varies accordingly to the direction of $\mathbf{v}$ emphasising the two-dimensional nature of the problem. The vertical axis points towards the compact object of mass M . Fig. 1(b).—The velocity sphere viewed from below. On the left the empty loss cone scenario; stellar density is nearly independent of J everywhere but $J \gtrsim J_{min}$ where it quickly declines. On the right the full loss cone regime; consumed stars are immediately replenished by stars from a region much more extended- and uniform density- than the loss cone itself (the “pinhole” limit). Picture from Lightman and Shapiro (1977).

By employing the same dimensional analysis carried out by Lightman and Shapiro (1977),

we expect that the stellar capture rate per unit energy $\Gamma_E(E_*)$ could be written as

$$\Gamma_E(E_*) \equiv \frac{d\Gamma}{dE} \approx \epsilon |E_*|^{-1} F_{max}(E_*) \quad (1.67)$$

where $F_{max}(E_*) \sim N(E_*)/P(E_*)$ denotes the maximum rate of loss cone replenishing from stars with specific energy E_* and ϵ is a correction factor helping us to account for the fraction of stars within the loss cone.

Owing to the 2-D nature of the loss-cone problem in the J-space, ϵ is expected to be an area correction factor, thus scaling as

$$\epsilon \sim \frac{J_{lc}^2}{(J_{lc} + j_2)^2}. \quad (1.68)$$

Therefore, in the full loss cone regime, where $J_{lc} \ll j_2 \approx J_{max}$, one easily finds that $\sim J_{lc}^2/J_{max}^2 \sim a/r$ and, by substitution into Equation (1.67),

$$\Gamma_E(E_*) \approx |E_*|^{-1} \frac{a}{r} \frac{N(E_*)}{P(E_*)}. \quad (1.69)$$

Equation (1.69) allows us to finally discuss how loss-cone refilling is accounted for in the Quinlan–Sesana hybrid model.

In his scattering experiments, Quinlan (1996) quantified the binary dimensionless hardening rate H and eccentricity growth rate K through the flux of stars

$$\Gamma_Q = \left(\frac{\rho}{m_*} \right) \Sigma v \quad (1.70)$$

across the binary effective cross section Σ , where ρ is the stellar mass density, m_* is the stellar mass, and v is the stellar velocity. Let b denote the impact parameter of a star initially unbound to the binary of mass M and undergoing a three-body interaction with it. The corresponding interaction cross section can then be defined as

$$\Sigma \equiv \pi b_{max}^2, \quad (1.71)$$

where b_{max} is the maximum impact parameter for which stellar capture occurs. If we assume this to take place whenever $r_p < a$, mindful of the gravitational focusing

$$b^2 = \frac{2GM r_p}{v^2} \quad (1.72)$$

relating impact parameter and pericenter distance, then we can pose $b_{max} = b(a)$ and express Γ_Q in Equation (1.70) as

$$\Gamma_Q = 2\pi \left(\frac{M}{m_*} \right) \frac{G\rho a}{v}, \quad (1.73)$$

with ρ standing for the mass density of the homogeneous stellar background.

On the other hand, Sesana (2010) estimated the rate Γ in Equation (1.67) under the assumption of a full loss cone, adopting an isothermal stellar sphere of total mass M_* , influence radius r_{inf} and constant stellar velocity dispersion σ .

If all stars have the same mass m_* , the total number of stars inside r is simply proportional to the stellar mass enclosed within this sphere, that is

$$N(< r) = \frac{M}{m_*} \frac{r}{r_{inf}}, \quad (1.74)$$

while, under the assumption of Keplerian orbits bound to the binary, the energy of each star depends on the semimajor only through the relation

$$E_*(r) = -\frac{GMm_*}{2r}. \quad (1.75)$$

In this regime, by exploiting the relation $dE = |E| d \ln r$, we can rewrite Equation (1.69) as (see, e.g., Perets, Hopman, et al. (2007); Perets and Alexander (2008))

$$\frac{d\Gamma}{d \log r} = \frac{a}{r} \frac{N(< r)}{P(r)}. \quad (1.76)$$

Finally, approximating the orbital period as $P(r) \sim r/\sigma$ and substituting the estimates for $N(< r)$ and $P(r)$ derived above into Equation (1.76), integration yields

$$\Gamma = 2\pi \frac{M}{m_*} \frac{G\rho_{\text{inf}} a}{\sigma}. \quad (1.77)$$

A comparison between Equations (1.70) and (1.77) shows that, by associating the velocity v of an unbound star at infinity with the velocity dispersion σ of the isothermal sphere, the prescriptions adopted for H and K effectively describe a system in which the loss cone is full at r_{inf} . However, by introducing the dimensionless factor LCN , it is possible to describe, within the hybrid model formulation, different regimes of the loss-cone simply by replacing ρ_{inf} with $LCN \cdot \rho_{\text{inf}}$ as will shown in chapter 3.

1.5 Gravitational Radiation from Black Hole Binaries

1.5.1 Linearized Gravitational Field Equation

The starting point of the theoretical derivation of GWs properties, such as the energy and angular momentum they transport, from BH binary and the associated back-reaction on the astrophysical source, is the linearization of field equations of General Relativity (GR)

$$R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}R = \frac{8\pi G}{c^4}T_{\mu\nu}. \quad (1.78)$$

These constitute ten coupled, non-linear, second-order partial differential equations for the metric tensor; nevertheless, they are not sufficient to uniquely set the ten independent components of $g_{\mu\nu}$ due to the four Bianchi identities, equivalent to the same number of differential constraints, resulting in six independent equations only. These four degrees of freedom are intimately connected to gauge symmetries, reflecting the arbitrariness of the four coordinates x^μ used to represent and solve Equation (1.78).

To show how gravitational radiation emerges from Equation (1.78), we need to expand it around the Minkowski metric $\eta_{\mu\nu} = \text{diag}(-1, +1, +1, +1)$; therefore we assume

$$g_{\mu\nu}(x) = \eta_{\mu\nu} + h_{\mu\nu}(x), \quad (1.79)$$

with $|h_{\mu\nu}| \ll 1$ being a symmetric tensor representing a small perturbation of the flat spacetime. The left-hand side of Equation (1.78) encodes the geometry and curvature of spacetime and contains all terms that are functions of the metric; in terms of the affine connection

$$\Gamma_{\mu\nu}^\lambda = \frac{1}{2}g^{\lambda\rho}(\partial_\mu g_{\nu\rho} + \partial_\nu g_{\rho\mu} - \partial_\rho g_{\mu\nu}) \quad (1.80)$$

and the Riemann tensor

$$R_{\lambda\mu}{}^{\rho}{}_{\nu} = \partial_{\lambda}\Gamma_{\mu\nu}^{\rho} - \partial_{\mu}\Gamma_{\lambda\nu}^{\rho} + \Gamma_{\lambda\sigma}^{\rho}\Gamma_{\mu\nu}^{\sigma} - \Gamma_{\mu\sigma}^{\rho}\Gamma_{\lambda\nu}^{\sigma}, \quad (1.81)$$

they can be expressed indeed as

$$R_{\mu\nu} = R_{\sigma\mu}{}^{\sigma}{}_{\nu} \quad (1.82)$$

and

$$R = g^{\alpha\beta}R_{\alpha\beta} = R_{\sigma}^{\sigma}. \quad (1.83)$$

On the right-hand side, instead, appears the Energy-Momentum tensor; it can be read as the source term of the gravitational field equations and turns out to be

$$T^{\mu\nu} = (\rho + p/c^2)u^{\mu}u^{\nu} + pg^{\mu\nu} \quad (1.84)$$

for a perfect fluid of density ρ , pressure p and four-velocity v^{μ} .

It straightforward to prove that, to the first order in the metric perturbation, they become:

$$R_{\mu\nu} = -\frac{1}{2}(\Box h_{\mu\nu} + \partial_{\mu}\partial_{\nu}h - \partial_{\lambda}\partial_{\mu}h_{\nu}^{\lambda} - \partial_{\nu}\partial_{\lambda}h_{\mu}^{\lambda}) \quad (1.85)$$

and

$$R = \partial_{\mu}\partial_{\nu}h^{\mu\nu} - \Box h, \quad (1.86)$$

where h denotes $h_{\mu\nu}$ trace.

Defining the so-called "trace reverse" of $h_{\mu\nu}$

$$\bar{h}_{\mu\nu} \equiv h_{\mu\nu} - \frac{1}{2}\eta_{\mu\nu}h \quad (1.87)$$

and plugging the terms of Equations (1.85) and (1.86) into Equation (1.78), the field equations take the form

$$\Box\bar{h}_{\mu\nu} + \eta_{\mu\nu}\partial_{\alpha}\partial_{\beta}\bar{h}^{\alpha\beta} - \partial_{\mu}\partial_{\lambda}\bar{h}_{\nu}^{\lambda} - \partial_{\nu}\partial_{\lambda}\bar{h}_{\mu}^{\lambda} = -\frac{16\pi G}{c^4}T_{\mu\nu}. \quad (1.88)$$

Taking advantage of the gauge freedom of Equation (1.78), it can be performed an infinitesimal coordinate transformation of the kind

$$x^{\mu} \rightarrow x'^{\mu} = x^{\mu} + \xi^{\mu}(x); \quad (1.89)$$

as tensors obey to precise law of transformation, this yields

$$\bar{h}'_{\mu\nu} = \frac{\partial x^{\alpha}}{\partial x'^{\mu}} \frac{\partial x^{\beta}}{\partial x'^{\nu}} \bar{h}_{\alpha\beta} \approx \bar{h}_{\mu\nu} - \partial_{\mu}\xi_{\nu} - \partial_{\nu}\xi_{\mu} + \eta_{\mu\nu}\partial_{\sigma}\xi^{\sigma}. \quad (1.90)$$

Yet, if the four functions ξ^{μ} are chosen so as to satisfy the so-called De Donder gauge condition, i.e. $\Box\xi^{\mu} = \partial_{\sigma}\bar{h}^{\sigma\mu}$, then $\partial_{\sigma}\bar{h}'^{\sigma\mu} = 0$ and, dropping the prime, Equation (1.88) simplifies to

$$\begin{cases} \Box\bar{h}_{\mu\nu} = -\frac{16\pi G}{c^4}T_{\mu\nu} \\ \partial_{\mu}\bar{h}^{\mu\nu} = 0. \end{cases} \quad (1.91)$$

We are now ready to address the problem of determining the general solution of the linearized Einstein's equations coupled with the De Donder gauge condition.

1.5.2 Generation and Propagation of Gravitational Waves

To study the formation of GWs we initially focus on the solution of the problem defined in Equation (1.91) inside the source where $T_{\mu\nu} \neq 0$.

Linear partial differential equations solution can be obtained by the method of Green's function; by definition, the Green's function G is the solution of the associated equation in case of a pont-like source, i.e.

$$\square_x G(x - x') = \delta^{(4)}(x - x'), \quad (1.92)$$

with $\delta^{(4)}(x - x')$ being the four-dimensional Dirac function.

It is relevant to note that, apart from the B.C. of the problem, G depends only on the linear operator; therefore, from the radiation problem in electromagnetism, we already know that the proper solution is the retarded Green's function

$$G(x - x') = -\frac{1}{4\pi|\mathbf{x} - \mathbf{x}'|} \delta^{(4)}(ct_{ret} - ct'), \quad (1.93)$$

where

$$t_{ret} = t - \frac{|\mathbf{x} - \mathbf{x}'|}{c} \quad (1.94)$$

is the time in past at which the radiation was emitted at $\mathbf{x}'$ in order to be observed at $\mathbf{x}$ at time t . Once the Green's function is found, the general solution will be the convolution between it and source term:

$$\bar{h}_{\mu\nu}(t, \mathbf{x}) = \frac{4G}{c^4} \int d^3x' \frac{T_{\mu\nu}(t_{ret}, \mathbf{x}')}{|\mathbf{x} - \mathbf{x}'|}. \quad (1.95)$$

Prior to examining the formula reported in Equation (1.95) and assessing which additional simplifications can be implemented and when they apply, it is appropriate to analyze how GWs propagate in vacuo.

Outside the GWs source, $T_{\mu\nu}$ vanishes and, owing to the linearity of the d'Alembert operator, the solution can be expressed as superposition of plane waves without loss of generality, that is

$$\bar{h}_{\mu\nu}(x) = \int d^3k (A_{\mu\nu}(\mathbf{k}) e^{ik_\rho x^\rho} + c.c.) \quad (1.96)$$

However, not all plane waves satisfy the linearized field equation in vacuo along with the gauge condition but linearity enables us to study the properties of a single plane wave separately; consequently, inserting the Ansatz $\bar{h}_{\mu\nu} \sim A_{\mu\nu} e^{ik_\rho x^\rho}$ yields the following two constraints:

1. ($k_\rho k^\rho = 0$): the four-vector $k^\mu = (\omega/c, \mathbf{k})$ is a light-like vector (GWs travel at light speed c in vacuo).
2. ($k_\mu A^{\mu\nu} = 0$): it implies $A^{0\nu} = A^{3\nu}$ if we deal with GWs propagating along the x^3 direction, hence $k^\mu = (k, 0, 0, k)$.

At this stage, the amplitude tensor $A^{\mu\nu}$ possesses six indipendet components — ten, due to symmetry, reduced by the four constraints imposed by the gauge conditions — and can be written as

$$A^{\mu\nu} = \begin{pmatrix} A^{00} & A^{10} & A^{20} & A^{00} \\ A^{10} & A^{11} & A^{12} & A^{10} \\ A^{20} & A^{12} & A^{22} & A^{20} \\ A^{00} & A^{10} & A^{20} & A^{00} \end{pmatrix}. \quad (1.97)$$

Out of the six polarizations shown in Equation (1.97), four are unphysical because they vanish when the residual gauge freedom associated with four arbitrary harmonic functions ξ^μ is exploited. One is free to adopt functions proportional to the plane wave solutions of the vacuo

problem, i.e. $\xi^\mu = \epsilon^\mu e^{ik_\rho x^\rho}$ are harmonic, and to set their amplitudes ϵ^μ such that

$$A_{TT}^{\mu\nu} = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & h_+ & h_x & 0 \\ 0 & h_x & -h_+ & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}. \quad (1.98)$$

The coordinate system where $A^{\mu\nu}$ is in the form of Equation (1.98), with two independent polarizations only, and defined by

$$\partial_\mu h^{\mu\nu} = 0, \quad h^{0\mu} = 0, \quad h_i^i = 0, \quad (1.99)$$

is referred to as "TT gauge"¹.

Having analyzed the propagation of GWs in vacuo and the properties of the TT gauge, we are now in the position to focus again on the source solution reported in Equation (1.95). One of the greatest simplifications to this integral consists of the so-called "Compact-source approximation"; it applies outside the source whenever its size is negligible with respect to the source-detector separation. In this regime, we can retain the leading term of the multipole expansion only, thus

$$h_{ij}^{TT}(t, \mathbf{x}) = \frac{4G}{c^4 r} \int d^3 x' T_{ij}(t_{ret}, \mathbf{x}'). \quad (1.100)$$

Furthermore, far from the source, the conservation of the Energy-Momentum tensor, namely $\nabla_\mu T^{\mu\nu} \approx \partial_\mu T^{\mu\nu} = 0$, yields

$$h_{ij}^{TT}(t, \mathbf{x}) = \frac{2G}{rc^4} \ddot{Q}_{ij}^{TT}(t - r/c) \quad (1.101)$$

at lowest order in the low-velocity expansion.

The term Q_{ij}^{TT} introduced in Equation (1.101)² is the representation of the quadrupole moment in the class of coordinate systems satisfying the TT gauge conditions; defining the quadrupole moment of $T^{00}/c^2 = \rho$ as

$$M^{ij} \equiv \frac{1}{c^2} \int d^3 x T^{00}(t, \mathbf{x}) x^i x^j, \quad (1.102)$$

it is thus possible to write

$$Q^{ij} \equiv M^{ij} - \frac{1}{3} \delta^{ij} M_{kk} = \int d^3 x \rho(t, \mathbf{x}) \left(x^i x^j - \frac{1}{3} r^2 \delta^{ij} \right). \quad (1.103)$$

Equation (1.100) shows that, unlike electromagnetic radiation, at lowest order the gravitational radiation arises from changes in the quadrupole moment; other important observables such as the energy and angular momentum carried by GWs, needed to estimate the back-reaction, will be derived from the quantities defined in Equation (1.103) indeed.

The energy and angular momentum conservation laws impose that the rates at which these quantities are carried away by GWs at time t and at large separation r from the source coincide with the corresponding rates of orbital energy —if the emitting system consists of a BH binary— and angular momentum, neglecting the spin contribute, loss evaluated at the appropriate retarded time; in formulae:

$$P_{quad} = \frac{dE_{gw}}{dt} = \frac{G}{5c^5} \langle \ddot{Q}_{ij} \ddot{Q}_{ij} \rangle = -\frac{dE_{source}}{dt}, \quad (1.104)$$

¹In the TT gauge, according to the definition provided, $\bar{h}_{ij}^{TT} = h_{ij}^{TT}$.

²The components with at least one time index are excluded from our analysis because they are constant and therefore do not contribute to GWs formation.

or equivalently in terms of M_{ij}

$$P_{quad} = \frac{G}{5c^5} \left\langle \ddot{M}_{ij} \ddot{M}_{ij} - \frac{1}{3} (\ddot{M}_{kk})^2 \right\rangle, \quad (1.105)$$

and

$$\frac{dL_{gw}^i}{dt} = \frac{2G}{5c^5} \epsilon^{ijk} \langle \ddot{Q}_{jl} \ddot{Q}_{kl} \rangle = -\frac{dL_{source}^i}{dt}. \quad (1.106)$$

Equations (1.104) and (1.106) will be of fundamental importance for estimating the back-reaction induced by GWs emission on the black hole binary.

1.5.3 Gravitational Waves from Black Hole Binaries and Back-Reaction

We now focus on the gravitational radiation emitted by two masses m_1 and m_2 moving along elliptic Keplerian orbit. Let $m = m_1 + m_2$ and $\mu = m_1 \cdot m_2 / m$ be the total mass and the reduced mass respectively.

Regardless of whether the orbit is bound or unbound, the motion of the two bodies takes place in a plane due to the conservation of the angular momentum $\mathbf{L}$; therefore, in this plane, the x-y plane, perpendicular to $\mathbf{L}$, it is convenient to introduce the polar coordinates (r, ψ) with origin at center-of-mass.

In this frame, as before denoted by S_{CM} ,

$$L_z = \mu r^2 \dot{\psi} \quad (1.107)$$

coincides with the modulus L of the constant vector while, as a consequence of the K oenig theorem,

$$E = \frac{1}{2} \mu \dot{r}^2 + \frac{L^2}{2\mu r^2} - \frac{Gm\mu}{r} \quad (1.108)$$

exhibits only the term of kinetic energy corresponding to the relative motion of the system, as in S_{CM} holds $K_{CM} = 0$ by definition.

The integration of Equation (1.108) requires, as in (Capozziello and De Laurentis 2008), the introduction of the auxiliary variable $w(\psi) = 1/r(\psi)$; plugging it into the energy conservation identity, Equation (1.108) becomes

$$w'^2 + w^2 - \frac{2Gm_1 m_2 \mu}{L^2} w = \frac{2\mu E}{L^2} \quad (1.109)$$

where $w' \equiv dw/d\psi$. Differentiating both sides of the last equation with respect to ψ leads to

$$w' \left(w'' + w - \frac{Gm_1 m_2 \mu}{L^2} \right) = 0. \quad (1.110)$$

Apart from the solution $w = \text{const}$, corresponding to a circular binary, the problem admits harmonic-oscillator-like and, upon solving for $r(\psi)$, it is found to be given by

$$r(\psi) = \frac{R}{1 + e \cos \psi}, \quad (1.111)$$

with $R = L^2 / (Gm\mu^2)$ being semi-latus rectum — the polar axis is aligned to the major axis $2a$, so that R corresponds to the distance between the origin and the point of the conic at $\psi = \pi/2$ — and is related to the eccentricity e by

$$e^2 = 1 - \frac{R}{a} = 1 + \frac{2EL^2}{G^2 m^2 \mu^3}. \quad (1.112)$$

It is worth of noticing that we are interested in the GWs emission process by a BH binary, which is by definition a gravitationally bound system; therefore, from this point onward it will be assumed $E < 0$ and thus, by virtue of Equation (1.112), $0 \leq e < 1$. Since the two semiaxes are given by

$$a = \frac{R}{1 - e^2} \quad b = \frac{R}{\sqrt{1 - e^2}}, \quad (1.113)$$

inserting Equation (1.112) into the first identity of Equation (1.113) gives the semi-major axis in terms of E ; explicitly:

$$a = -\frac{Gm\mu}{2E} \quad (1.114)$$

It follows that fixing the energy means setting the semi-major axis.

The functions $r(t)$ and $\psi(t)$ can be obtained by direct integration of Equations (1.107) and (1.108); they are usually given in the parametric form

$$r(t) = a[1 - e \cos u(t)] \quad \psi(t) = \frac{\cos u(t) - e}{1 - e \cos u(t)} \quad (1.115)$$

in terms of the "eccentric anomaly" u defined in the famous Kepler equation

$$u(t) - e \sin u(t) = \left(\frac{Gm}{a^3} \right)^{1/2} t \equiv \omega_0 t. \quad (1.116)$$

The transformation from polar coordinates (r, ψ) to Cartesian ones (x, y) is illustrated below

$$\begin{cases} x = r \cos \psi = a [\cos u(t) - e] \\ y = r \sin \psi = b \sin u(t). \end{cases} \quad (1.117)$$

We now possess all the necessary ingredients to proceed with the computation of the rates of energy and angular momentum carried by GWs, which will play a fundamental role in assessing the back-reaction on the binary system.

In the reference system S_{CM} holds $\rho(t, \mathbf{x}) = \mu \delta(x - r \cos \psi) \delta(y - r \sin \psi) \delta(z)$ and, according to Equation (1.102), it is immediate to get

$$M_{ij} = \mu r^2(t) \begin{bmatrix} \cos^2(\psi) & \sin(\psi) \cos(\psi) \\ \sin(\psi) \cos(\psi) & \sin^2(\psi) \end{bmatrix}. \quad (1.118)$$

The time derivatives are easier computed when r , which is a function of time itself, is expressed as a function of ψ ; this is accomplished by replacing this variable with the general solution of the two-body problem provided by Equation (1.111). A simple computation returns the third time derivatives needed to derive the power radiated in the quadrupole approximation, i.e.

$$\ddot{M}_{11} = \beta(1 + e \cos \psi)^2 [2 \sin 2\psi + 3e \sin \psi \cos^2 \psi], \quad (1.119)$$

$$\ddot{M}_{22} = \beta(1 + e \cos \psi)^2 [-2 \sin 2\psi - e \sin \psi (1 + 3 \cos^2 \psi)], \quad (1.120)$$

$$\ddot{M}_{12} = \beta(1 + e \cos \psi)^2 [-2 \cos 2\psi + e \cos \psi (1 - 3 \cos^2 \psi)], \quad (1.121)$$

where the factor β is defined as

$$\beta^2 \equiv \frac{4G^3 \mu^2 m^3}{a^5 (1 - e^2)^5}. \quad (1.122)$$

By means of Equation (1.104) we get in conclusion

$$P(\psi) = \frac{8G^4}{15c^5} \frac{\mu^2 m^3}{a^5(1-e^2)^5} (1 + e \cos(\psi))^4 [12(1 + e \cos(\psi))^2 + e^2 \sin^2(\psi)]. \quad (1.123)$$

However, we recall that this instantaneous quantity is not constant over one orbital period for elliptic Keplerian orbits as in general M_{ij} will suffer from different rate of variation from pericenter to apocenter; therefore, taking the temporal average over one orbital period, $T \equiv 2\pi/\omega_0$ returns

$$P \equiv \frac{1}{T} \int_0^T dt P(\psi) = \frac{\omega_0}{2\pi} \int_0^{2\pi} d\psi \frac{P(\psi)}{\dot{\psi}} = \frac{32G^4 \mu^2 m^3}{5c^5 a^5} F(e), \quad (1.124)$$

with

$$F(e) = \frac{1}{(1-e^2)^{7/2}} \left(1 + \frac{73}{24}e^2 + \frac{37}{96}e^4 \right) \quad (1.125)$$

being a function strongly depending on the eccentricity, which reduces to unity in case of circular binary.

Throughout the inspiral phase, the binary radiates not only energy but angular momentum as well, carried by GWS, and undergoes secular changes in both semimajor axis and ellipticity until coalescence eventually occurs. Therefore, another key ingredient in evaluating the back-reaction on the binary evolution is the determination of the angular momentum loss rate. It is quantified in Equation (1.106) which becomes

$$\frac{dL_{source}^i}{dt} = -\frac{2G}{5c^5} \epsilon^{ikl} \langle \ddot{M}_{ka} \ddot{M}_{la} \rangle \quad (1.126)$$

as the terms of the form $\epsilon^{ijk} \delta_{jl} \ddot{Q}_{kl}$ vanish, since they correspond to the contraction of an antisymmetric tensor with a symmetric one.

Taking into account that, in S_{CM} , all components M_{i3} are identically zero and that $L = L_z = L_{source}^3$, one can rewrite Equation (1.126) in a more convenient form consistent with the previous arguments:

$$\frac{dL}{dt} = \frac{4G}{5c^5} \langle \ddot{M}_{12} (\ddot{M}_{11} - \ddot{M}_{22}) \rangle. \quad (1.127)$$

The last identity provides dL/dt in terms of the second and third derivatives of the quadrupole moment of ρ ; although the second derivative $\ddot{M}_{12}$ has not been computed explicitly yet, it follows straightforwardly from the same procedure used to derive $\ddot{M}_{ij}$, hence

$$\ddot{M}_{12} = \frac{G\mu m}{a(1-e^2)} \sin \psi \left[-4(1 + e \cos \psi)^2 \cos \psi + 2e (3 \cos^2 \psi - 1 + 2e \cos^3 \psi) \right]. \quad (1.128)$$

Yet, adopting the expressions for reported in Equations (1.119), (1.120) and (1.128) respectively, we get the angular momentum loss, corresponding to minus the angular momentum carried by the gravitational radiation in the quadrupole approximation; then, the temporal average over a single orbital period T , transformed into an integral over the angular position ψ , returns

$$\frac{dL}{dt} = -\frac{32}{5} \frac{G^{7/2} \mu^2 m^{5/2}}{c^5 a^{7/2}} \frac{1}{(1-e^2)^2} \left(1 + \frac{7}{8}e^2 \right) \quad (1.129)$$

Lastly, the semi-major axis and eccentricity evolution of the BH binary results by imposing the conservation of the energy and angular momentum of the system composed of the binary and GWs; differentiating Equations (1.112) and (1.114) with respect to the time yields

$$\frac{da}{dt} = \frac{Gm\mu}{2E^2(a)} (-P) \quad (1.130)$$

and

$$2e \frac{de}{dt} = \frac{2}{G^2 m^2 \mu^3} \left[-PL^2(a, e) + 2E(a)L(a, e) \frac{dL}{dt} \right]. \quad (1.131)$$

By rearranging all the terms and performing the appropriate substitutions, Peters and Mathews (1963) gave

$$\frac{da}{dt} = -\frac{64}{5} \frac{G^3 \mu m^2}{c^5 a^3} \frac{1}{(1-e^2)^{7/2}} \left(1 + \frac{73}{24}e^2 + \frac{37}{96}e^4 \right) \quad (1.132)$$

along with

$$\frac{de}{dt} = -\frac{304}{15} \frac{G^3 \mu m^2}{c^5 a^4} \frac{e}{(1-e^2)^{5/2}} \left(1 + \frac{121}{304}e^2 \right) \quad (1.133)$$

that has been employed by Quinlan (1996) and Sesana (2010) in order to reconstruct evolutionary tracks of BH binaries through a "Hybrid model", which will be described later.

Equations (1.132) and (1.133) prove that GWs emission is efficient, more precisely at little binary separation due to a^{-3} and a^{-4} dependence respectively, not only in shrinking the BHs pair, by extracting orbital energy, but in promoting the orbit circularization as well, thus making this mechanism dominant over stellar scattering at late stage of binary inspiral.

Furthermore, another key piece of information that can be derived from the theory is the expected coalescence time τ of the binary in the case of gravitational-wave emission alone. This should reproduce the characteristic hardening timescale of the final stage, when orbital shrinking driven by gravitational radiation takes over stellar scattering. This is obtained by combining Equations (1.132) and (1.133); this mathematical operation leads to a differential equation for $a = a(e)$, which reads

$$\frac{da}{de} = \frac{12}{19} \frac{1 + \frac{73}{24}e^2 + \frac{37}{96}e^4}{e(1-e^2) \left(1 + \frac{121}{304}e^2 \right)}. \quad (1.134)$$

The last equation admits analytic solution of the kind

$$a(e) = a_0 \frac{g(e)}{g(e_0)} \quad (1.135)$$

where

$$g(e) = \frac{e^{12/19}}{1-e^2} \left(1 + \frac{121}{304}e^2 \right)^{870/2299} \quad (1.136)$$

and (a_0, e_0) denotes the initial point of the of binary evolutionary track.

Last but not least, given the BH binary initial condition (a_0, e_0) , the coalescence time is found by integration over the eccentricity and by noticing that, if we set $(a(0), e(0)) = (a_0, e_0)$, $e(\tau) = 0$ due to orbit circularization:

$$\tau(a_0, e_0) = \int_0^{\tau(a_0, e_0)} dt = \int_{e_0}^0 de \frac{dt}{de} = \int_{e_0}^0 de \frac{a^4(e)(1-e^2)^{5/2}}{e \left(1 + \frac{121}{304}e^2 \right)}. \quad (1.137)$$

By substituting Equation (1.135) into Equation (1.137) one gets the final expression

$$\tau_{\text{coal}}(a_0, e_0) = \frac{12}{19} \frac{c^5}{G^3 \mu m^2} \frac{a_0^4}{\int_0^{e_0} \frac{e^{29/19} \left(1 + \frac{121}{304}e^2 \right)^{1181/2299}}{(1-e^2)^{3/2}} de}. \quad (1.138)$$

2

A HYBRID APPROACH TO BINARY EVOLUTION

In light of the forthcoming GW observatories, LISA and ET, investigating the properties and evolution of GW-loud sources is of particular importance. Among these, IMBH binaries are especially compelling and studying their dynamical evolution, along with the associated coalescence time, is essential for assessing their prospects as GW sources.

This chapter aims to review the physical framework underlying the Quinlan-Sesana hybrid model for the dynamical evolution of black hole binaries and to present the main results of the numerical experiments performed using this approach. The chapter begins with a detailed description of the hybrid model, focusing on its main parameters and on the numerical integration of the differential equations governing the evolution of the binary semimajor axis and eccentricity. The model is then applied to physically motivated stellar backgrounds. The second section is devoted to numerical experiments in stellar-cusp environments. Finally, the chapter ends by reporting the main predictions of the model for nucleated dwarf galaxies and for non-nucleated, artificially evolved galaxies at redshift z , according to the observed cosmic evolution of galaxy sizes.

2.1 Quinlan-Sesana Hybrid Model

As already discussed in Chapter 1, the coalescence time of a BH binary driven solely by GW emission depends strongly on several parameters, including the initial binary separation a_0 and eccentricity e_0 . However, the GW-driven regime is reached only after a sequence of earlier hardening mechanisms, primarily dynamical friction and three-body stellar scattering, which dominate the early stages of the binary evolution. It is therefore essential to model the dynamical evolution of the binary by consistently accounting for the different hardening processes from its formation to final coalescence in order to obtain a reliable description of the binary evolutionary track, $a(t)$ and $e(t)$ and its coalescence time.

Although direct N -body simulations provide a self-consistent approach to carrying out this analysis, they remain computationally too demanding to efficiently explore the broad parameter space governing binary evolution. This parameter space, for our purposes, includes the BH masses, the binary separation and eccentricity at pairing, and the properties of the stellar background. In this context, the self-consistent hybrid model developed by Quinlan (1996) and Sesana (2010) provides an efficient alternative, overcoming the computational limitations and long runtimes associated with direct N -body experiments while retaining the key physical processes driving the binary evolution.

Let us begin by introducing the Quinlan formalism and the main model parameters were determined through restricted three-body scattering experiment, eventually leading to the coupled differential equation for $(a(t), e(t))$. Adopting the original notation, we denote by m_1 and $m_2 < m_1$ the primary and secondary BH masses, respectively, by $M_{12} = m_1 + m_2$ the total binary mass, $q = m_2/m_1$ the BHs mass ratio, and by $m_* \ll m_2$ the mass of an individual star, assuming all stars have the same mass. In his work, Quinlan (1996) studied the dynamical evolution of a BH binary embedded in a fixed, homogeneous background of unbound stars with density ρ and a Maxwellian velocity distribution characterised by a constant velocity dispersion σ . The description relies on three dimensionless quantities, H , K , and J , which denote the dimensionless binary hardening rate, eccentricity growth rate, and stellar mass ejection rate, respectively. They are defined as follows:

$$H = \frac{\sigma}{G\rho} \frac{d}{dt} \left(\frac{1}{a} \right), \quad (2.1)$$

$$K = \frac{de}{d \ln(1/a)} \quad (2.2)$$

and

$$J = \frac{1}{M_{12}} \frac{dM_{ej}}{d \ln(1/a)}. \quad (2.3)$$

Let us focus on Equations (2.1) and (2.2). Both dimensionless rates and the derivatives of the binary parameters account for the various, independent mechanisms driving its evolution, i.e.

$$\begin{cases} \frac{da}{dt} = \sum_i \left. \frac{da}{dt} \right|_i = -\frac{a^2 G \rho}{\sigma} \sum_i H_i \\ \frac{de}{dt} = \sum_i \left. \frac{de}{dt} \right|_i = -\frac{a G \rho}{\sigma} \sum_i H_i K_i \end{cases} \quad (2.4)$$

and

$$\begin{cases} H_i = \frac{\sigma}{a^2 G \rho} \left. \frac{da}{dt} \right|_i \\ K_i = a \left. \frac{de}{dt} \right|_i \left(\sum_j \left. \frac{da}{dt} \right|_j \right)^{-1} \end{cases} \quad (2.5)$$

where the index i identifies a precise dynamical process (Sesana 2010).

As in Quinlan (1996) and Sesana, Haardt, et al. (2006), in our numerical experiments based on this hybrid model, we account for the binary evolution driven close encounters with unbound, loss-cone stars and by GW emission, as described by the equations derived by Peters and Mathews (1963). Therefore, we assume in our numerical experiments

$$\begin{cases} \frac{da}{dt} = \frac{da}{dt}\bigg|_u + \frac{da}{dt}\bigg|_{GW} = -\frac{a^2 G \rho}{\sigma} (H_u + H_{GW}) \\ \frac{de}{dt} = \frac{de}{dt}\bigg|_u + \frac{de}{dt}\bigg|_{GW} = -\frac{a G \rho}{\sigma} (H_u K_u + H_{GW} K_{GW}) \end{cases} \quad (2.6)$$

The latter, derived in § 1.5, is reported here for completeness and to ensure consistency with the notation adopted in this chapter:

$$\frac{da}{dt}\bigg|_{GW} = -\frac{64 G^3 \mu m^2}{5 c^5 a^3} \frac{1}{(1-e^2)^{7/2}} \left(1 + \frac{73}{24} e^2 + \frac{37}{96} e^4\right), \quad (2.7)$$

$$\frac{de}{dt}\bigg|_{GW} = -\frac{304 G^3 \mu m^2}{15 c^5 a^4} \frac{e}{(1-e^2)^{5/2}} \left(1 + \frac{121}{304} e^2\right). \quad (2.8)$$

The former, instead, was independently derived by Quinlan (1996) and Sesana, Haardt, et al. (2006) using two different approaches, as shown below. Indeed, with the aim of deriving the dimensionless rates, both works employ suites of three-body scattering experiments designed to characterize the response of a BH binary to repeated close encounters with stars over its lifetime. However, while Quinlan (1996) fix the incoming stellar velocity v and vary the binary separation a , Sesana, Haardt, et al. (2006) adopt the opposite approach, varying v at fixed a . It is worth noting that the two approaches are physically equivalent, but lead to different parametrizations of the functions H_u and K_u . When numerically integrating Equations (2.6), the latter parametrization will be adopted, as it directly involves the quantities being integrated. Let us now briefly review how H_u and K_u were originally estimated. The underlying idea is to infer the variations in the binary orbital energy and angular momentum from the corresponding changes experienced by the scattered star, which are equal in magnitude and opposite in sign as a consequence of energy and angular momentum conservation. The first step is to define the dimensionless energy change

$$C \equiv \frac{a \Delta E_*}{G \mu}, \quad (2.9)$$

introduced by Hills (1983), and the dimensionless impact parameter

$$x \equiv \frac{b}{b_0} \quad (2.10)$$

with $b_0^2 = 2GM_{12}a/v^2$.

Denoting by $\langle C \rangle$ the average over all angular variables, the “single-velocity” hardening rate $H_1(v)$, derived from restricted three-body scattering experiments in which all stars have the same velocity at infinity, is given by

$$H_1(v) = \frac{1}{\pi b_0^2} \int db, (2\pi b) 4\pi \langle C \rangle = 8\pi \int dx, \langle C \rangle x \equiv 8\pi I_x(C). \quad (2.11)$$

where the operator $I_x(C)$ is implicitly defined by Equation (2.11). Finally, the hardening rate $H_u(\sigma)$ is obtained by averaging the single-velocity hardening rate $H_1(v)$ over the Maxwellian distribution of stellar velocities:

$$H_u(\sigma) = \int_0^\infty dv 4\pi v^2 H_1(v) \frac{\sigma}{v} f(v, \sigma) \quad (2.12)$$

where

$$f(v, \sigma) = \frac{1}{(2\pi\sigma^2)^{3/2}} e^{-\frac{v^2}{2\sigma^2}} \quad (2.13)$$

is the Maxwellian velocity distribution associated with the velocity dispersion σ . Similarly, one finds K_u from the “single-velocity” dimensionless eccentricity growth rate K_1 , which has the same meaning of H_1 . It can be derived starting from the expression for the binary eccentricity,

$$e^2 = 1 + \frac{2EL^2}{\mu(GM_{12})^2}, \quad (2.14)$$

which relates the eccentricity to the binary orbital energy E and angular momentum L . Differentiating Equation (2.14) and expressing the resulting variations in terms of finite differences yields

$$\frac{\Delta e}{\Delta \ln(1/a)} = \frac{1-e^2}{2e} \left[-2 \left(\frac{E}{\Delta E} \right) \left(\frac{\Delta L_z}{L_z} \right) - 1 \right] = \frac{1-e^2}{2e} (B - C), \quad (2.15)$$

where

$$B \equiv -\frac{M_{12}}{m_*} \frac{\Delta L_z}{L_z}. \quad (2.16)$$

is the dimensionless angular momentum change. Finally, by integrating over all possible impact parameters and angular variables and normalizing the result to B , the single-velocity eccentricity growth rate K_1 turns out to be

$$K_1 = \frac{\int dx, xC(x) \left\langle \frac{\Delta e}{\Delta \ln(1/a)} \right\rangle}{\int dx, xC(x)} = \frac{1-e^2}{2e} \left[\frac{I_x(B-C)}{I_x(C)} \right]. \quad (2.17)$$

The best-fitting functions found by Sesana, Haardt, et al. (2006) from numerical experiments to H_u and K_u are

$$H(a) = A_H \left(1 + \frac{a}{a_{0,H}} \right)^{\gamma_H} \quad (2.18)$$

and

$$K(a, e) = A_K \left(1 + \frac{a}{a_{0,K}} \right)^{\gamma_K} + B_K, \quad (2.19)$$

with best-fitting parameters listed in Tables 2.1 and 2.2.

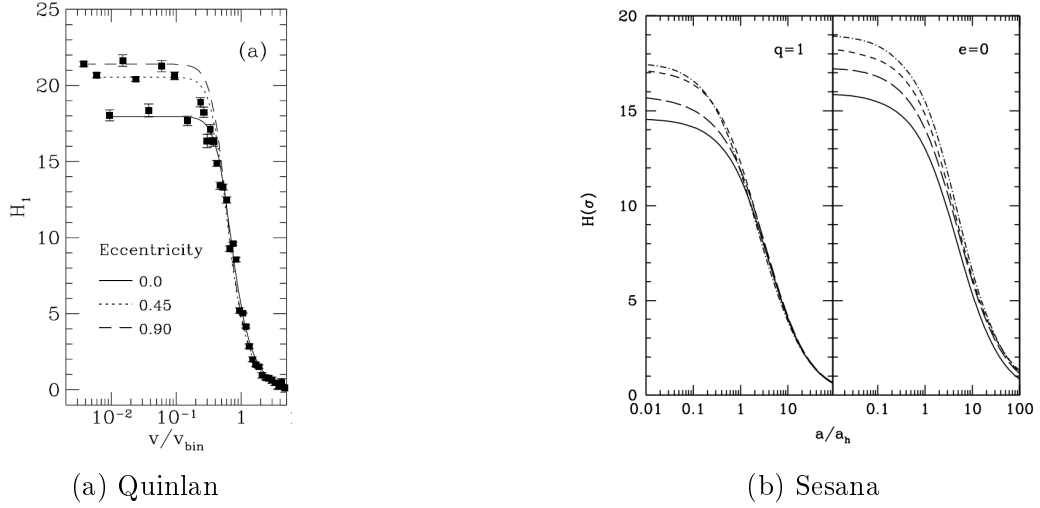
q	A_H	$a_{0,H}$	γ_H
1	14.55	3.48	-0.95
1/3	15.82	4.18	-0.90
1/9	17.17	3.59	-0.79
1/27	18.15	3.32	-0.77
1/81	18.81	3.87	-0.82
1/243	19.16	4.16	-0.86

Table 2.1: H values

e	A_K	$a_{0,K}$	γ_K	B_K
$q = 1$				
0.15	0.037	0.339	-3.335	-0.012
0.3	0.075	0.151	-1.548	-0.008
0.45	0.105	0.088	-0.893	-0.005
0.6	0.121	0.090	-0.895	-0.008
0.75	0.134	0.064	-0.544	-0.006
0.9	0.082	0.085	-0.663	-0.004
$q = 1/3$				
0.15	0.082	0.042	-0.168	-0.048
0.3	0.095	0.213	-1.152	-0.012
0.45	0.129	0.137	-0.655	-0.006
0.6	0.166	0.081	-0.546	-0.006
0.75	0.159	0.079	-0.497	-0.010
0.9	0.095	0.122	-0.716	-0.008
$q = 1/9$				
0.15	0.051	0.385	-0.891	-0.011
0.3	0.111	0.307	-1.107	-0.007
0.45	0.172	0.526	-1.174	-0.016
0.6	0.181	0.251	-1.169	-0.007
0.75	0.179	0.195	-0.846	-0.004
0.9	0.117	0.400	-1.170	-0.001
$q = 1/27$				
0.15	0.064	0.284	-1.206	0.021
0.3	0.143	1.033	-1.537	-0.021
0.45	0.212	0.722	-1.257	-0.022
0.6	0.216	0.430	-1.163	-0.014
0.75	0.173	0.771	-1.934	-0.014
0.9	0.129	0.329	-1.125	-0.020

Table 2.2: Best-fit parameters.

The absence of a column dedicated to the eccentricity sampling in Table 2.2 reflects one of the main findings of the scattering experiments by Quinlan (1996) and Sesana, Haardt, et al. (2006): the dimensionless hardening rate H_u is almost independent of the BH binary eccentricity. This result, shown in Figure 2.1, is in excellent agreement with the main statement of Heggie’s law, discussed in § 1.3.

Figure 2.1: Comparison of the H function according to Quinlan and Sesana.

In contrast, the dimensionless eccentricity growth rate K_u depends on both the binary eccentricity e and the separation a . Following the procedure illustrated by Sesana (2010), at a given binary separation, $K_u(a, e)$ is obtained from Equation (2.19) by linearly interpolating in the eccentricity domain the best-fitting parameters reported in Table 2.2. The interpolation is supplemented by the boundary conditions $K(a, e = 0) = 0$, as theoretically predicted in the test-particle limit for circular orbits, since circular binaries remain circular, and $K(a, e = 1) = 0$. Thus far, this Section has focused on introducing the main quantities and equations that characterize the self-consistent hybrid model for the dynamical evolution of BH binaries in stellar environments. We are now in a position to illustrate how the model can be applied to an arbitrary stellar background and how the corresponding evolution of the binary separation and eccentricity, $a(t)$ and $e(t)$, is obtained numerically.

As discussed by Quinlan (1996) and Sesana, Haardt, et al. (2006), the three-body scattering experiments underlying the hybrid model consider stars that are initially unbound to the binary and whose velocities follow a Maxwellian distribution. Since these calculations are performed in a specific stellar background, a prescription is required to extend their results to the diverse stellar distributions hosting MBHs. A useful calibration is provided by Sesana and Khan (2015), who showed that the hardening rate of a MBHB embedded in a spherically symmetric and isotropic stellar system, characterized by a mass density $\rho(r)$ and velocity dispersion $\sigma(r)$ measured from direct N -body simulations, agrees to within approximately 30% with that predicted by the hybrid model for a homogeneous stellar background characterized by a constant density ρ_{inf} and velocity dispersion σ_{inf} , denoting the stellar mass density and velocity dispersion values at the binary influence radius respectively. Indeed, once the binary has ejected most of the stars initially bound within its radius of influence, the loss cone is predominantly replenished by stars coming from just outside r_{inf} , thus having locally a nearly homogeneous spatial distribution with $\rho = \rho_{inf}$ and velocity dispersion $\sigma = \sigma_{inf}$. In summary, once the BH masses and the properties of the stellar background are specified, the dynamical evolution of the binary is governed by the following system of coupled differential equations:

$$\begin{cases} \frac{da}{dt} = -\frac{a^2 G \rho_{inf}}{\sigma_{inf}} H_u(a) - \frac{64}{5} \frac{G^3 \mu M_{12}^2}{c^5 a^3} \frac{1}{(1-e^2)^{7/2}} \left(1 + \frac{73}{24} e^2 + \frac{37}{96} e^4 \right) \\ \frac{de}{dt} = -\frac{a G \rho_{inf}}{\sigma_{inf}} H_u(a) K_u(a, e) - \frac{304}{15} \frac{G^3 \mu M_{12}^2}{c^5 a^4} \frac{e}{(1-e^2)^{5/2}} \left(1 + \frac{121}{304} e^2 \right) \end{cases} \quad (2.20)$$

which can be solved numerically for a given set of initial conditions (a_0, e_0) .

The system of Equations (2.20) constitutes a set of first-order autonomous ordinary differential

equations (ODEs), which can be written in the general form

$$\frac{d\mathbf{u}}{dt} = \mathbf{f}(\mathbf{u}), \quad (2.21)$$

where $\mathbf{u} = (a, e)$ is the vector of dynamical variables.

The numerical integration of this system with the adopted fourth-order Runge–Kutta (RK4) method requires the time step to satisfy an appropriate stability condition. The stability analysis to the first order of RK4 applied to a system of N ODEs in the N -dimensional vector $\mathbf{u}$ yields the following criterion (Butcher 2016):

$$\Delta t \leq \min \left\{ \frac{2.79}{|\lambda_i|} \right\} \quad \forall i = 1, N \quad (2.22)$$

where λ_i denotes the i -th eigenvalue of the Jacobian matrix $\mathbf{J} = \partial \mathbf{f} / \partial \mathbf{u}$. For the system considered here, the Jacobian elements can be derived directly from Equations (2.20) as

$$J_{11} = \left. \frac{da}{dt} \right|_u \left(\frac{2}{a} + \frac{\gamma}{a + a_0 a_h} \right) - \left. \frac{da}{dt} \right|_{gw} \left(\frac{3}{a} \right), \quad (2.23)$$

$$J_{12} = \left. \frac{da}{dt} \right|_{GW} \left(\frac{7e}{1 - e^2} + \frac{73e/12 + 37e^3/24}{1 + 73e^2/24 + 37e^4/96} \right), \quad (2.24)$$

$$J_{21} = \left. \frac{de}{dt} \right|_u \left(\frac{1}{a} + \frac{\gamma_H}{a + a_{0,H} a_h} + \frac{\gamma_K}{a + a_{0,K} a_h} \right) - \left. \frac{de}{dt} \right|_{GW} \left(\frac{4}{a} \right), \quad (2.25)$$

$$J_{22} = \left. \frac{de}{dt} \right|_u \left(\frac{\partial \ln K}{\partial e} \right) + \left. \frac{de}{dt} \right|_{GW} \left(\frac{5e}{1 - e^2} + \frac{1 + 363e^2/304}{e + 121e^3/304} \right). \quad (2.26)$$

These expressions allow the two eigenvalues, λ_1 and λ_2 , of the Jacobian matrix to be evaluated at each point along the binary evolutionary track and the adaptive time-step to be calculated as

$$\Delta t \leq 2.79 \cdot \min \left\{ \frac{1}{|\lambda_1|}, \frac{1}{|\lambda_2|} \right\}. \quad (2.27)$$

The following sections, § 2.2-2.3, present the results of the numerical experiments performed with the hybrid model described above, considering different IMBH binaries masses, initial conditions and host stellar systems.

2.2 Stellar Cusps IMBH Binary Evolution

Observational evidences and direct N -body simulations suggest that galaxies hosting dynamically evolved BH binaries may undergo stellar scouring during binary hardening, significantly altering the original stellar distribution and leading to the formation of a shallower inner cusp (see, e.g., Milosavljević and Merritt (2001); Merritt and Milosavljević (2005)). As the binary orbit shrinks within the galactic core, it transfers energy to the surrounding stars, progressively depleting the central stellar distribution and producing a stellar mass deficit in the innermost region of the host galaxy.

In light of these results, the first stellar background considered here for studying the dynamical evolution of an IMBH binary, from pairing to coalescence, is thus a power-law density profile within the binary influence radius, matched to a singular isothermal sphere (SIS) outside it. The stellar density $\rho(r)$ of this spherically symmetric distribution is given by

$$\begin{cases} \rho(r) = \rho_{inf} \left(\frac{r}{r_{inf}} \right)^{-\gamma} & \text{if } r < r_{inf} \\ \rho(r) = \frac{\sigma^2}{2\pi G r^2} & \text{if } r > r_{inf} \end{cases} \quad (2.28)$$

where γ is the exponent of the inner power-law stellar cusp and σ denotes the position-independent velocity dispersion of the SIS ($\gamma = 2$) profile.

Adopting the notation introduced in the previous Section, we assume that σ is uniquely determined by the binary mass M_{12} through the $M - \sigma$ relation

$$M_6 = 0.84\sigma_{70}^4. \quad (2.29)$$

provided by Tremaine et al. (2002), with σ_{70} indicating the stellar velocity dispersion in units of 70 km/s and M_6 the binary mass in units of $10^6 M_\odot$.

Imposing the continuity of the stellar profile in Equation (2.28) at the binary influence radius r_{inf} returns the following analytic expressions for r_{inf} and ρ_{inf} :

$$r_{inf} = (3 - \gamma) \frac{GM_{12}}{\sigma^2}, \quad (2.30)$$

$$\rho_{inf} = \frac{\sigma^2}{2\pi G r_{inf}^2}. \quad (2.31)$$

As already mentioned in Section..., dynamical friction exerted by the stellar background on an intruding secondary BH efficiently drags it towards the galactic center, where it eventually forms a gravitationally bound binary with the primary BH, with semimajor axis a_0 . This separation is defined as the radius enclosing a stellar mass equal to the secondary BH mass M_2 and, for the adopted stellar cusp profile, is given analytically by

$$a_0 = (3 - \gamma) \frac{GM}{\sigma^2} \left(\frac{q}{1 + q} \right)^{1/(3-\gamma)} = r_{inf} \left(\frac{q}{1 + q} \right)^{1/(3-\gamma)}, \quad (2.32)$$

This separation is adopted as the initial condition for the numerical integration of Equations (2.20) using RK4 method, with the initial eccentricity specified later in this Section. The integration is terminated when the binary separation falls below the Schwarzschild radius of the system,

$$R_s = \frac{2GM_{12}}{c^2}. \quad (2.33)$$

Before presenting and discussing the main results of the numerical experiments for this stellar environment, it is useful to introduce a set of characteristic distances, associated to different binary separations, that allow us to distinguish between the different stages of binary evolution and identify the dominant physical mechanism governing each stage. Starting the binary evolution from a_0 , the binary is initially too “soft”, in the sense of Heggie’s law, for scattering interactions with unbound stars to efficiently contribute to its hardening. Therefore, once dynamical friction has become inefficient at a_0 , the subsequent binary evolution is driven by interactions with bound stars orbiting within the binary sphere of influence (Sesana, Haardt, et al. 2008). As the binary hardens, these stars are progressively ejected from the central region, carrying away the orbital energy extracted from the binary and causing its separation to decrease towards the hardening radius

$$a_h = \frac{Gm_2}{4\sigma^2}, \quad (2.34)$$

below which the stellar-scattering term included in Equations (2.20) becomes effective.

Provided that a sufficient supply of loss-cone stars is available, the binary eventually enters the GW-dominated regime, in which GW emission drives the orbital decay towards final coalescence. The transition occurs at the separation

$$a_{GW} = 4 \left[\frac{128\pi(3 - \gamma)F(e)}{5H} \right]^{1/5} \frac{\sigma}{c} q^{-4/5} (1 + q)^{3/5} a_h, \quad (2.35)$$

for a stellar cusp like stellar background and defined by the condition $H_u = H_{GW}$ (Sesana 2010).

In conclusion, for the equal-mass BH binaries considered in this work, the initial binary separation a_0 given by Equation (2.32) and the hardening radius a_h defined in Equation (2.34) are of the same order of magnitude. Therefore, the contribution of hardening due to the erosion of the bound stellar cusp is negligible over the range of separations considered here and can be omitted from our analysis (Sesana and Khan 2015).

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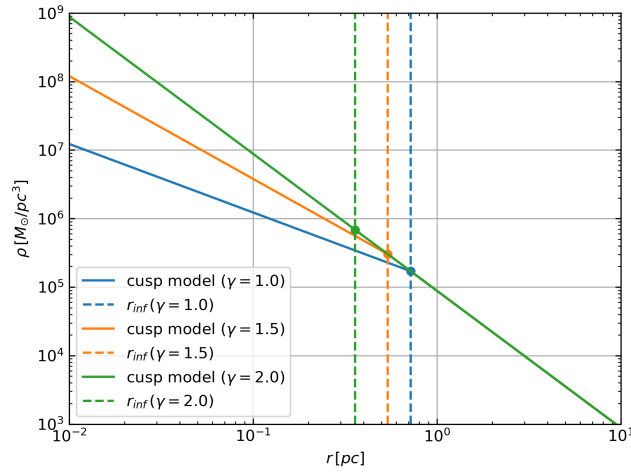


Figure 2.2: ...

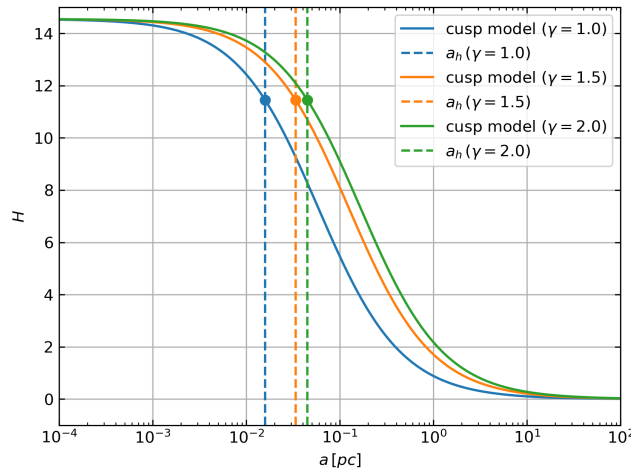


Figure 2.3: ...

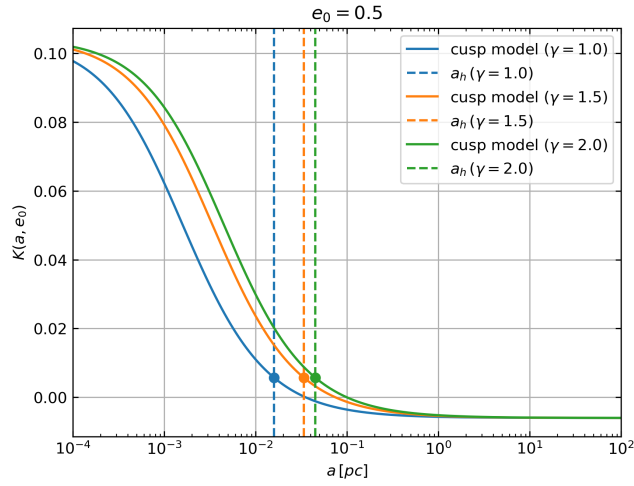


Figure 2.4: ...

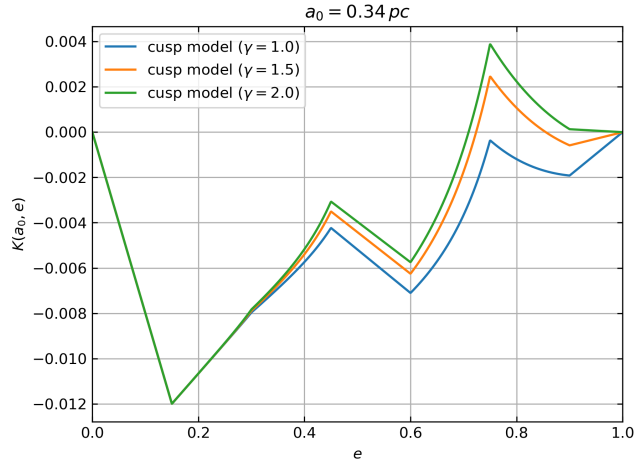


Figure 2.5: ...

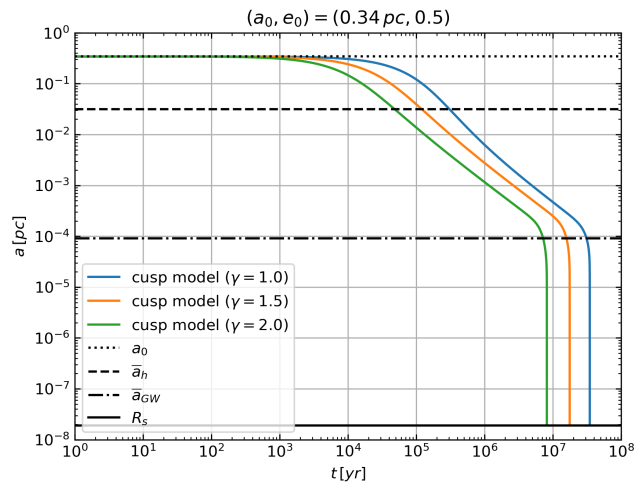


Figure 2.6: ...

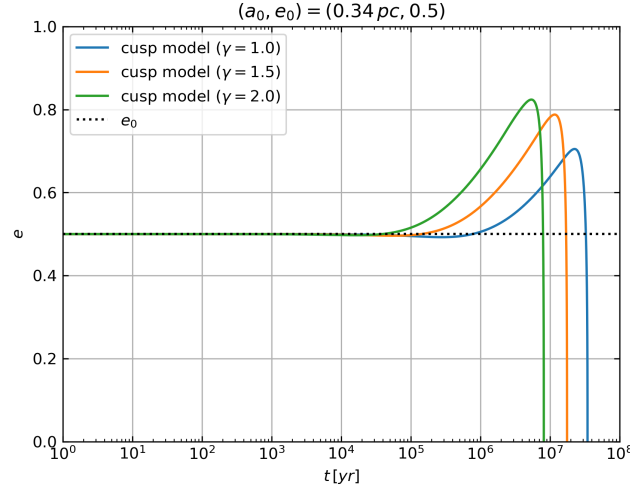


Figure 2.7: ...

HERE

Bound cusp erosion:

$$\left. \frac{da}{dt} \right|_b = -\frac{2a^2}{GM_1M_2} \int_0^\infty \Delta \mathcal{E} \frac{d^2 N_{ej}}{da_\star dt} da_\star \quad (2.36)$$

$$\left. \frac{de}{dt} \right|_b = \int_0^\infty \Delta e \frac{d^2 N_{ej}}{da_\star dt} da_\star \quad (2.37)$$

This work Sesana, Haardt, et al. 2006 is aimed at unveiling by full three-body scattering experiments how a population of HVS form as a result of the interaction between a MBHB and ambient stars, unbound to the binary, drawn from a MB distribution . Other than the dynamical properties of these HVS, binary hardening, mass ejection rate and eccentricity growth rate (H, J, K) are tracked by numerical simulations.

Dynamical friction is efficient (especially in wet major mergers) in dragging two massive BHs towards the innermost region of the new formed galaxy; over the time scale t_{df} the binary forms and hardens through gravitational slingshot, hence ejecting low angular momentum stars passing close to the binary.

Following Q96 approach, the equations defining (H, J, K) are (2.1),(2.3) and (2.2).

At odds with Q96, where three-body scattering experiments are conducted for a fixed impact velocity, varying the binary separation and then averaging over the Maxwellian distribution, now the separation is fixed, i.e. $V_c = (GM_{12}/a)^{1/2} = const$, and the quantity v/V_c is sampled over four decades, corresponding to a sampling of about four decades in binary separation.

Each run is set by the initial configuration, corresponding to a point in the parameter space (q, e_0); the combination of binary masses ratio $q = 1, 1/3, 1/9, 1/27, 1/81, 1/243$ and initial eccentricity $e_0 = 0.01, 0.3, 0.6, 0.9$ leads to 24 runs, involving $4 \cdot 10^6$ stars each.

The (Maxwellian averaged) rates H, J , and K derived from scattering experiments are fitted by the following functions

$$H = A(1 + a/a_0)^\gamma \quad (2.38)$$

$$J = A(a/a_0)^\alpha [(1 + (a/a_0)^\beta)^\gamma] \quad (2.39)$$

$$K = A(1 + a/a_0)^\gamma + B. \quad (2.40)$$

shows that the hardening rate is nearly constant when $a < a_h$, in perfect agreement with Q96 results. Therefore hard binary can be characterised by a constant hardening rate. Additionally,

H decreases with q (asymmetric binaries harden faster) and increases with eccentricity (circular binaries harden more slowly).

As reported in Figure ??, K is close to zero at any binary separation for circular binary; if the binary is eccentric at a_h already, then K (and e) increases monotonically with binary shrinking.

NOTE: in this paper the binary shrinking is studied taking into account gravitational slingshot on UNBOUND stars passing close to binary and ejected at higher velocity. In particular:

1. the ejection criterion relies on the stellar background model (here SIS stellar Bulge is considered)
2. The stellar background is assumed to be fixed; this approximation holds if the ejected mass does not exceed a trashold
3. Binary shrinking requires low angular momentum stars as they have to pass closer and closer to the binary BHs for slingshot to be effective; however from $a < a_h$ the fraction of stars with these properties, populating the so called "Loss cone", is very small. Being depleted by three-body encounters, binary hardening when binary is hard depends on the efficiency of loss cone refilling.

Bound stars Sesana, Haardt, et al. 2008 ,Sesana 2010

and is adopted as starting point of the numerical integration.

Hereafter, three competing mechanisms are involved:

Being independent of each other, by superposition, we can put the pieces together and the evolution of the binary obeys to

Besides, ; lengthscale boundaries are a_0 ,

$$a_h \approx \frac{1}{4(3-\gamma)} \left(\frac{q}{1+q} \right)^{(2-\gamma)(3-\gamma)} \quad (2.41)$$

and

respectively.

We recall that

$$\rho_{inf} = \frac{\sigma^2}{2\pi G r_{inf}^2} \quad (2.42)$$

and that setting $\rho = \rho_{inf}$ inside Equation (2.55) is equivalent to impose a full loss-cone outside the binary sphere of influence during the unbound phase; the effects due to different normalizations could be explored.

Sesana Sesana 2010 explored a relatively wide region of the phase space related to the initial configuration of the binary; here, we report the comparison between Sesana and our evolutionary track for $(q, \gamma, e_0, M) = (1, 1.5, 0.1, 10^5 M_\odot)$.

This kind of configuration, among the analyzed ones, is quite relevant to our studies on IMBH binaries at high z as they are expected to be more symmetric than their local universe cousins. As pointed out at the end of Section 1.4, a_h and r_{inf} are of the order of $\sim GM/\sigma^2$ and the bound cusp erosion process can be suppressed at first instance.

The estimate of the hardening radius, below which a_h is nearly constant, and the best fitting parameters in Table 1 Sesana, Haardt, et al. 2006 allows us to find

$$H = A(1 + a/a_0)^\gamma; \quad (2.43)$$

the function is shown in Figure ??.

Therefore, at odds with H , K is a function of (a, e) ; we list below the plots of the 1-D functions $K(a) = K(a, e_0)$ and $K(e) = K(a_h, e)$.

In order to compare Sesana predictions with our model we set (t_0, a_0, e_0) for the unbound phase to be effective and selecting these values directly from Sesana plot: $(t_0, a_0, e_0) = (12Myr, 0.05 \cdot a_0, 0.131)$.

indicate a good agreement between our model and Sesana one. In the first plot a decreases more slowly than Sesana prediction as we are neglecting the bound cusp erosion contribute, which hardens the binary even when the unbound phase takes over at $\sim a_h$. On the other hand eccentricity reaches a lower peak compared to the reference model because of bound cusp erosion leads to e growth as well. Furthermore, we test the feedback of the various shrinking mechanisms by assuming them working in isolation, hence suppressing all the others.

Figure refers to the evolution under the effect of slingshot of unbound stars; as the initial separation is below a_h we can appreciate the expected linear decay of a and the overall eccentricity growth.

Figure refers to the evolution under GWs production; this becomes effective only at very close separation and causes both the fast binary coalescence once effective and binary circularization. The coalescence time agrees perfectly to the theoretical prediction of Equation 1.138

The dataset used above help us to construct another curve, precisely $(e(t, a(t)))$ which can be compared to the close form solution provided by Equation (1.135) as shown in Figure 2.8 and 2.9.

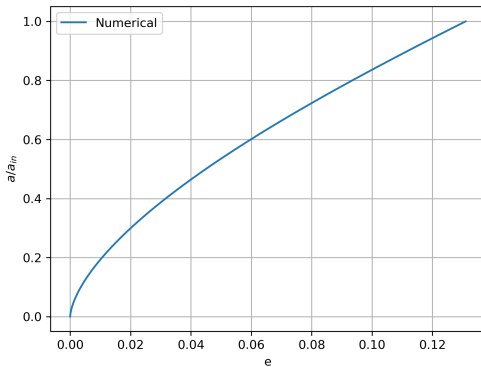


Figure 2.8

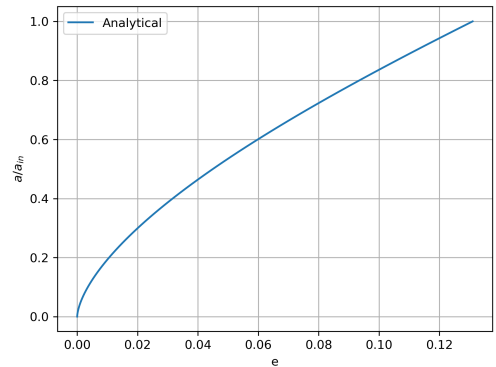


Figure 2.9

Below, the Figure 2.10 gives us the possibility of visualizing the relative error of the numerical solution with respect to the analytical one; it turns out to be within one part over 10^{12} .

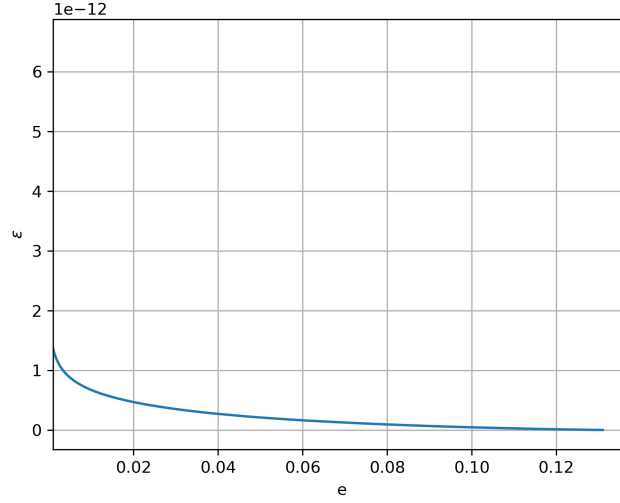


Figure 2.10

2.3 NGC 205 IMBH Binary evolution

NGC 205 is a nearby dwarf galaxy with a NSC residing at its innermost region; it is believed to host an IMBH binary, whose estimated mass is $\sim 10^4 M_\odot$.

The 2-D surface brightness profile of the dwarf, both the bulge and NSC treated as independent stellar components, is translated into 3-D Sersic density profile using Prugniel & Simien **Prugniel**:

$$\rho(r) = \rho_0 \left(\frac{r}{R_e} \right)^{-p} e^{-b \left(\frac{r}{R_e} \right)^{1/n}}. \quad (2.44)$$

Here, ρ_0 is the density normalization unambiguously set by total galaxy mass, effective radius and Sersic index n ; from **Terzic**:

$$\rho_0 = \frac{M_*}{4\pi R_e^3 n \Gamma[n(3-p)]} b^{n(3-p)}. \quad (2.45)$$

The parameters p and b can be expressed as a function of n as follows: $p = 1.0 - 0.6097/n + 0.05563/n^2$, and $b \approx 2n - 1/3 + 0.009876/n$, for $0.5 < n < 10$.

Observations constrain the quantities (M, R_{eff}, n) for both bulge and NSC as reported in Figure (2.11) of Khan and Holley-Bockelmann 2021.

Component	n	$r_{\text{eff}}(\text{pc})$	M_* ($10^7 M_\odot$)	a_2/a_1
IMBH	–	0.14	0.004	–
NSC	1.6	1.3	0.18	0.95
Bulge	1.4	516	97.2	0.90

Figure 2.11: Column 1: Galaxy component. Column 2: Sersic index. Column 3: Effective radius of the galaxy component (note: for the IMBH, we quote the influence radius instead). Column 4: Stellar mass. Column 5: 2D axial ratio as measured by GALFIT.

It has (with standard substitution $Z = b \left(\frac{r}{R_e} \right)^{1/n}$)

$$M(< r) = 4\pi \rho_0 R_e^3 n b^{n(p-3)} \gamma \left[n(3-p), b \left(\frac{r}{R_e} \right)^{1/n} \right] = M_* \frac{\gamma \left[n(3-p), b \left(\frac{r}{R_e} \right)^{1/n} \right]}{\Gamma[n(3-p)]} \quad (2.46)$$

with

$$\Gamma(a) = \int_0^\infty dt e^{-t} t^{a-1} \quad (2.47)$$

the "Complete gamma function",

$$\Gamma(a, x) = \int_x^\infty dt e^{-t} t^{a-1} \quad (2.48)$$

the Upper incomplete gamma function and

$$\gamma(a, x) = \int_0^x dt e^{-t} t^{a-1} \quad (2.49)$$

the Lower incomplete gamma function. By definition follows:

$$\Gamma(a) = \Gamma(a, x) + \gamma(a, x) \quad (2.50)$$

The spatial distribution is spherical and untruncated but the mass profile converges to the finite value M_* , so the potential can be found from

$$\Phi(r) = \Phi(r_0) + \frac{GM(< r_0)}{r_0} - \frac{GM(< r)}{r} - 4\pi G \int_r^{r_0} dt t \rho(t) \quad (2.51)$$

Setting $r_0 \rightarrow \infty$ and $\Phi(r_0) = 0$, it becomes

$$\Phi(r) = -\frac{GM(< r)}{r} - 4\pi G \int_r^\infty dt t \rho(t) \quad (2.52)$$

and computation returns

$$\Phi(r) = -4\pi G \rho_0 R_e^2 n b^{n(p-2)} \left\{ \frac{b^{-n}}{r/R_e} \gamma[n(3-p), Z] + \Gamma[n(2-p), Z] \right\} \quad (2.53)$$

or equivalently

$$\Phi(r) = -\frac{GM_*}{r} \left\{ \frac{\gamma[n(3-p), Z]}{\Gamma[n(3-p)]} + Z^n \frac{\Gamma[n(2-p), Z]}{\Gamma[n(3-p)]} \right\} \quad (2.54)$$

Numerically it is found

- Bulge: $\rho_0 = 4.80 \cdot 10^5 M_\odot / pc^3$
- NSC: $\rho_0 = 2.95 M_\odot / pc^3$

and NGC 205 density profile is reconstructed in Figure (2.12).

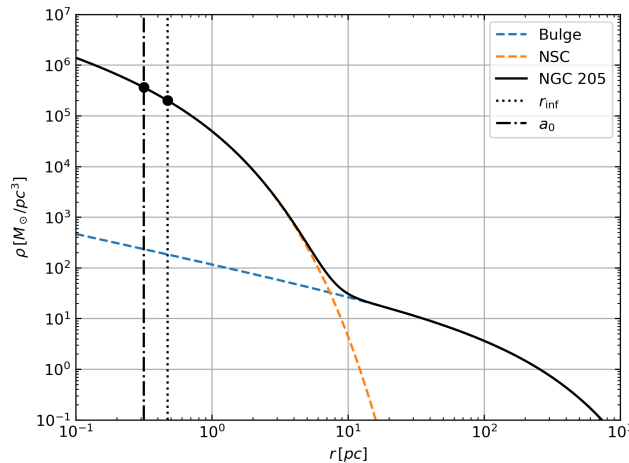


Figure 2.12: NGC 205 bulge, NSC and total mass density profile of stellar distribution.

Adopting the same prescription of Khan and Holley-Bockelmann 2021 and Holley-Bockelmann et al. 2025, a symmetric IMBH binary of masses $M_1 = M_2 = 4 \cdot 10^4 M_\odot$ is introduced into NGC 205 core and its evolution, due to stellar and GWs hardening, is tracked combining 3-body scattering experiment hybrid model Quinlan 1996, Sesana, Haardt, et al. 2006 and Peters and Mathews 1963 equations.

It is quite relevant to point out that in this astrophysical system the hardening due to bound stars Sesana, Haardt, et al. 2008 is negligible in light of the symmetry of the binary and the arguments exposed at the end of Sec.(1.4).

Therefore, combining the set of equations related to Unbound evolution

$$\frac{da}{dt}|_u = -\frac{a^2 G \rho_{\text{inf}}}{\sigma} H \quad (2.55)$$

$$\frac{de}{dt}|_u = \frac{a G \rho_{\text{inf}}}{\sigma} H K \quad (2.56)$$

and the equations describing GWs driven hardening 1.132 1.133, the set of differential equations must be solved reads

$$\frac{da}{dt} = \frac{da}{dt}|_u + \frac{da}{dt}|_{gw} \quad (2.57)$$

$$\frac{de}{dt} = \frac{de}{dt}|_u + \frac{de}{dt}|_{gw} \quad (2.58)$$

Although the original formulation of the hybrid model relies on a SIS profile, with very peculiar features (i.e. constant velocity dispersion), Sesana Sesana and Khan 2015 showed how to use it for a more general stellar background. The vast majority of stars, unbound to the binary, refilling the loss cone (i.e. having pericenter $\sim a$) come from a region that is just beyond the binary influence radius, hence r_{inf} is the relevant physical scale of the model and σ appearing in Equations 2.55 and 2.56, which is in general a function of the radius for spherically symmetric systems, must be computed at the influence radius (here defined as the radius enclosing stellar mass that is twice the binary one).

Apart from setting the initial point in the (a, e) space, we need to determine several quantities; we start with binary influence radius and $\sigma_{\text{inf}} = \sigma(r_{\text{inf}})$; they both require the knowledge of the cumulative mass $M(< r)$ associated to the Sersic profile. It results from the integration of $\rho(r)$ and admits analytic solution **Terzic**

$$M(< R) = M_{\text{tot}} \frac{\gamma\left(2n, b_n \left(\frac{R}{R_e}\right)^{1/n}\right)}{\Gamma(2n)}. \quad (2.59)$$

The root of the equation

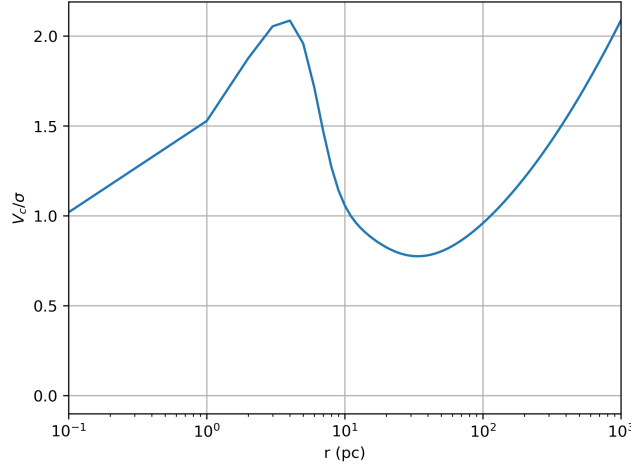
$$M(< r_{\text{inf}}) - 2(M_1 + M_2) = 0 \quad (2.60)$$

is obtained numerically by bisection and holds $r_{\text{inf}} = 0.47 pc$.

On the other hand, Jeans equation provides a way to compute the velocity dispersion and in case of spherical distributions it greatly simplifies to

$$\sigma^2(r) = \frac{1}{\rho(r)} \int_r^\infty \rho(s) \frac{GM(< s)}{s^2} ds. \quad (2.61)$$

The integral leads to $\sigma_{\text{inf}} = 32.6 km/s$. As both $\sigma(r)$ and the circular speed $V_c(r) = \sqrt{GM(< r)/r}$ depend on $M(< r)$, previously obtained, we can derive them and Figure 2.13 displays their ratio as a function of the radius.

Figure 2.13: NGC 205 V_c/σ vs r

The last quantity we have to specify in order to implement the hybrid model is the binary hardening radius a_h ; indeed Sesana, Haardt, et al. 2006 found out from 3-body scattering experiments that H and K functions can be fitted to

$$H = A(1 + a/a_0)^\gamma \quad (2.62)$$

and

$$K = A(1 + a/a_0)^\gamma + B \quad (2.63)$$

The parameters of the fits to H and J are listed in Tables 1 and 2 of Sesana, Haardt, et al. 2006 for different binary mass ratios and eccentricity; there, a_0 is given in a_h units, thus forcing us to determine it. In literature several definitions have been proposed; hereafter, we will be using

$$a_h \approx \frac{GM_2}{4\sigma^2}, \quad (2.64)$$

unless otherwise stated. This definition leads to $a_h \approx 0.04pc$.

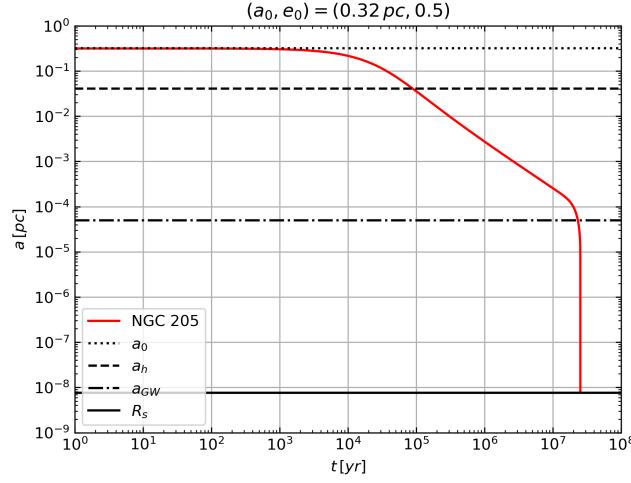
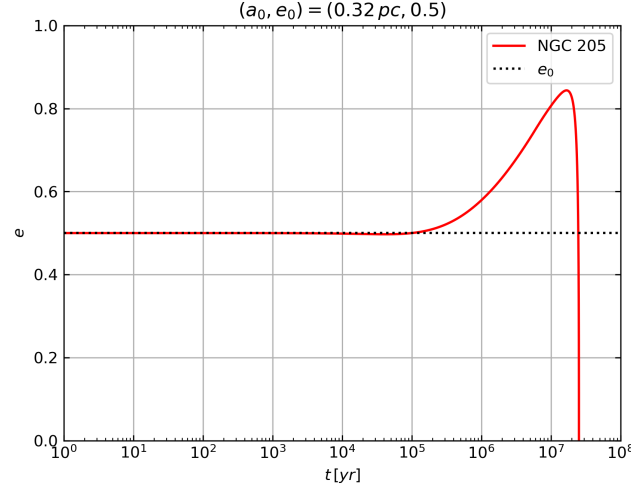
We can finally proceed and numerically integrate the set of equations 2.57 and 2.58 by means of R-K(4). For the temporal evolution to be properly resolved from the initial configuration to the coalescence of the binary, the time step adjusts itself at each iteration, thus taking care of whether the dominant timescale is set by stellar hardening or GWs hardening at that time.

By defining

$$\Delta t_i = \epsilon \cdot \min \left(\frac{a}{|da/dt|_u}, \frac{a}{|da/dt|_{gw}} \right), \quad (2.65)$$

where $\epsilon \sim 10^{-3-4}$, and the initial conditions $(a_0, e_0) = (15pc, 0.7)$, the integration returns $a(t)$ and $e(t)$.

They are shown in Figure 2.14 and 2.15.

Figure 2.14: NGC 205 a vs t Figure 2.15: NGC 205 e vs t

The estimated coalescence time is $t_{coal} = 13.15 Myr$; it is of the same order of magnitude of Khan and Holley-Bockelmann 2021 and Holley-Bockelmann et al. 2025 predictions of $t_{coal} = 74 Myr$.

NOTE: Khan estimate comes from N-body simulations and in the special case of NGC 205 a constant value of eccentricity $e = 0.39$ is assumed for the late evolution of the binary.

Assuming a reliable time (universal time) at which the binary is in the configuration (a_0, e_0) it might be interesting to get an estimate of the corresponding z_{coal} from

$$t(z) = \int_z^\infty \frac{dz'}{(1+z')H(z')} \quad (2.66)$$

and compare it with LISA upper limit ($z \sim 20$), hence addressing the problem of a possible detection.

If the binary is in the initial configuration at $z \approx 10$, it will coalesce at $z \approx 9.4$.

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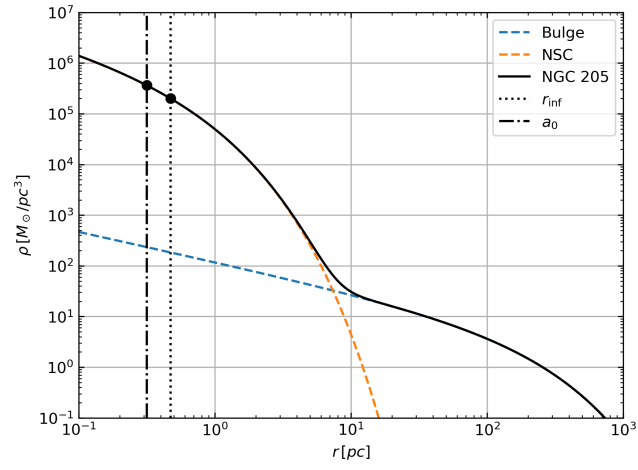


Figure 2.16: ...

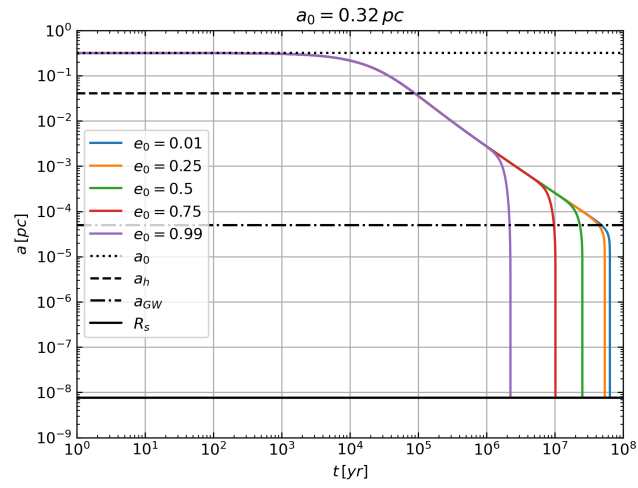


Figure 2.17: ...

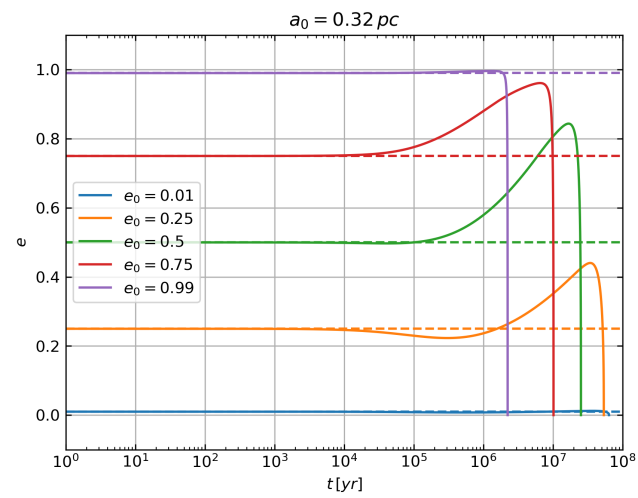


Figure 2.18: ...

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2.3.1 Cosmic Size Evolution Down to NGC 205

Here, we explore the importance of the existence of NSC to the binary evolution and coalescence time. A such structure is expected to drastically reduce the time taken by system to merge as a consequence of the larger number of stars capable of passing close to the binary.

In this spirit, we will be referring to the same parameters table a procedure of the prevois SubSection but the NSC which is artificially removed from the dwarf galaxy; hence, the only stellar background affecting binary evolution is the stellar bulge.

By repeating the same analysis and setting $(t_0, a_0, e_0) = (0yr, 15pc, 0.7)$, we show Figure 2.19 and 2.20.

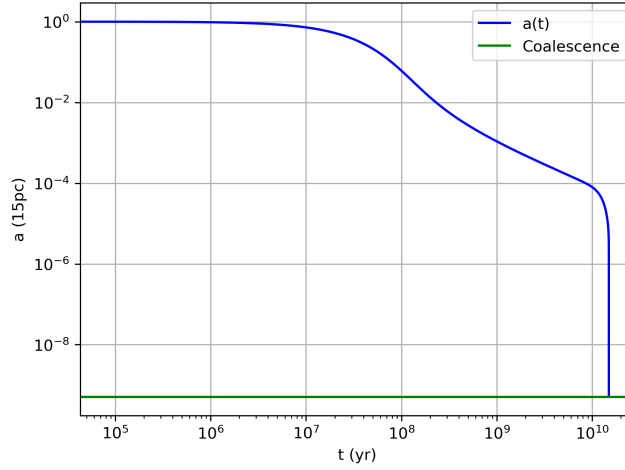


Figure 2.19

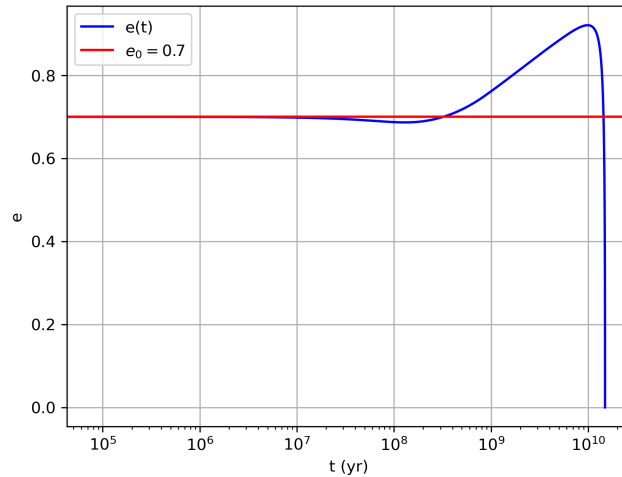


Figure 2.20

Now $t_{coal} = 14.93Gyr$, almost three orders of magnitude longer than the NSC scenario. Following Ono cosmic size evolution relation $R_{eff}(z) = R_{eff}(z=0)(1+z)^{-1.22}$ Ono et al. 2023, we take NGC 205 as a $z=0$ reference model and we discuss the evolution of the same IMBH binary inside its plausible progenitors distributed between $z=1$ and $z=20$. All galaxies share same bulge mass and Sersic index; density profiles differ from each other only because of the different effective radius as displayed in Figure 2.21.

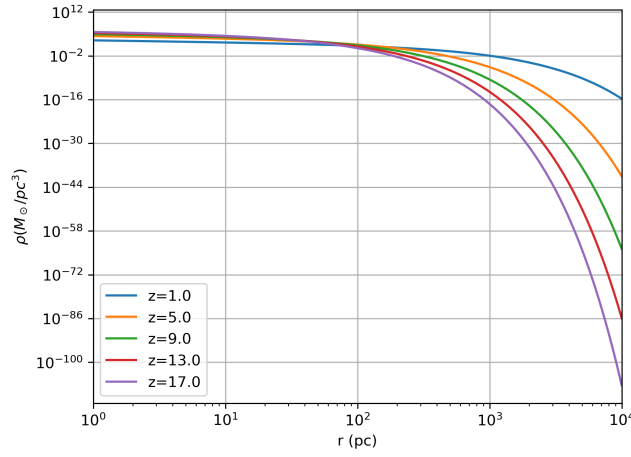


Figure 2.21

High redshift galaxies tend to be more compact than low redshift counterparts. Hardening radii are easily derived from these profiles according to Equation 2.64 and a comparison between different hosts H and K functions- see Figure 2.22 and 2.23- suggest that IMBH binary becomes "hard" at lower separation if embedded in a denser environment.

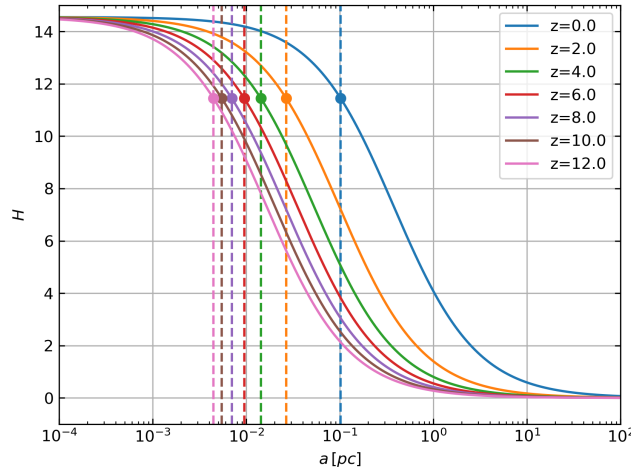


Figure 2.22

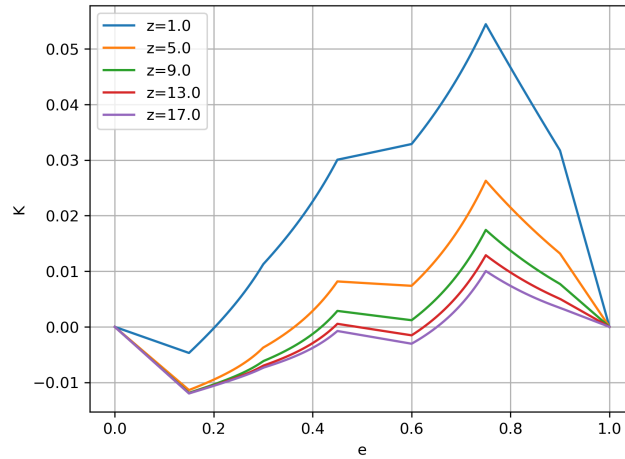


Figure 2.23: This family of functions is obtained evaluating $K(a, e)$ of each galaxy at their mean hardening radius.

We present in Figure 2.24 and 2.25 the evolutionary tracks of a sample of galaxies obtained by applying RK(4) to the Equations (2.57) and (2.58) and initial condition $(t, a_0, e_0) = (0 \text{ yr}, 15 \text{ pc}, 0.7)$.

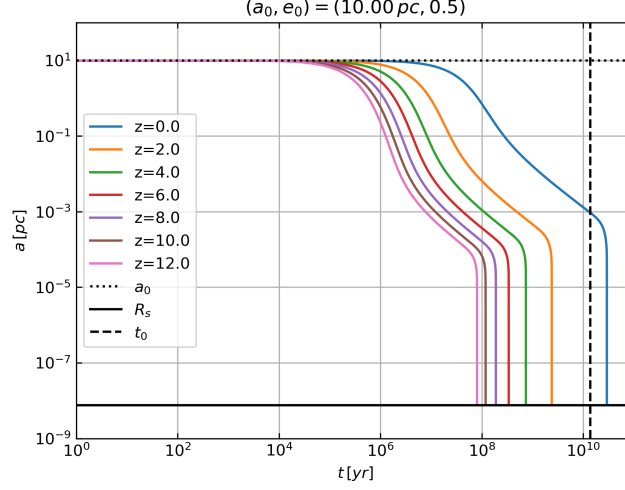


Figure 2.24

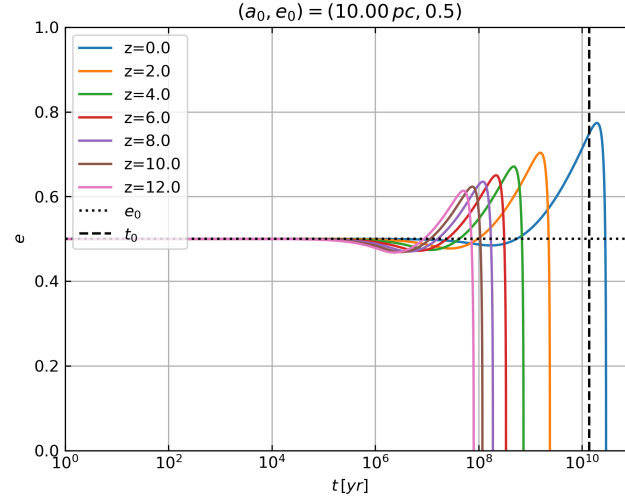


Figure 2.25

We note that higher compactness tends to compensate the effects of NSC removal in terms of the coalescence time.

Lastly, we modify the initial time t_0 of each run, turning it into $t(z)$, thus focusing on the possibility of coalescence within the present time and consequently a potential GWs observation.

Table 2.3: Galaxies coalescence time grid (Gyr)

z	t_{coal} (Gyr)
1	3.24728387
2	1.334011
3	0.7072763
4	0.43164087
5	0.28805534
6	0.20450559
7	0.15193439
8	0.11686942
9	0.09240003
10	0.07469761
11	0.06150781
12	0.05143637
13	0.04358488
14	0.03735426
15	0.03233322
16	0.02823207
17	0.02484224
18	0.02201059
19	0.01962278
20	0.01759202

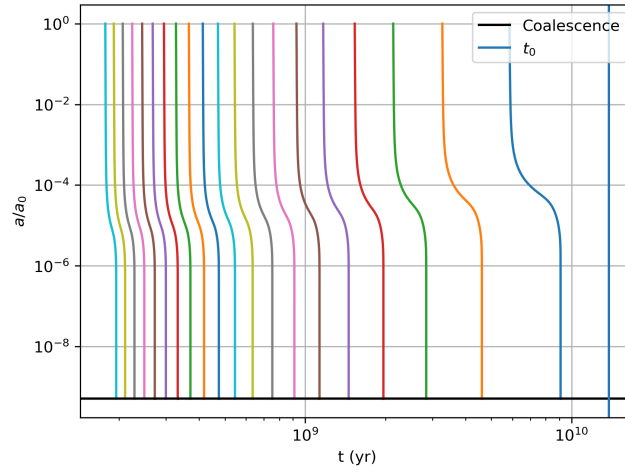


Figure 2.26

Therefore, forming in the past at $z > 0$, they all merge before the present time.

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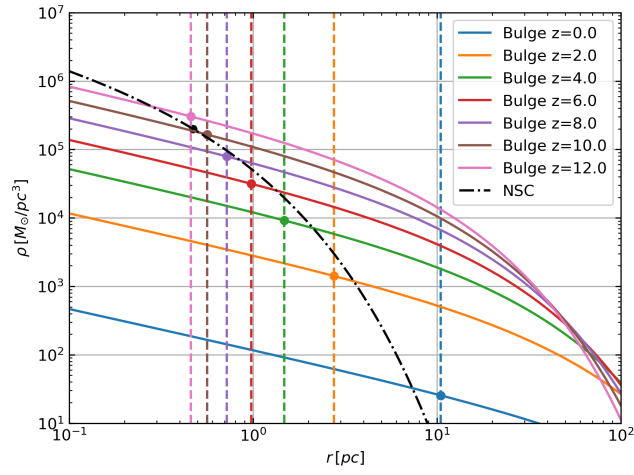


Figure 2.27: ...

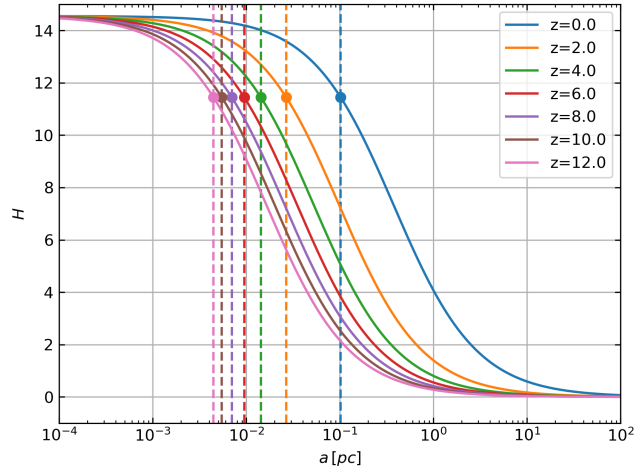


Figure 2.28: ...

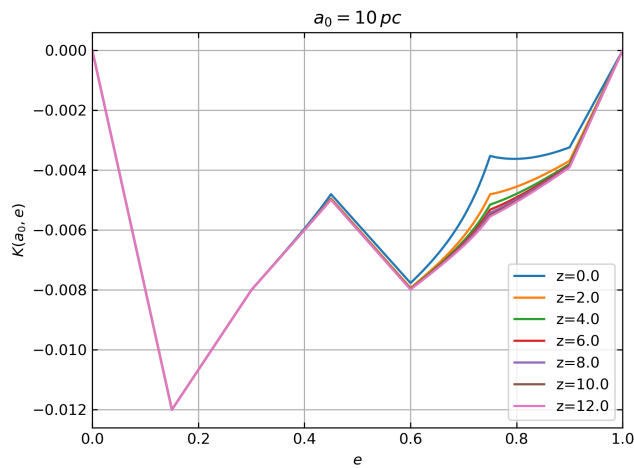


Figure 2.29: ...

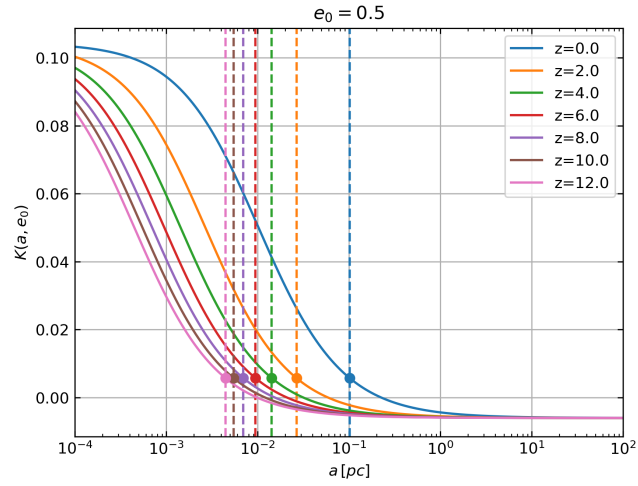


Figure 2.30: ...

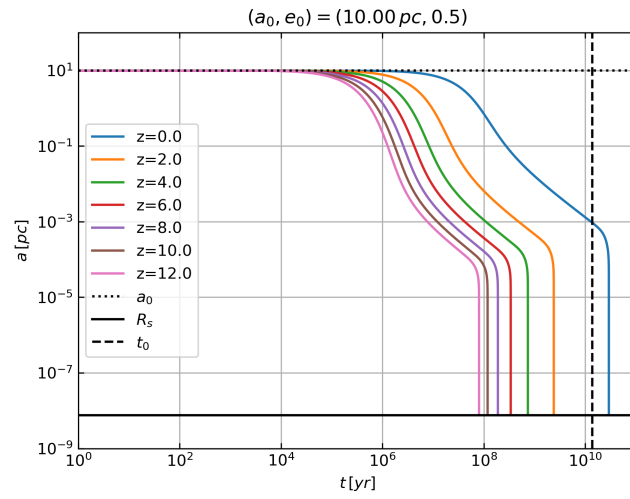


Figure 2.31: ...

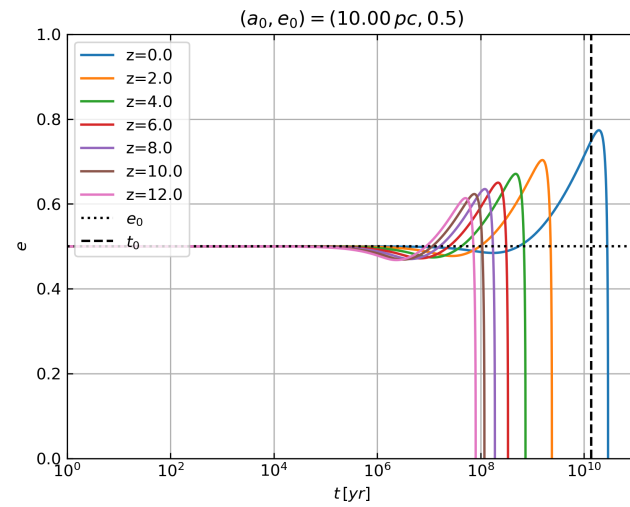


Figure 2.32: ...

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3

GENERATION OF INITIAL CONDITIONS FOR N-BODY STELLAR SYSTEMS

3.1 Dynamical Equilibrium of Stellar Systems

as anticipated in Section 1.1.

As long as the stellar component is observed within t_{2b} , granularity is unimportant, collisions are negligible and stars can be described as particles moving in a smooth potential generated by a continuous mass density ρ , proxy of the actual stellar distribution. In this regime, when dealing with stars that are all identical to one another, the DF will be unique and will satisfy the Collisionless Boltzmann Equation (CBE). By denoting with $\mathbf{w}$ an arbitrary set of canonical coordinates of PS, this conservation equation descends from differentiating with respect to the time—along the orbits of the potential $\Phi(\mathbf{x}, t)$ associated to $\rho(\mathbf{x}, t)$ —of the normalization condition

$$\int_{\mathbb{R}^6} d\mathbf{w} f(\mathbf{w}) = 1. \quad (3.1)$$

By extending the Lagrangian derivative notion D/Dt to the coordinates of the phase-space, the CBE reads:

$$\frac{Df}{Dt} = \frac{\partial f}{\partial t} + \dot{\mathbf{w}} \cdot \frac{\partial f}{\partial \mathbf{w}} = \frac{\partial f}{\partial t} + [f, H] = 0, \quad (3.2)$$

where the square bracket indicates the Poisson bracket, which is defined as

$$[f, H] = \sum_i \left(\frac{\partial f}{\partial q_i} \frac{\partial H}{\partial p_i} - \frac{\partial f}{\partial p_i} \frac{\partial H}{\partial q_i} \right), \quad (3.3)$$

and H is the Hamiltonian function of the system.

The arbitrariness of the canonical coordinates $\mathbf{w}$ enables us to specialize Equation (3.2) to the inertial Cartesian coordinates $(\mathbf{x}, \mathbf{v})$. With this choice, the Hamiltonian becomes

$$H(\mathbf{x}, \mathbf{v}, t) = \frac{1}{2}v^2 + \Phi(\mathbf{x}, t) \quad (3.4)$$

and the CBE turns into

$$\frac{\partial f}{\partial t} + \mathbf{v} \cdot \frac{\partial f}{\partial \mathbf{x}} - \frac{\partial \Phi}{\partial \mathbf{x}} \cdot \frac{\partial f}{\partial \mathbf{v}} = 0. \quad (3.5)$$

Being interested in studying the equilibrium condition of galaxies and in particular of spherically symmetric models, it is useful the following definition: a dynamical system is said to be

stationary if $\partial f / \partial t = 0$. Therefore, collisionless systems that have achieved equilibrium will be provided of a DF obeying to the steady-state CBE

$$\frac{Df}{Dt} = \dot{\mathbf{w}} \cdot \frac{\partial f}{\partial \mathbf{w}} = [f, H] = 0. \quad (3.6)$$

Noteworthy, for this condition to apply it is sufficient to have a steady-state potential $\Phi(\mathbf{x}, \mathbf{v})$. Solutions to Equation (3.6) are referred to as integrals of motion. Specifically, the function $I(\mathbf{x}, \mathbf{v})$ is an integral of motion for the potential $\Phi(\mathbf{x}, \mathbf{v})$ if it satisfies

$$\frac{D}{Dt}[I(\mathbf{x}(t), \mathbf{v}(t))] = 0 \quad (3.7)$$

along any orbit of the potential and a relevant example is represented by the Hamiltonian itself of the system, provided

$$[H, H] = 0. \quad (3.8)$$

The comparison between Equation (3.7) and Equation (3.6) reveals that solving the steady-state CBE is equivalent to being an integral of motion; this leads to the formulation of the Jeans theorem.

Jeans Theorem. – *Any steady-state solution of the collisionless Boltzmann equation depends on the phase-space coordinates only through integrals of motion in the given potential, and any function of the integrals yields a steady-state solution of the collisionless Boltzmann equation.*

The remaining quantities to be defined, which correspond to observables that can be inferred from kinematic measurements, are the first- and second-order velocity moments of the distribution function. These are

$$\overline{v_i}(\mathbf{x}, t) \equiv \frac{1}{\rho(\mathbf{x}, t)} \int_{\mathbb{R}^6} d^3\mathbf{v} f v_i \quad (3.9)$$

and

$$\sigma_{ij}^2(\mathbf{x}, t) \equiv \frac{1}{\rho(\mathbf{x}, t)} \int_{\mathbb{R}^6} d^3\mathbf{v} f (v_i - \overline{v_i})(v_j - \overline{v_j}) = \overline{v_i v_j} - \overline{v_i} \overline{v_j}, \quad (3.10)$$

respectively the mean velocity and the velocity dispersion tensor.

We are now finally in a position to recall the most peculiar kinematical properties of an important subclass of equilibrium models, namely Ergodic systems. The Jeans Theorem imposes that the DF f of systems at equilibrium will be any non-negative function of H ; in case f depends on H only, i.e. $f = f(H)$, then systems with this type of DF are said Ergodic.

Let us focus on first-order moment of the DF. In the inertial Cartesian frame where stars move under the action of a steady-state potential $\Phi(\mathbf{x})$, the Hamiltonian of the system is an even function of $\mathbf{v}$, causing $\overline{v_x}(\mathbf{x}) = \overline{v_y}(\mathbf{x}) = \overline{v_z}(\mathbf{x})$ to vanish.

Similarly, it can be demonstrated that the velocity-dispersion tensor is isotropic; thus it holds

$$\sigma_{ij}^2(\mathbf{x}) = \sigma^2(\mathbf{x}) \delta_{ij} = \delta_{ij} \frac{4\pi}{3\rho(\mathbf{x})} \int_0^{+\infty} dv v^4 f \left(\frac{1}{2} v^2 + \Phi(\mathbf{x}) \right) \quad (3.11)$$

Therefore, every system characterized by an ergodic DF is isotropic. In the next Section, the properties of the generated steady-state isotropic stellar distributions will be presented and compared with the theoretical predictions summarized above.

So far, we have been concerned with the velocity-moments of the distribution function of Ergodic systems; however, the generation of initial conditions (ICs) for N-body, equilibrium models relies on the correct sampling of the DF itself. Recovering it from ρ for globally isotropic stellar systems can be done by means of the Eddington's formula

$$f(\varepsilon) = \frac{1}{\sqrt{8\pi^2}} \frac{d}{d\varepsilon} \int_{\varepsilon_t}^{\varepsilon} \frac{d\rho}{d\Psi} \frac{d\Psi}{\sqrt{\varepsilon - \Psi}} \quad (3.12)$$

with $\varepsilon = -H$ and $\Psi = -\Phi$ being the relative energy and potential respectively and ε_t the truncation energy of the system.

Finally, the last tool introduced in this Section is the virial ratio, closely connected with the Virial theorem, which provides a fast diagnostic for determining whether the generated model is in equilibrium or is either evolving toward collapse or expansion. By definition a self-gravitating system, constituted by N particles, with total kinetic energy

$$K = \sum_{i=1}^N \frac{1}{2} m_i v_i^2 \quad (3.13)$$

and gravitational potential energy

$$U = -\frac{1}{2} \sum_{i=1}^N \sum_{j \neq i} \frac{G m_i m_j}{|\mathbf{x}_i - \mathbf{x}_j|} \quad (3.14)$$

is virialized if it holds

$$2K' + U = 0 \quad \forall t \quad (3.15)$$

in the barycentric inertial reference system S_{CM} . For these systems the virial ratio

$$Q \equiv \frac{2K'}{|U|} \quad (3.16)$$

should be close to unity. In presence of a non vanishing velocity of the system center of mass, which flags that the time evolution is studied in a frame S moving with respect to S_{CM} , Equation (3.16) must be adjusted as follows:

$$Q \equiv \frac{2(K - K_{CM})}{|U|}. \quad (3.17)$$

3.2 OpOpGadget and Galaxy Modeling

Once recasted the main aspects of the theory underlying the dynamical equilibrium configuration of stellar systems, with focus on Ergodic DFs, we can now address the problem of generative N-body ICs for stellar populations which are compatible with the inferred quantities: total mass, density profile and associated structural parameters.

Therefore, stellar positions and velocities are generated by means of `OpOpGadget`. This Python module offers a number of tools for both generating N-body ICs and analyzing N-body simulations outputs. `OpOpGadget` is equipped with an extensive library of analytic models for spherically symmetric stellar distributions. The first step in the workflow initialization is to specify the desired model, assign the values of the parameters appearing in its functional form, and choose the number of particles to represent the stellar environment. Then, particles spherical coordinates are assigned by randomly picked the cumulative mass profile

$$M_*(< r) = 4\pi \int_0^r dy y^2 \rho(y) \quad (3.18)$$

through the variable $u = np.random.uniform(M_{min} = 0, M_{max} = M_*, N)$. Because $M(< r)$ is a monotonically increasing function of radius, it can be inverted so that N radii of kind $r = M_*^{-1}(u)$ are finally obtained. Yet, two additional independent random variables, uniformly distributed in the intervals $[-1, 1]$ and $[0, 2\pi]$ respectively, are used to sample the direction cosine and the azimuthal angle, thereby obtaining N sets of spherical coordinates (r, θ, ϕ) , which

are subsequently transformed into Cartesian coordinates.

While particle positions are drawn from the cumulative mass distribution, the velocity components are obtained by sampling the DF, assuming that the gravitational potential Φ is generated by the mass distribution associated with the previously initialized particles. Three random numbers $\mathbf{v}$ lying in the range $[-1, 1]$ allows to temporary assign a normalized velocity to each particle, which will found to have the binding energy $\varepsilon = \Psi(1 - v^2)$. Finally, the Von Neumann rejection method is implemented and the normalized velocity is accepted if

$$u \leq \frac{f(\varepsilon)}{f(\varepsilon = \Psi)}, \quad (3.19)$$

where u is another independent random number in $[0, 1]$. If the Inequality (3.19) is true, the final velocity vector will be obtained by taking the product between the normalized one and the factor $\sqrt{2\Psi}$. At the end of the process, each particle will have its own *ID*, mass $m = M_*/N$, position $\mathbf{x}$ and $\mathbf{v}$.

Let us denote with p the variable containing all this information, *OpOpGadget* provides already-implemented methods to this objects and the most relevant are:

- ***NbodyModel***: generates a spherically symmetric, multi-component, equilibrium model as mentioned above
- ***Analysis(NbodyModel)***: takes as argument an *NbodyModel* object and is furnished with the following methods, used in our analysis:
 1. ***Analysis.com()***: returns Cartesian and velocity of *NbodyModel*
 2. ***Analysis.softening_scale(kernel=Gadget)***: calculates the softening scale ϵ using the spline kernel of *Gadget2* (Springel 2005). This softening replaces the Dirac delta function in $\rho(r) = m\delta(r)$ with the normalized spline kernel

$$W(r, h) = \begin{cases} 1 - 6q^2 + 6q^3 & \text{if } 0 \leq q < 1/2 \\ 2(1 - q)^3 & \text{if } 1/2 \leq q < 1, \\ 0 & \text{if } q \geq 1. \end{cases} \quad q \equiv \frac{r}{h}, \quad (3.20)$$

having a compact support which extends to $h = 2.8\epsilon$, beyond which the gravitational potential reverts to the standard Newtonian form.

3. ***Analysis.tdyn(mq=50)***: provides the dynamical time of *NbodyModel* as $T_{dyn} = 0.5\pi\sqrt{R_h^3/(GM_h)}$, where R_h is the radius that contains the fraction $M_h = M_* \cdot mq\%$. This estimate has been corrected to make it compatible with the adopted definition of dynamical time t_{dyn} at half-mass radius r_{half} , which is

$$t_{dyn} = \frac{1}{\sqrt{G\bar{\rho}}} \quad (3.21)$$

with $\bar{\rho}$ being, for a spherical system,

$$\bar{\rho} = \frac{M/2}{\frac{4}{3}\pi r_{half}^3}. \quad (3.22)$$

- ***Profile(NbodyModel, xmin, xmax)***: has two methods that allows to reproduce between $xmin$ and $xmax$ the radial profile of relevant quantities:
 1. ***Profile.dens()***: outputs both the radial and density grids

2. *Profile.mass()*: outputs both the radial and cumulative mass grids

The dwarf galaxies, along with their nuclear star clusters, that have been generated are the same as those studied within the Sesana (2010) hybrid model framework. The comparison between the results of this semi-analytical, shown in Section (?), and the numerical merger simulations is intended to assess whether a model based on an already-formed binary black hole embedded in a single galaxy can effectively capture the properties that, in our case, emerge self-consistently from the full merger process.

In Khan and Holley-Bockelmann (2021), galaxies bulge and NSCs density profiles are expressed in terms of a Sersic profile with de-projection formula analyzed by **Prugniel** and reported in Equation (2.44). Nevertheless, the Sersic analytic model implemented in OpOpGadget takes as a reference Lima Neto et al. (1999) de-projection

$$\rho(r) = \rho_0 \left(\frac{r}{a}\right)^{-p} e^{-(r/a)^\nu}, \quad (3.23)$$

where $\nu = n^{-1}$, the inverse of the Sersic index, and a , denoting the Sersic profile scalelength at which the surface density $\Sigma(a) = \Sigma_0$, are known as primary parameters. Therefore, it is evident that the same density distribution can be described through different combinations of independent parameters and a proper conversion between primary and secondary, i.e. depending on primary ones, is required before continuing. Here, to pass from a to R_{eff} , we need to use the following analytical approximation

$$\ln R_{eff}/a = \frac{0.6950 - \ln \nu}{\nu} - 0.1789. \quad (3.24)$$

Moreover, particular attention should be paid to the fact that the stellar particles to be initialized are required to be in dynamical equilibrium within the combined gravitational potential of both stars and central black hole. The presence of such a massive and compact object is expected to modify the equilibrium configuration of the stars with respect to the isolated case, primarily by increasing their velocity dispersion; this can be done by passing a Python list of OpOpGadget analytic models to NbdoyModel. Despite being a single object rather than a particles distribution, a given model can be realized even with just one particle and the adopted one for generating BHs is the Plummer sphere. According to this choice, density and cumulative mass are respectively

$$\rho(r) = \frac{3M}{4\pi a^3} \left(1 + \frac{r^2}{a^2}\right)^{-5/2} \quad (3.25)$$

and

$$M(< r) = M \frac{r^3}{(r^2 + a^2)^{3/2}}; \quad (3.26)$$

here, a is the Plummer scalelength and is linked to the half-mass radius $r_{1/2}$ of the distribution by

$$a \simeq 0.766 r_{1/2}. \quad (3.27)$$

The use of gravitational softening in N-body experiments sets an intrinsic spatial resolution scale, below which the system cannot be accurately resolved. Accordingly, the Plummer scale radius a is taken to be the mean interparticle distance at the half-mass radius r_h

$$\lambda = r_h \left(\frac{2}{N}\right)^{1/3}, \quad (3.28)$$

which serves as a proxy for the gravitational softening length of the galaxy prior to its generation. Finally, Equation (2.51) shows that the Plummer potential

$$\Phi(r) = -\frac{GM}{\sqrt{r^2 + a^2}} \quad (3.29)$$

reduces to the Keplerian form at distances much larger than the scale radius, $r \gg a$. This asymptotic behavior explains why the Plummer model is a convenient description of BHs.

3.3 Analysis of the Generated Initial Conditions

3.3.1 NCG 205 N-body Initial Conditions

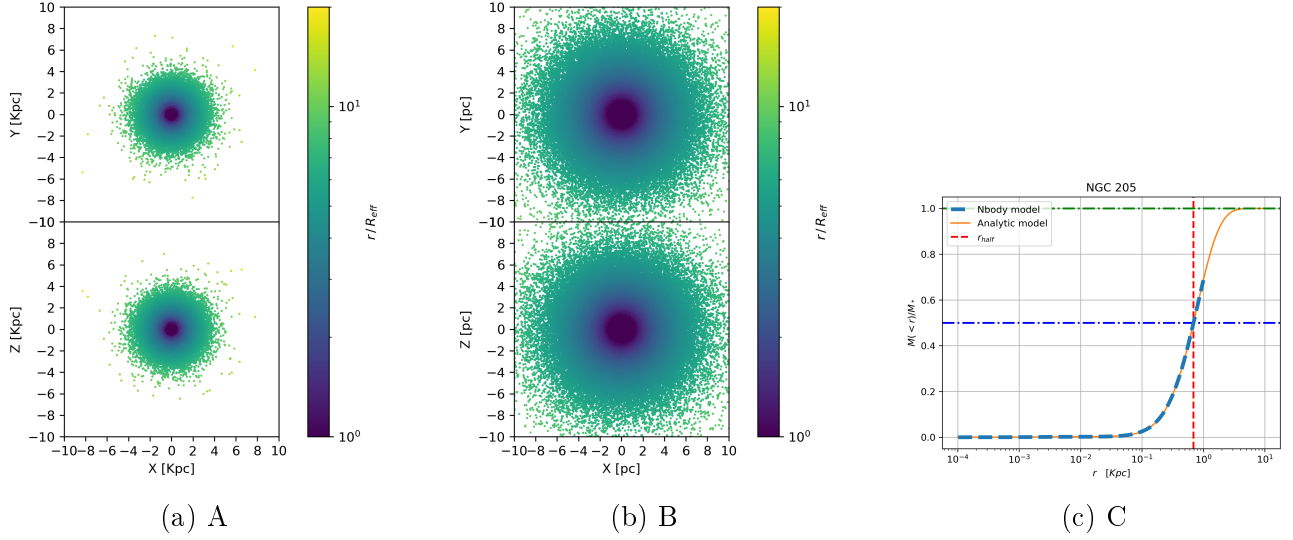


Figure 3.1: aaa

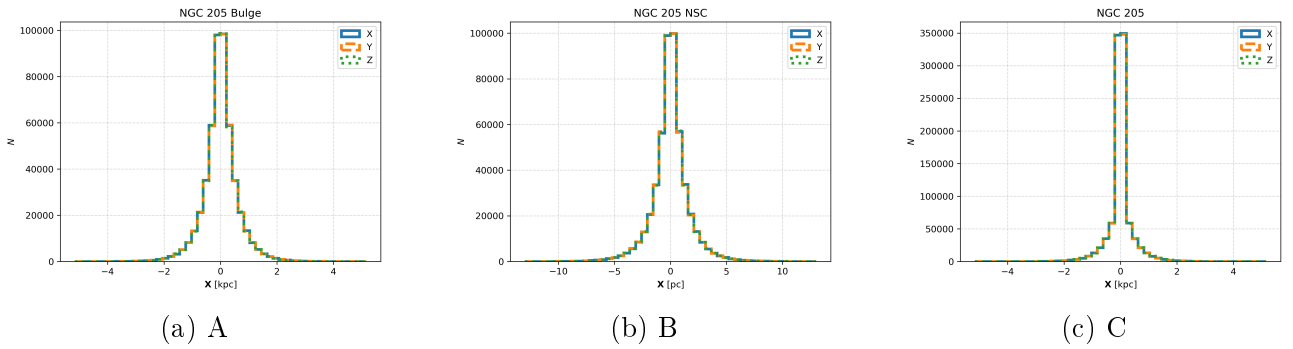


Figure 3.2: aaa

COM data:

- position:
- velocity:

Numerically is found $r_{half} = \dots = R_{eff}$.

Yet, we get $t_{dyn} = \dots$, which agrees within $x\%$ the estimate obtained from the Analysis tool provided by OpOpgadget, which measures this quantity directly from the generated model. Waiting for evolution with Ketju.

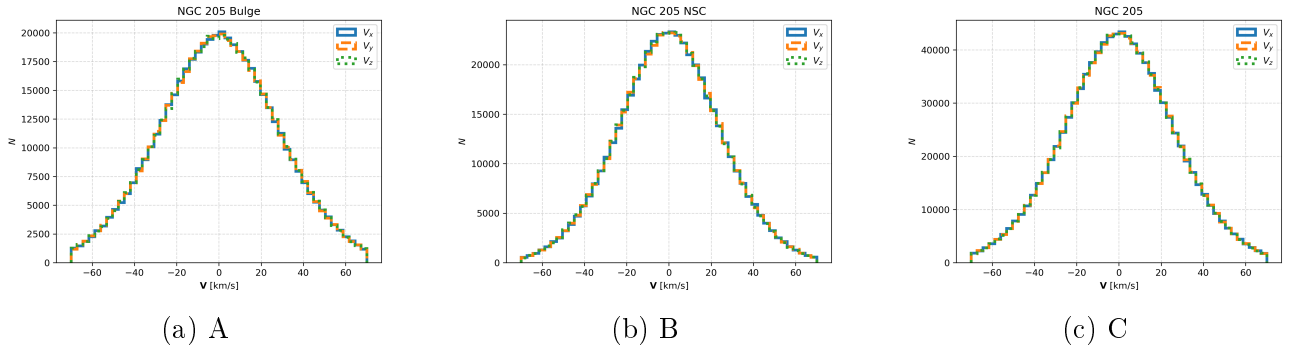


Figure 3.3: aaa

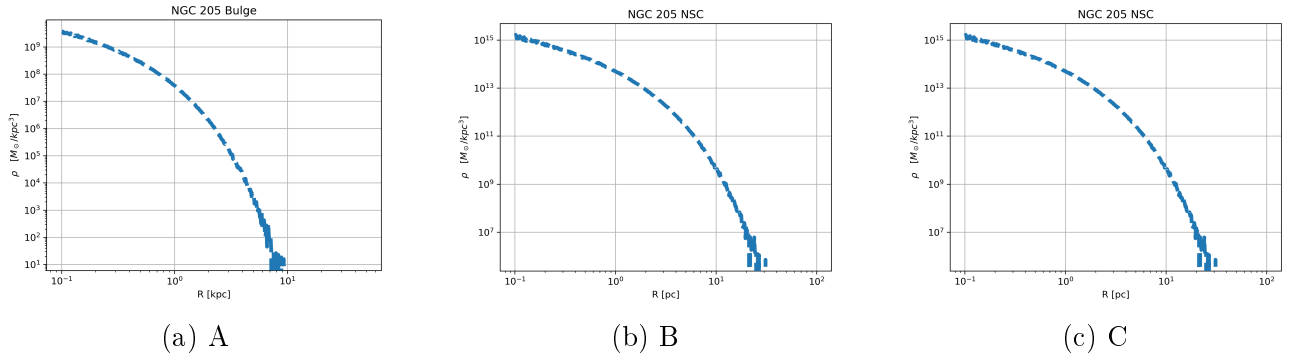


Figure 3.4: aaa

3.3.2 NCG 5206 N-body Initial Conditions

3.4 Ketju Numerical Simulations

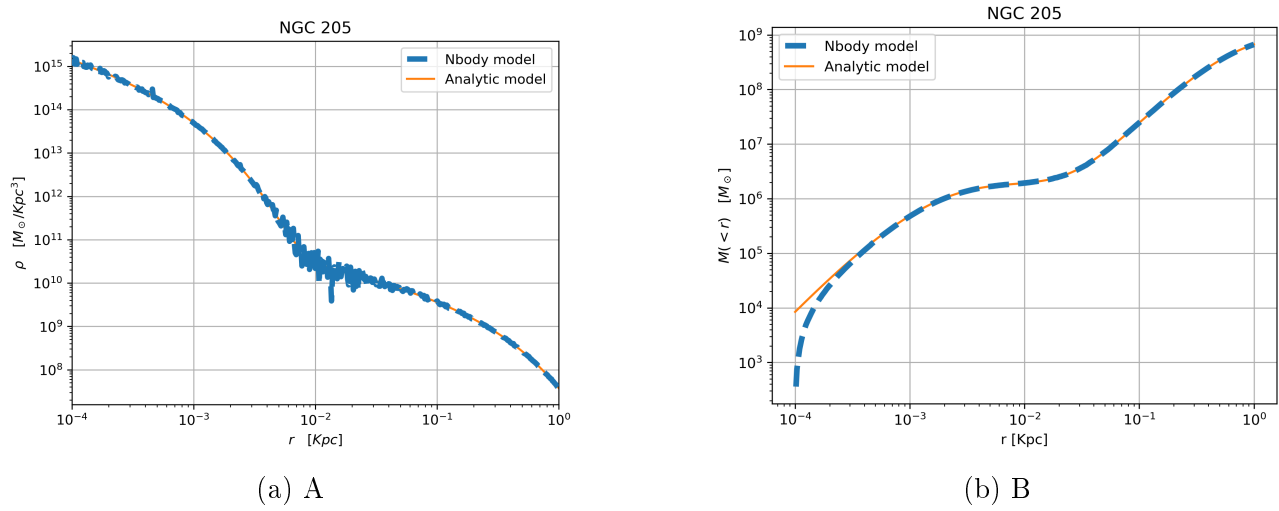


Figure 3.5: aaa

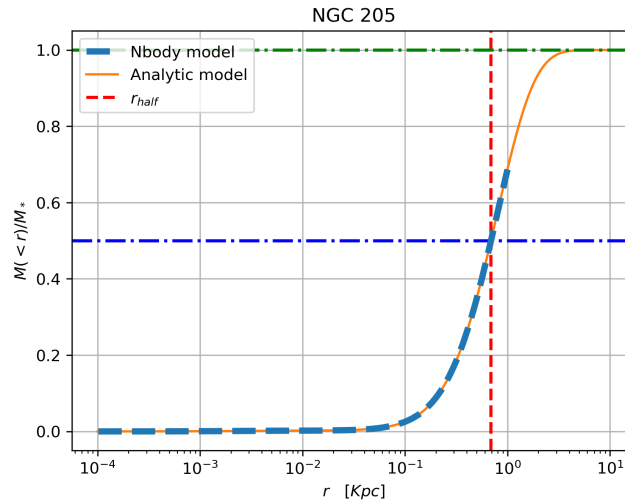


Figure 3.6: aaa

Things done:

- created double galaxy
- modified BHs position and velocity
- modified their position and velocity according to Nipoti paper (r_{90})
- built Hists to check position and velocity distribution
- verified dyn equilibrium
- computed the softening length in several ways
- computed dyn time
- ketju run
- BHs analysis with ketju-utils
- snapshot analysis with OpOpgadget
- method of shrinking spheres to get the density peak position

BIBLIOGRAPHY

- Abac, Adrian et al. (Mar. 2026). “The Science of the Einstein Telescope”. In: *JCAP* 2026.3, 081, p. 081. DOI: 10.1088/1475-7516/2026/03/081. arXiv: 2503.12263 [gr-qc].
- Abbott, B. P. et al. (Feb. 2016). “Observation of Gravitational Waves from a Binary Black Hole Merger”. In: *Phys. Rev. Lett.* 116 (6), p. 061102. DOI: 10.1103/PhysRevLett.116.061102. URL: <https://link.aps.org/doi/10.1103/PhysRevLett.116.061102>.
- Amaro-Seoane, Pau, Jeff Andrews, et al. (Mar. 2023). “Astrophysics with the Laser Interferometer Space Antenna”. English. In: *Living Reviews in Relativity* 26.1. ISSN: 2367-3613. DOI: 10.1007/s41114-022-00041-y.
- Amaro-Seoane, Pau, Heather Audley, et al. (Feb. 2017). “Laser Interferometer Space Antenna”. In: *arXiv e-prints*, arXiv:1702.00786, arXiv:1702.00786. DOI: 10.48550/arXiv.1702.00786. arXiv: 1702.00786 [astro-ph.IM].
- Bañados, Eduardo et al. (Jan. 2018). “An 800-million-solar-mass black hole in a significantly neutral Universe at a redshift of 7.5”. In: *Nature* 553.7689, pp. 473–476. DOI: 10.1038/nature25180. arXiv: 1712.01860 [astro-ph.GA].
- Barnes, Joshua E. and Lars Hernquist (Dec. 1992). “Formation of dwarf galaxies in tidal tails”. In: *Nature* 360.6406, pp. 715–717. DOI: 10.1038/360715a0.
- Baumgardt, H. (2017). “N-body modelling of globular clusters: masses, mass-to-light ratios and intermediate-mass black holes”. In: *Monthly Notices of the Royal Astronomical Society* 464.2, pp. 2174–2202. DOI: 10.1093/mnras/stw2488. arXiv: 1609.08794 [astro-ph.GA].
- Baumgardt, Holger, Alessia Gualandris, and Simon Portegies Zwart (2006). “Ejection of hypervelocity stars from the Galactic Centre by intermediate-mass black holes”. In: *Monthly Notices of the Royal Astronomical Society* 372.1, pp. 174–182.
- Bennett, DP et al. (2002). “Gravitational microlensing events due to stellar-mass black holes”. In: *The Astrophysical Journal* 579.2, pp. 639–659.
- Binney, James and Scott Tremaine (2011). *Galactic dynamics*. Princeton university press.
- Bradač, Maruša et al. (Dec. 2025). “Star Formation under a Cosmic Microscope: Highly Magnified $z = 11$ Galaxy behind the Bullet Cluster”. In: *The Astrophysical Journal Letters* 995.2, p. L74. DOI: 10.3847/2041-8213/ae27d2. URL: <https://doi.org/10.3847/2041-8213/ae27d2>.
- Butcher, John Charles (2016). *Numerical methods for ordinary differential equations*. John Wiley & Sons.
- Capozziello, Salvatore and Mariafelicia De Laurentis (2008). “Gravitational waves from stellar encounters”. In: *Astroparticle Physics* 30.2, pp. 105–112.
- Chandrasekhar, S. (Mar. 1943). “Dynamical Friction. I. General Considerations: the Coefficient of Dynamical Friction.” In: *The Astrophysical Journal* 97, p. 255. DOI: 10.1086/144517.
- Cimatti, Andrea, Filippo Fraternali, and Carlo Nipoti (2019). *Introduction to Galaxy Formation and Evolution: From Primordial Gas to Present-Day Galaxies*. Cambridge University Press. DOI: 10.1017/9781108610998.
- Ciotti, Luca (2021). *Introduction to Stellar Dynamics*. Cambridge: Cambridge University Press.

- Colpi, Monica et al. (Feb. 2024). “LISA Definition Study Report”. In: *arXiv e-prints*, arXiv:2402.07571, arXiv:2402.07571. DOI: 10.48550/arXiv.2402.07571. arXiv: 2402.07571 [astro-ph.CO].
- Devecchi, B. and M. Volonteri (Mar. 2009). “Formation of the First Nuclear Clusters and Massive Black Holes at High Redshift”. In: *The Astrophysical Journal* 694.1, pp. 302–313. DOI: 10.1088/0004-637X/694/1/302. arXiv: 0810.1057 [astro-ph].
- Farmer, R et al. (2019). “Mind the gap: the location of the lower edge of the pair-instability supernova black hole mass gap”. In: *The Astrophysical Journal* 887.1, p. 53.
- Fryer, C. L., S. E. Woosley, and A. Heger (Mar. 2001). “Pair-Instability Supernovae, Gravity Waves, and Gamma-Ray Transients”. In: *The Astrophysical Journal* 550.1, pp. 372–382. DOI: 10.1086/319719. arXiv: astro-ph/0007176 [astro-ph].
- Gebhardt, Karl, R. M. Rich, and Luis C. Ho (Dec. 2005). “An Intermediate-Mass Black Hole in the Globular Cluster G1: Improved Significance from New Keck and Hubble Space Telescope Observations”. In: *The Astrophysical Journal* 634.2, pp. 1093–1102. ISSN: 1538-4357. DOI: 10.1086/497023.
- Grebel, Eva K, John S Gallagher III, and Daniel Harbeck (2003). “The progenitors of dwarf spheroidal galaxies”. In: *The Astronomical Journal* 125.4, pp. 1926–1939.
- Greene, Jenny E., Jay Strader, and Luis C. Ho (Aug. 2020). “Intermediate-Mass Black Holes”. In: *Annual Review of Astronomy and Astrophysics* 58, pp. 257–312. DOI: 10.1146/annurev-astro-032620-021835. arXiv: 1911.09678 [astro-ph.GA].
- Gualandris, Alessia et al. (Jan. 2022). “Eccentricity evolution of massive black hole binaries from formation to coalescence”. In: *Monthly Notices of the Royal Astronomical Society* 511.4, pp. 4753–4765. ISSN: 1365-2966. DOI: 10.1093/mnras/stac241. URL: <http://dx.doi.org/10.1093/mnras/stac241>.
- Heggie, D. C. (Dec. 1975). “Binary evolution in stellar dynamics.” In: *mnras* 173, pp. 729–787. DOI: 10.1093/mnras/173.3.729.
- Heggie, Douglas C. and Piet Hut (Apr. 1993). “Binary–Single-Star Scattering. IV. Analytic Approximations and Fitting Formulae for Cross Sections and Reaction Rates”. In: *apjs* 85, p. 347. DOI: 10.1086/191768.
- Hills, J. G. (Aug. 1983). “The effect of low-velocity, low-mass intruders (collisionless gas) on the dynamical evolution of a binary system”. In: *aj* 88, pp. 1269–1283. DOI: 10.1086/113418.
- Hilz, Michael et al. (Oct. 2012). “Relaxation and stripping – The evolution of sizes, dispersions and dark matter fractions in major and minor mergers of elliptical galaxies”. In: *Monthly Notices of the Royal Astronomical Society* 425.4, pp. 3119–3136. ISSN: 0035-8711. DOI: 10.1111/j.1365-2966.2012.21541.x. eprint: <https://academic.oup.com/mnras/article-pdf/425/4/3119/4916245/425-4-3119.pdf>. URL: <https://doi.org/10.1111/j.1365-2966.2012.21541.x>.
- Holley-Bockelmann, Kelly et al. (Dec. 2025). “Handy Relation between Binary Black Hole Merger Times and Host Galaxy Properties”. In: *Astrophysical Journal Letters* 995.1, L32, p. L32. DOI: 10.3847/2041-8213/ae1ccd. arXiv: 2508.14253 [astro-ph.GA].
- Jahnke, Knud (Jan. 2025). “The Soltan argument at redshift 6: UV-luminous quasars contribute less than 10% to early black hole mass growth”. In: *The Open Journal of Astrophysics* 8, 9, p. 9. DOI: 10.33232/001c.129063. arXiv: 2411.03184 [astro-ph.GA].
- Khan, Fazeel Mahmood and Kelly Holley-Bockelmann (Nov. 2021). “Extremely efficient mergers of intermediate-mass black hole binaries in nucleated dwarf galaxies”. In: *MNRAS* 508.1, pp. 1174–1188. DOI: 10.1093/mnras/stab2646. arXiv: 2109.12129 [astro-ph.GA].
- Khan, Fazeel Mahmood, Fiza Javed, et al. (Nov. 2024). “The Potential for Long-lived Intermediate-mass Black Hole Binaries in the Lowest Density Dwarf Galaxies”. In: *The Astrophysical Journal* 976.1, 22, p. 22. DOI: 10.3847/1538-4357/ad8082. arXiv: 2408.14541 [astro-ph.GA].

- Khochfar, S. and J. Silk (July 2009). “Dry mergers: a crucial test for galaxy formation”. In: *MNRAS* 397.1, pp. 506–510. DOI: 10.1111/j.1365-2966.2009.14958.x. arXiv: 0809.1734 [astro-ph].
- Kikuchihara, Shotaro et al. (Apr. 2020). “Early Low-mass Galaxies and Star-cluster Candidates at $z \sim 6-9$ Identified by the Gravitational-lensing Technique and Deep Optical/Near-infrared Imaging”. In: *The Astrophysical Journal* 893.1, 60, p. 60. DOI: 10.3847/1538-4357/ab7dbe. arXiv: 1905.06927 [astro-ph.GA].
- Kormendy, John and Douglas Richstone (Jan. 1995). “Inward Bound—The Search For Super-massive Black Holes In Galactic Nuclei”. In: *Annual Review of Astronomy and Astrophysics* 33, p. 581. DOI: 10.1146/annurev.aa.33.090195.003053.
- Lightman, Alan P and Stuart L Shapiro (1977). “The distribution and consumption rate of stars around a massive, collapsed object”. In: *Astrophysical Journal*, vol. 211, Jan. 1, 1977, pt. 1, p. 244-262. 211, pp. 244–262.
- Lima Neto, G. B., D. Gerbal, and I. Marquez (Oct. 1999). “The specific entropy of elliptical galaxies: an explanation for profile-shape distance indicators?” In: *MNRAS* 309.2, pp. 481–495. ISSN: 1365-2966. DOI: 10.1046/j.1365-8711.1999.02849.x. URL: <http://dx.doi.org/10.1046/j.1365-8711.1999.02849.x>.
- Lodato, Giuseppe and Priyamvada Natarajan (Oct. 2006). “Supermassive black hole formation during the assembly of pre-galactic discs”. In: *mnras* 371.4, pp. 1813–1823. DOI: 10.1111/j.1365-2966.2006.10801.x. arXiv: astro-ph/0606159 [astro-ph].
- Lynden-Bell, D. (Jan. 1967). “Statistical mechanics of violent relaxation in stellar systems”. In: *mnras* 136, p. 101. DOI: 10.1093/mnras/136.1.101.
- Maggiore, Michele (2007). *Gravitational Waves: Volume 1: Theory and Experiments*. DOI: 10.1093/acprof:oso/9780198570745.001.0001.
- Magorrian, John et al. (June 1998). “The Demography of Massive Dark Objects in Galaxy Centers”. In: *Astrophysical Journal* 115.6, pp. 2285–2305. DOI: 10.1086/300353. arXiv: astro-ph/9708072 [astro-ph].
- Maiolino, Roberto et al. (Nov. 2024). “JADES: The diverse population of infant black holes at $4 < z < 11$: Merging, tiny, poor, but mighty”. In: *aap* 691, A145, A145. DOI: 10.1051/0004-6361/202347640. arXiv: 2308.01230 [astro-ph.GA].
- Mannerkoski, Matias et al. (Sept. 2023). “KETJU - resolving small-scale supermassive black hole dynamics in GADGET-4”. In: *MNRAS* 524.3, pp. 4062–4082. DOI: 10.1093/mnras/stad2139. arXiv: 2306.04963 [astro-ph.IM].
- Meneghetti, Massimo (2022). *Introduction to Gravitational Lensing: With Python Examples*.
- Merritt, David (Nov. 2013). “Loss-cone dynamics”. In: *Classical and Quantum Gravity* 30.24, p. 244005. ISSN: 1361-6382. DOI: 10.1088/0264-9381/30/24/244005. URL: <http://dx.doi.org/10.1088/0264-9381/30/24/244005>.
- Merritt, David and Miloš Milosavljević (2005). “Massive Black Hole Binary Evolution”. In: *Living Reviews in Relativity* 8. ISSN: 1433-8351. DOI: 10.12942/lrr-2005-8. URL: <http://dx.doi.org/10.12942/lrr-2005-8>.
- Mikkola, Seppo and Mauri J. Valtonen (Nov. 1992). “Evolution of binaries in the field of light particles and the problem of two black holes”. In: *Monthly Notices of the Royal Astronomical Society* 259.1, pp. 115–120. ISSN: 0035-8711. DOI: 10.1093/mnras/259.1.115. eprint: <https://academic.oup.com/mnras/article-pdf/259/1/115/3513810/mnras259-0115.pdf>. URL: <https://doi.org/10.1093/mnras/259.1.115>.
- Milosavljević, Miloš (2003). “The Final Parsec Problem”. In: *AIP Conference Proceedings*. Vol. 686. AIP, pp. 201–210. DOI: 10.1063/1.1629432. URL: <http://dx.doi.org/10.1063/1.1629432>.

- Milosavljević, Miloš and David Merritt (Dec. 2001). “Formation of Galactic Nuclei”. In: *The Astrophysical Journal* 563.1, p. 34. DOI: 10.1086/323830. URL: <https://doi.org/10.1086/323830>.
- Nakane, Minami et al. (2026). *VENUS: A Strongly Lensed Clumpy Galaxy at $z \sim 11 - 12$ behind the Galaxy Cluster MACS J0257.1-2325*. arXiv: 2511.14483 [astro-ph.GA]. URL: <https://arxiv.org/abs/2511.14483>.
- Neumayer, Nadine, Anil Seth, and Torsten Böker (2020). “Nuclear star clusters”. In: *The Astronomy and Astrophysics Review* 28.1. ISSN: 1432-0754. DOI: 10.1007/s00159-020-00125-0. URL: <http://dx.doi.org/10.1007/s00159-020-00125-0>.
- Nguyen, Dieu D. et al. (May 2018). “Nearby Early-type Galactic Nuclei at High Resolution: Dynamical Black Hole and Nuclear Star Cluster Mass Measurements”. In: *The Astrophysical Journal* 858.2, p. 118. DOI: 10.3847/1538-4357/aabe28. URL: <https://doi.org/10.3847/1538-4357/aabe28>.
- Nipoti, Carlo, Pasquale Londrillo, and Luca Ciotti (Oct. 2007). “Galaxy merging in modified Newtonian dynamics”. In: *Monthly Notices of the Royal Astronomical Society: Letters* 381.1, pp. L104–L108. ISSN: 1745-3925. DOI: 10.1111/j.1745-3933.2007.00378.x. URL: <http://dx.doi.org/10.1111/j.1745-3933.2007.00378.x>.
- Ono, Yoshiaki et al. (July 2023). “Morphologies of Galaxies at $z \gtrsim 9$ Uncovered by JWST/NIRCam Imaging: Cosmic Size Evolution and an Identification of an Extremely Compact Bright Galaxy at $z \sim 12$ ”. In: *Astrophysical Journal* 951.1, 72, p. 72. DOI: 10.3847/1538-4357/acd44a. arXiv: 2208.13582 [astro-ph.GA].
- Perets, Hagai B. and Tal Alexander (Apr. 2008). “Massive Perturbers and the Efficient Merger of Binary Massive Black Holes”. In: *The Astrophysical Journal* 677.1, p. 146. DOI: 10.1086/527525. URL: <https://doi.org/10.1086/527525>.
- Perets, Hagai B., Clovis Hopman, and Tal Alexander (Feb. 2007). “Massive Perturber-driven Interactions between Stars and a Massive Black Hole”. In: *The Astrophysical Journal* 656.2, pp. 709–720. DOI: 10.1086/510377. arXiv: astro-ph/0606443 [astro-ph].
- Peters, P. C. and J. Mathews (July 1963). “Gravitational Radiation from Point Masses in a Keplerian Orbit”. In: *Physical Review* 131.1, pp. 435–440. DOI: 10.1103/PhysRev.131.435.
- Peterson, Bradley M (1997). *An introduction to active galactic nuclei*. Cambridge University Press.
- Quinlan, Gerald D. (July 1996). “The dynamical evolution of massive black hole binaries I. Hardening in a fixed stellar background”. In: *New Astronomy* 1.1, pp. 35–56. DOI: 10.1016/S1384-1076(96)00003-6. arXiv: astro-ph/9601092 [astro-ph].
- Reines, Amy E., Jenny E. Greene, and Marla Geha (Oct. 2013). “Dwarf Galaxies with Optical Signatures of Active Massive Black Holes”. In: *Astrophysical Journal* 775.2, 116, p. 116. DOI: 10.1088/0004-637X/775/2/116. arXiv: 1308.0328 [astro-ph.CO].
- Rodriguez, Carl L. et al. (Aug. 2019). “Black holes: The next generation—repeated mergers in dense star clusters and their gravitational-wave properties”. In: *Physical Review D* 100.4. ISSN: 2470-0029. DOI: 10.1103/physrevd.100.043027. URL: <http://dx.doi.org/10.1103/PhysRevD.100.043027>.
- Saslaw, William C. (Oct. 1970). “Violent Relaxation in Galaxies and Stellar Systems”. In: *Monthly Notices of the Royal Astronomical Society* 150.3, pp. 299–309. ISSN: 0035-8711. DOI: 10.1093/mnras/150.3.299. URL: <https://doi.org/10.1093/mnras/150.3.299>.
- Schutte, Zachary, Amy E. Reines, and Jenny E. Greene (Dec. 2019). “The Black Hole-Bulge Mass Relation Including Dwarf Galaxies Hosting Active Galactic Nuclei”. In: *The Astrophysical Journal* 887.2, 245, p. 245. DOI: 10.3847/1538-4357/ab35dd. arXiv: 1908.00020 [astro-ph.GA].
- Sesana, Alberto (Aug. 2010). “Self Consistent Model for the Evolution of Eccentric Massive Black Hole Binaries in Stellar Environments: Implications for Gravitational Wave Observa-

- tions". In: *Astrophysical Journal* 719.1, pp. 851–864. DOI: 10.1088/0004-637X/719/1/851. arXiv: 1006.0730 [astro-ph.CO].
- Sesana, Alberto, Francesco Haardt, and Piero Madau (Nov. 2006). "Interaction of Massive Black Hole Binaries with Their Stellar Environment. I. Ejection of Hypervelocity Stars". In: *Astrophysical Journal* 651.1, pp. 392–400. DOI: 10.1086/507596. arXiv: astro-ph/0604299 [astro-ph].
- (Oct. 2008). "Interaction of Massive Black Hole Binaries with Their Stellar Environment. III. Scattering of Bound Stars". In: *Astrophysical Journal* 686.1, pp. 432–447. DOI: 10.1086/590651. arXiv: 0710.4301 [astro-ph].
- Sesana, Alberto and Fazeel Mahmood Khan (Nov. 2015). "Scattering experiments meet N-body - I. A practical recipe for the evolution of massive black hole binaries in stellar environments". In: *MNRAS* 454.1, pp. L66–L70. DOI: 10.1093/mnrasl/slv131. arXiv: 1505.02062 [astro-ph.GA].
- Springel, Volker (Dec. 2005). "The cosmological simulation code gadget-2". In: *Monthly Notices of the Royal Astronomical Society* 364.4, pp. 1105–1134. ISSN: 0035-8711. DOI: 10.1111/j.1365-2966.2005.09655.x. eprint: <https://academic.oup.com/mnras/article-pdf/364/4/1105/18657201/364-4-1105.pdf>. URL: <https://doi.org/10.1111/j.1365-2966.2005.09655.x>.
- Taylor, Stephen R. (2021). *The Nanohertz Gravitational Wave Astronomer*. arXiv: 2105.13270 [astro-ph.HE]. URL: <https://arxiv.org/abs/2105.13270>.
- Toomre, Alar and Juri Toomre (Dec. 1972). "Galactic Bridges and Tails". In: *ApJ* 178, pp. 623–666. DOI: 10.1086/151823.
- Tremaine, Scott et al. (Aug. 2002). "The Slope of the Black Hole Mass versus Velocity Dispersion Correlation". In: *Astrophysical Journal* 574.2, pp. 740–753. DOI: 10.1086/341002. arXiv: astro-ph/0203468 [astro-ph].
- White, S. D. M. and M. J. Rees (May 1978). "Core condensation in heavy halos: a two-stage theory for galaxy formation and clustering." In: *mnras* 183, pp. 341–358. DOI: 10.1093/mnras/183.3.341.